

Formal Book

formalizing “Proofs from THE BOOK” by Martin Aigner and Günter M. Ziegler

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Chapter 1

Six proofs of the infinity of primes

Theorem 1.1 (Euclid's proof). *A finite set $\{p_1, \dots, p_r\}$ cannot be the collection of all prime numbers.*

Proof. For any finite set $\{p_1, \dots, p_r\}$ of primes, consider the number $n = p_1 p_2 \cdots p_r + 1$. This n has a prime divisor p . But p is not one of the p_i 's: otherwise p would be a divisor of n and of the product $p_1 p_2 \cdots p_r$, and thus also of the difference $n - p_1 p_2 \cdots p_r = 1$, which is impossible. So a finite set $\{p_1, \dots, p_r\}$ cannot be the collection of *all* prime numbers. $\square$

Theorem 1.2 (Second Proof). *Any two Fermat numbers $F_n := 2^{2^n} + 1$ are relatively prime.*

Proof. Let us first look at the Fermat numbers $F_n = 2^{2^n} + 1$ for $n = 0, 1, 2, \dots$. We will show that any two Fermat numbers are relatively prime; hence there must be infinitely many primes. To this end, we verify the recursion

$$\prod_{k=0}^{n-1} F_k = F_n - 2,$$

from which our assertion follows immediately. Indeed, if m is a divisor of, say, F_k and F_n (with $k < n$), then m divides 2, and hence $m = 1$ or 2. But $m = 2$ is impossible since all Fermat numbers are odd. To prove the recursion we use induction on n . For $n = 1$, we have $F_0 = 3$ and $F_1 - 2 = 3$. With induction we now conclude

$$\prod_{k=0}^n F_k = \left(\prod_{k=0}^{n-1} F_k \right) F_n = (F_n - 2) F_n = (2^{2^n} - 1)(2^{2^n} + 1) = 2^{2^{n+1}} - 1 = F_{n+1} - 2.$$

$\square$

Theorem 1.3 (Third Proof). *There is no largest prime.*

Proof. Suppose $\mathbb{P}$ is finite and p is the largest prime. We consider the so-called *Mersenne number* $2^p - 1$ and show that any prime factor q of $2^p - 1$ is bigger than p , which will yield the desired conclusion. Let q be a prime dividing $2^p - 1$, so we have $2^p \equiv 1 \pmod{q}$. Since p is prime, this means that the element 2 has order p in the multiplicative group $\mathbb{Z}_q \setminus \{0\}$ of the field $\mathbb{Z}_q$. This group has $q - 1$ elements. By Lagrange's theorem, we know that the order of every element divides the size of the group, that is, we have $p \mid q - 1$, and hence $p < q$. $\square$

Theorem 1.4 (Fourth Proof). *The prime counting function is unbounded*

Proof. Let $\pi(x) := \#\{p \leq x : p \in \mathbb{P}\}$ be the number of primes that are less than or equal to the real number x . We number the primes $\mathbb{P} = \{p_1, p_2, p_3, \dots\}$ in increasing order. Consider the natural logarithm $\log x$, defined as

$$\log x = \int_1^x \frac{1}{t} dt.$$

Now we compare the area below the graph of $f(t) = \frac{1}{t}$ with an upper step function. (See also the appendix for this method.) Thus for $n \leq x < n+1$ we have

$$\log x \leq 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n-1} + \frac{1}{n} \leq \sum \frac{1}{m},$$

where the sum extends over all $m \in \mathbb{N}$ which have only prime divisors $p \leq x$.

Since every such m can be written in a unique way as a product of the form $\prod_{p \leq x} p^{k_p}$, we see that the last sum is equal to

$$\prod_{p \in \mathbb{P}, p \leq x} \left(\sum_{k \geq 0} \frac{1}{p^k} \right).$$

The inner sum is a geometric series with ratio $\frac{1}{p}$, hence

$$\log x \leq \prod_{p \leq x} \frac{1}{1 - \frac{1}{p}} = \prod_{p \leq x} \frac{p}{p-1} = \prod_{k=1}^{\pi(x)} \frac{p_k}{p_k-1}.$$

Now clearly $p_k \geq k+1$, and thus

$$\frac{p_k}{p_k-1} = 1 + \frac{1}{p_k-1} \leq 1 + \frac{1}{k} = \frac{k+1}{k},$$

and therefore

$$\log x \leq \prod_{k=1}^{\pi(x)} \frac{k+1}{k} = \pi(x) + 1.$$

Everybody knows that $\log x$ is not bounded, so we conclude that $\pi(x)$ is unbounded as well, and so there are infinitely many primes. $\square$

Theorem 1.5 (Fifth Proof). *The set of primes $\mathbb{P}$ is infinite.*

Proof. Consider the following curious topology on the set $\mathbb{Z}$ of integers. For $a, b \in \mathbb{Z}, b > 0$, we set

$$N_{a,b} = \{a + nb : n \in \mathbb{Z}\}.$$

Each set $N_{a,b}$ is a two-way infinite arithmetic progression. Now call a set $O \subseteq \mathbb{Z}$ open if either O is empty, or if to every $a \in O$ there exists some $b > 0$ with $N_{a,b} \subseteq O$. Clearly, the union of open sets is open again. If O_1, O_2 are open, and $a \in O_1 \cap O_2$ with $N_{a,b_1} \subseteq O_1$ and $N_{a,b_2} \subseteq O_2$, then $a \in N_{a,b_1 b_2} \subseteq O_1 \cap O_2$. So we conclude that any finite intersection of open sets is again open. Therefore, this family of open sets induces a bona fide topology on $\mathbb{Z}$.

Let us note two facts:

- (A) Any nonempty open set is infinite.
- (B) Any set $N_{a,b}$ is closed as well.

Indeed, the first fact follows from the definition. For the second, we observe

$$N_{a,b} = \mathbb{Z} \setminus \bigcup_{i=1}^{b-1} N_{a+i,b},$$

which proves that $N_{a,b}$ is the complement of an open set and hence closed.

So far, the primes have not yet entered the picture — but here they come. Since any number $n \neq 1, -1$ has a prime divisor p , and hence is contained in $N_{0,p}$, we conclude

$$\mathbb{Z} \setminus \{1, -1\} = \bigcup_{p \in \mathbb{P}} N_{0,p}.$$

Now if $\mathbb{P}$ were finite, then $\bigcup_{p \in \mathbb{P}} N_{0,p}$ would be a finite union of closed sets (by (B)), and hence closed. Consequently, $\{1, -1\}$ would be an open set, in violation of (A). $\square$

Theorem 1.6 (Sixth Proof). *The series $\sum_{p \in \mathbb{P}} \frac{1}{p}$ diverges.*

Proof. Our final proof goes a considerable step further and demonstrates not only that there are infinitely many primes, but also that the series $\sum_{p \in \mathbb{P}} \frac{1}{p}$ diverges. The first proof of this important result was given by Euler (and is interesting in its own right), but our proof, devised by Erdős, is of compelling beauty.

Let $p_1, p_2, p_3, \dots$ be the sequence of primes in increasing order, and assume that $\sum_{p \in \mathbb{P}} \frac{1}{p}$ converges. Then there must be a natural number k such that $\sum_{i \geq k+1} \frac{1}{p_i} < \frac{1}{2}$. Let us call $p_1, \dots, p_k$ the small primes, and $p_{k+1}, p_{k+2}, \dots$ the big primes. For an arbitrary natural number N , we therefore find

$$\sum_{i \geq k+1} \frac{N}{p_i} < \frac{N}{2}. \quad (1)$$

Let N_b be the number of positive integers $n \leq N$ which are divisible by at least one big prime, and N_s the number of positive integers $n \leq N$ which have only small prime divisors. We are going to show that for a suitable N

$$N_b + N_s < N,$$

which will be our desired contradiction, since by definition $N_b + N_s$ would have to be equal to N .

To estimate N_b , note that $\left\lfloor \frac{N}{p_i} \right\rfloor$ counts the positive integers $n \leq N$ which are multiples of p_i . Hence by (1) we obtain

$$N_b \leq \sum_{i \geq k+1} \left\lfloor \frac{N}{p_i} \right\rfloor < \frac{N}{2}. \quad (2)$$

Let us now look at N_s . We write every $n \leq N$ which has only small prime divisors in the form $n = a_n b_n^2$, where a_n is the square-free part. Every a_n is thus a product of *different* small primes, and we conclude that there are precisely 2^k different square-free parts. Furthermore, as $b_n^2 \leq n \leq N$, we find that there are at most $\sqrt{N}$ different square parts, and so

$$N_s \leq 2^k \sqrt{N}.$$

Since (2) holds for any N , it remains to find a number N with $2^k \sqrt{N} < \frac{N}{2}$, or $2^{k+1} < \sqrt{N}$, and for this $N = 2^{2k+2}$ will do. $\square$

1.1 Appendix: Infinitely many more proofs

Theorem 1.7. *If the sequence $S = (s_1, s_2, s_3, \dots)$ is almost injective and of subexponential growth, then the set $\mathbb{P}_S$ of primes that divide some member of S is infinite.*

Proof. We may assume that $f(n)$ is monotonely increasing. Otherwise, replace $f(n)$ by $F(n) = \max_{i \leq n} f(i)$; you can easily check that with this $F(n)$ the sequence S again satisfies the subexponential growth condition.

Let us suppose for a contradiction that $\mathbb{P}_S = \{p_1, \dots, p_k\}$ is finite. For $n \in \mathbb{N}$, let

$$s_n = \varepsilon_n p_1^{\alpha_1} \cdots p_k^{\alpha_k}, \quad \text{with } \varepsilon_n \in \{1, 0, -1\}, \alpha_i \geq 0,$$

where the $\alpha_i = \alpha_i(n)$ depend on n . (For $s_n = 0$ we can put $\alpha_i = 0$ for all i .) Then

$$2^{\alpha_1 + \cdots + \alpha_k} \leq |s_n| \leq 2^{2^{f(n)}} \quad \text{for } s_n \neq 0,$$

and thus by taking the binary logarithm

$$0 \leq \alpha_i \leq \alpha_1 + \cdots + \alpha_k \leq 2^{f(n)} \quad \text{for } 1 \leq i \leq k.$$

Hence there are not more than $2^{f(n)} + 1$ different possible values for each $\alpha_i = \alpha_i(n)$. Since f is monotone, this gives a first estimate

$$\#\{\text{distinct } |s_n| \neq 0 \text{ for } n \leq N\} \leq (2^{f(N)} + 1)^k \leq 2^{(f(N)+1)k}.$$

On the other hand, since S is almost injective only c terms in the sequence can be equal to 0, and each nonzero absolute value can occur at most $2c$ times, so we get the lower estimate

$$\#\{\text{distinct } |s_n| \neq 0 \text{ for } n \leq N\} \geq \frac{N - c}{2c}.$$

Altogether, this gives

$$\frac{N - c}{2c} \leq 2^{k(f(N)+1)}.$$

Taking again the logarithm with base 2 on both sides, we obtain

$$\log_2(N - c) - \log_2(2c) \leq k(f(N) + 1) \quad \text{for all } N.$$

This, however, is plainly false for large N , as k and c are constants, so $\frac{\log_2(N-c)}{\log_2 N}$ goes to 1 for $N \rightarrow \infty$, while $\frac{f(N)}{\log_2 N}$ goes to 0. □

Theorem 1.8 (Infinity of primes). *There are infinitely many primes. (Six + infinitely many proofs)*

Proof. See theorems in this chapter. □

Chapter 2

Bertrand's postulate

Theorem 2.1 (Bertrand's postulate). *For any positive natural number, there is a prime which is greater than it, but no more than twice as large.*

Proof. We will estimate the size of the binomial coefficient $\binom{2n}{n}$ carefully enough to see that if it didn't have any prime factors in the range $n < p \leq 2n$, then it would be “too small.” Our argument is in five steps.

1. We first prove Bertrand's postulate for $n \leq 511$. For this one does not need to check 511 cases: it suffices (this is “Landau's trick”) to check that

$$2, 3, 5, 7, 13, 23, 43, 83, 163, 317, 521$$

is a sequence of prime numbers, where each is smaller than twice the previous one. Hence every interval $\{y : n < y \leq 2n\}$, with $n \leq 511$, contains one of these 11 primes.

2. Next we prove that

$$\prod_{p \leq x} p \leq 4^{x-1} \quad \text{for all real } x \geq 2, \quad (2.1)$$

where our notation — here and in the following — is meant to imply that the product is taken over all prime numbers $p \leq x$. The proof that we present for this fact uses induction on the number of these primes. It is not from Erdős' original paper, but it is also due to Erdős, and it is a true Book Proof. First we note that if q is the largest prime with $q \leq x$, then

$$\prod_{p \leq x} p = \prod_{p \leq q} p \quad \text{and} \quad 4^{q-1} \leq 4^{x-1}.$$

Thus it suffices to check (2.1) for the case where $x = q$ is a prime number. For $q = 2$ we get “ $2 \leq 4$,” so we proceed to consider odd primes $q = 2m + 1$. (Here we may assume, by induction, that (2.1) is valid for all integers x in the set $\{2, 3, \dots, 2m\}$.) For $q = 2m + 1$ we split the product and compute

$$\prod_{p \leq 2m+1} p = \prod_{p \leq m+1} p \cdot \prod_{m+1 < p \leq 2m+1} p \leq 4^m \binom{2m+1}{m} \leq 4^m 2^{2m} = 4^{2m}.$$

All the pieces of this “one-line computation” are easy to see. In fact,

$$\prod_{p \leq m+1} p \leq 4^m$$

holds by induction. The inequality

$$\prod_{m+1 < p \leq 2m+1} p \leq \binom{2m+1}{m}$$

follows from the observation that $\binom{2m+1}{m} = \frac{(2m+1)!}{m!(m+1)!}$ is an integer, where the primes that we consider all are factors of the numerator $(2m+1)!$, but not of the denominator $m!(m+1)!$. Finally

$$\binom{2m+1}{m} \leq 2^{2m}$$

holds since

$$\binom{2m+1}{m} \quad \text{and} \quad \binom{2m+1}{m+1}$$

are two (equal!) summands that appear in

$$\sum_{k=0}^{2m+1} \binom{2m+1}{k} = 2^{2m+1}.$$

3. From Legendre's theorem we get that $\binom{2n}{n} = \frac{(2n)!}{n!n!}$ contains the prime factor p exactly

$$\sum_{k \geq 1} \left(\left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \left\lfloor \frac{n}{p^k} \right\rfloor \right)$$

times. Here each summand is at most 1, since it satisfies

$$\left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \left\lfloor \frac{n}{p^k} \right\rfloor < \frac{2n}{p^k} - 2 \left(\frac{n}{p^k} - 1 \right) = 2,$$

and it is an integer. Furthermore the summands vanish whenever $p^k > 2n$. Thus $\binom{2n}{n}$ contains p exactly

$$\sum_{k \geq 1} \left(\left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \left\lfloor \frac{n}{p^k} \right\rfloor \right) \leq \max\{r : p^r \leq 2n\}$$

times. Hence the largest power of p that divides $\binom{2n}{n}$ is not larger than $2n$. In particular, primes $p > \sqrt{2n}$ appear at most once in $\binom{2n}{n}$.

Furthermore — and this, according to Erdős, is the key fact for his proof — primes p that satisfy $\frac{2}{3}n < p < n$ do not divide $\binom{2n}{n}$ at all! Indeed, $3p > 2n$ implies (for $n \geq 3$, and hence $p \geq 3$) that p and $2p$ are the only multiples of p that appear as factors in the numerator of $\frac{(2n)!}{n!n!}$, while we get two p -factors in the denominator.

4. Now we are ready to estimate $\binom{2n}{n}$, benefitting from a suggestion by Raimund Seidel, which nicely improves Erdős' original argument. For $n \geq 3$, using an estimate for the lower bound, we get

$$\frac{4^n}{2n} \leq \binom{2n}{n} \leq \prod_{p \leq \sqrt{2n}} 2n \cdot \prod_{\sqrt{2n} < p \leq \frac{2}{3}n} p \cdot \prod_{n < p \leq 2n} p.$$

Now, there are no more than $\sqrt{2n}$ primes in the first factor; hence using (1) for the second factor and letting $P(n)$ denote the number of primes between n and $2n$ we get

$$\frac{4^n}{2n} < ((2n)^{\sqrt{2n}}) \cdot (4^{\frac{2}{3}n}) \cdot (2n)^{P(n)},$$

that is,

$$4^{\frac{n}{3}} < (2n)^{\sqrt{2n}+1+P(n)}. \quad (2)$$

5. Taking the logarithm to base 2, the last inequality is turned into

$$P(n) > \frac{2n}{3 \log_2(2n)} - (\sqrt{2n} + 1). \quad (3)$$

It remains to verify that the right-hand side of (3) is positive for n large enough. We show that this is the case for $n = 2^9 = 512$ (actually, it holds for $n = 468$ onward). By writing $2n - 1 = (\sqrt{2n} - 1)(\sqrt{2n} + 1)$ and cancelling the $(\sqrt{2n} + 1)$ -factor it suffices to show

$$\sqrt{2n} - 1 > 3 \log_2(2n) \quad \text{for } n \geq 2^9. \quad (4)$$

For $n = 2^9$, (4) becomes $31 > 30$, and comparing the derivatives $(\sqrt{x} - 1)' = \frac{1}{2\sqrt{x}}$ and $(3 \log_2 x)' = \frac{3}{\log 2} \frac{1}{x}$ we see that $\sqrt{x} - 1$ grows faster than $3 \log_2 x$ for $x > (\frac{6}{\log 2})^2 \approx 75$ and thus certainly for $x \geq 2^{10} = 1024$. $\square$

2.1 Appendix: Some estimates

Theorem 2.2. *For all $n \in \mathbb{N}$*

$$\log n + \frac{1}{n} < H_n < \log n + 1.$$

Proof. There is a very simple-but-effective method of estimating sums by integrals. For estimating the harmonic numbers

$$H_n = \sum_{k=1}^n \frac{1}{k}$$

we draw the figure and derive from it

$$H_n - 1 = \sum_{k=2}^n \frac{1}{k} < \int_1^n \frac{1}{t} dt = \log n$$

by comparing the area below the graph of $f(t) = \frac{1}{t}$ ($1 \leq t \leq n$) with the area of the dark shaded rectangles, and

$$H_n - \frac{1}{n} = \sum_{k=1}^{n-1} \frac{1}{k} > \int_1^n \frac{1}{t} dt = \log n$$

by comparing with the area of the large rectangles (including the lightly shaded parts). Taken together, this yields

$$\log n + \frac{1}{n} < H_n < \log n + 1. \quad \square$$

Theorem 2.3. *For all $n \in \mathbb{N}$*

$$e \left(\frac{n}{e} \right)^n < n! < en \left(\frac{n}{e} \right)^n.$$

Proof. The same method applied to

$$\log(n!) = \log 2 + \log 3 + \cdots + \log n = \sum_{k=2}^n \log k$$

yields

$$\log((n-1)!) < \int_1^n \log t \, dt < \log(n!),$$

where the integral is easily computed:

$$\int_1^n \log t \, dt = [t \log t - t]_1^n = n \log n - n + 1.$$

Thus we get a lower estimate on $n!$

$$n! > e^{n \log n - n + 1} = e \left(\frac{n}{e} \right)^n$$

and at the same time an upper estimate

$$n! = n(n-1)! < n e^{n \log n - n + 1} = e n \left(\frac{n}{e} \right)^n. \quad \square$$

Theorem 2.4.

$$\binom{n}{k} \leq \frac{n^k}{k!} \leq \frac{n^k}{2^{k-1}}$$

Proof.

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!} \leq \frac{n^k}{k!} \leq \frac{n^k}{2^{k-1}}. \quad \square$$

Chapter 3

Binomial coefficients are (almost) never powers

Theorem 3.1 (Sylvester's theorem). *For all positive natural n, k such that $n \geq 2k$, at least one of the numbers $n, n-1, \dots, n-k+1$ has a prime divisor p greater than k , or, equivalently the binomial coefficient $\binom{n}{k}$ always has a prime factor $p > k$.*

Proof. TODO □

Theorem 3.2 (Binomial coefficients are (almost) never powers). *The equation $\binom{n}{k} = m^l$ has no integer solutions with $l \geq 2$ and $4 \leq k \leq n-4$.*

Proof. Note first that we may assume $n \geq 2k$ because of $\binom{n}{k} = \binom{n}{n-k}$. Suppose the theorem is false, and that $\binom{n}{k} = m^\ell$. The proof, by contradiction, proceeds in the following four steps.

1. By Sylvester's theorem 3.1, there is a prime factor p of $\binom{n}{k}$ greater than k , hence p^ℓ divides $n(n-1) \dots (n-k+1)$. Clearly, only one of the factors $n-i$ can be a multiple of p (because $p > k$), and we conclude $p^\ell \mid n-i$, and therefore

$$n \geq p^\ell > k^\ell \geq k^2.$$

2. Consider any factor $n-j$ of the numerator and write it in the form $n-j = a_j m_j^\ell$, where a_j is not divisible by any nontrivial ℓ -th power. We note by (1) that a_j has only prime divisors less than or equal to k . We want to show next that $a_i \neq a_j$ for $i \neq j$. Assume to the contrary that $a_i = a_j$ for some $i < j$. Then $m_i \geq m_j + 1$ and

$$\begin{aligned} k &> (n-i) - (n-j) = a_j(m_i^\ell - m_j^\ell) \geq a_j((m_j+1)^\ell - m_j^\ell) \\ &> a_j \ell m_j^{\ell-1} \geq \ell(a_j m_j^\ell)^{1/2} \geq \ell(n-k+1)^{1/2} \\ &\geq \ell \left(\frac{n}{2} + 1 \right)^{1/2} > n^{1/2}, \end{aligned}$$

which contradicts $n > k^2$ from above.

3. Next we prove that the a_i 's are the integers $1, 2, \dots, k$ in some order. (According to Erdős, this is the crux of the proof.) Since we already know that they are all distinct, it suffices to prove that

$$a_0 a_1 \dots a_{k-1} \text{ divides } k!.$$

Substituting $n - j = a_j m_j^\ell$ into the equation $\binom{n}{k} = m^\ell$, we obtain

$$a_0 a_1 \cdots a_{k-1} (m_0 m_1 \cdots m_{k-1})^\ell = k! m^\ell.$$

Canceling the common factors of $m_0 \cdots m_{k-1}$ and m yields

$$a_0 a_1 \cdots a_{k-1} u^\ell = k! v^\ell$$

with $\gcd(u, v) = 1$. It remains to show that $v = 1$. If not, then v contains a prime divisor p . Since $\gcd(u, v) = 1$, p must be a prime divisor of $a_0 a_1 \cdots a_{k-1}$ and hence is less than or equal to k . By the theorem of Legendre, we know that $k!$ contains p to the power $\sum_{i \geq 1} \left\lfloor \frac{k}{p^i} \right\rfloor$. We now estimate the exponent of p in $n(n-1) \cdots (n-k+1)$. Let i be a positive integer, and let $b_1 < b_2 < \cdots < b_s$ be the multiples of p^i among $n, n-1, \dots, n-k+1$. Then $b_s = b_1 + (s-1)p^i$ and hence

$$(s-1)p^i = b_s - b_1 \leq n - (n-k+1) = k-1,$$

which implies

$$s \leq \left\lfloor \frac{k-1}{p^i} \right\rfloor + 1 \leq \left\lfloor \frac{k}{p^i} \right\rfloor + 1.$$

So for each i , the number of multiples of p^i among $n, \dots, n-k+1$, and hence among the a_j 's, is bounded by $\left\lfloor \frac{k}{p^i} \right\rfloor + 1$. This implies that the exponent of p in $a_0 a_1 \cdots a_{k-1}$ is at most

$$\sum_{i=1}^{\ell-1} \left(\left\lfloor \frac{k}{p^i} \right\rfloor + 1 \right)$$

with the reasoning that we used for Legendre's theorem in Chapter 2. The only difference is that this time the sum stops at $i = \ell - 1$, since the a_j 's contain no ℓ -th powers.

Taking both counts together, we find that the exponent of p in v^ℓ is at most

$$\sum_{i=1}^{\ell-1} \left(\left\lfloor \frac{k}{p^i} \right\rfloor + 1 \right) - \sum_{i \geq 1} \left\lfloor \frac{k}{p^i} \right\rfloor \leq \ell - 1,$$

and we have our desired contradiction, since v^ℓ is an ℓ -th power.

This suffices already to settle the case $\ell = 2$. Indeed, since $k \geq 4$, one of the a_i 's must be equal to 4, but the a_i 's contain no squares. So let us now assume that $\ell \geq 3$.

4. Since $k \geq 4$, we must have $a_{i_1} = 1$, $a_{i_2} = 2$, $a_{i_3} = 4$ for some i_1, i_2, i_3 , that is,

$$n - i_1 = m_1^\ell, \quad n - i_2 = 2m_2^\ell, \quad n - i_3 = 4m_3^\ell.$$

We claim that $(n - i_2)^2 \neq (n - i_1)(n - i_3)$. If not, put $b = n - i_2$ and $n - i_1 = b - x$, $n - i_3 = b + y$, where $0 < |x|, |y| < k$. Hence

$$b^2 = (b - x)(b + y) \quad \text{or} \quad (y - x)b = xy,$$

where $x = y$ is plainly impossible. Now we have by part (1)

$$|xy| = b|y - x| \geq b > n - k > (k - 1)^2 \geq |xy|,$$

which is absurd.

So we have $m_2^2 \neq m_1 m_3$, where we assume $m_2^2 > m_1 m_3$ (the other case being analogous), and proceed to our last chain of inequalities. We obtain

$$\begin{aligned} 2(k-1)n &> n^2 - (n-k+1)^2 > (n-i_2)^2 - (n-i_1)(n-i_3) \\ &= 4[m_2^{2\ell} - (m_1 m_3)^\ell] \geq 4[(m_1 m_3 + 1)^\ell - (m_1 m_3)^\ell] \\ &\geq 4\ell m_1^{\ell-1} m_3^{\ell-1}. \end{aligned}$$

Since $\ell \geq 3$ and $n > k^\ell \geq k^3 > 6k$, this yields

$$\begin{aligned} 2(k-1)n m_1 m_3 &> 4\ell m_1^\ell m_3^\ell = \ell(n-i_1)(n-i_3) \\ &> \ell(n-k+1)^2 > 3 \left(n - \frac{n}{6}\right)^2 > 2n^2. \end{aligned}$$

Now since $m_i \leq n^{1/\ell} \leq n^{1/3}$ we finally obtain

$$kn^{2/3} \geq km_1 m_3 > (k-1)m_1 m_3 > n,$$

or $k^3 > n$. With this contradiction, the proof is complete. $\square$

Chapter 4

Representing numbers as sums of two squares

Lemma 4.1 (Lemma 1). *For primes $p = 4m + 1$ the equation $s^2 \equiv -1 \pmod{p}$ has two solutions $s \in \{1, 2, \dots, p-1\}$, for $p = 2$ there is one such solution, while for primes of the form $p = 4m + 3$ there is no solution.*

Proof. For $p = 2$ take $s = 1$. For odd p , we construct the equivalence relation on $\{1, 2, \dots, p-1\}$ that is generated by identifying every element with its additive inverse and with its multiplicative inverse in $\mathbb{Z}_p$. Thus the “general” equivalence classes will contain four elements

$$\{x, -x, \bar{x}, -\bar{x}\}$$

since such a 4-element set contains both inverses for all its elements. However, there are smaller equivalence classes if some of the four numbers are not distinct:

- $x \equiv -x$ is impossible for odd p .
- $x \equiv \bar{x}$ is equivalent to $x^2 \equiv 1$. This has two solutions, namely $x = 1$ and $x = p-1$, leading to the equivalence class $\{1, p-1\}$ of size 2.
- $x \equiv -\bar{x}$ is equivalent to $x^2 \equiv -1$. This equation may have no solution or two distinct solutions $x_0, p-x_0$: in this case the equivalence class is $\{x_0, p-x_0\}$.

The set $\{1, 2, \dots, p-1\}$ has $p-1$ elements, and we have partitioned it into quadruples (equivalence classes of size 4), plus one or two pairs (equivalence classes of size 2). For $p-1 = 4m+2$ we find that there is only the one pair $\{1, p-1\}$, the rest is quadruples, and thus $s^2 \equiv -1 \pmod{p}$ has no solution. For $p-1 = 4m$ there has to be the second pair, and this contains the two solutions of $s^2 \equiv -1$ that we were looking for. $\square$

Lemma 4.2 (Lemma 2). *No number $n = 4m + 3$ is a sum of two squares.*

Proof. The square of any even number is $(2k)^2 = 4k^2 \equiv 0 \pmod{4}$, while squares of odd numbers yield $(2k+1)^2 = 4(k^2+k) + 1 \equiv 1 \pmod{4}$. Thus any sum of two squares is congruent to 0, 1 or 2 $\pmod{4}$. $\square$

Proposition 4.3 (First proof). *Every prime of the form $p = 4m + 1$ is a sum of two squares, that is, it can be written as $p = x^2 + y^2$ for some natural numbers $x, y \in \mathbb{N}$.*

Proof. Consider the pairs (x', y') of integers with $0 \leq x', y' \leq \sqrt{p}$, that is, $x', y' \in \{0, 1, \dots, \lfloor \sqrt{p} \rfloor\}$. There are $(\lfloor \sqrt{p} \rfloor + 1)^2$ such pairs. Using the estimate $\lfloor x \rfloor + 1 > x$ for $x = \sqrt{p}$, we see that we have more than p such pairs of integers. Thus for any $s \in \mathbb{Z}$, it is impossible that all the values $x' - sy'$ produced by the pairs (x', y') are distinct modulo p . That is, for every s there are two distinct pairs

$$(x', y'), (x'', y'') \in \{0, 1, \dots, \lfloor \sqrt{p} \rfloor\}^2$$

with $x' - sy' \equiv x'' - sy'' \pmod{p}$. Now we take differences: We have $x' - x'' \equiv s(y' - y'') \pmod{p}$. Thus if we define $x := |x' - x''|$, $y := |y' - y''|$, then we get

$$(x, y) \in \{0, 1, \dots, \lfloor \sqrt{p} \rfloor\}^2 \quad \text{with} \quad x \equiv \pm sy \pmod{p}.$$

Also we know that not both x and y can be zero, because the pairs (x', y') and (x'', y'') are distinct.

Now let s be a solution of $s^2 \equiv -1 \pmod{p}$, which exists by Lemma 1. Then $x^2 \equiv s^2 y^2 = -y^2 \pmod{p}$, and so we have produced

$$(x, y) \in \mathbb{Z}^2 \quad \text{with} \quad 0 < x^2 + y^2 < 2p \quad \text{and} \quad x^2 + y^2 \equiv 0 \pmod{p}.$$

But p is the only number between 0 and $2p$ that is divisible by p . Thus $x^2 + y^2 = p$: done! $\square$

Proposition 4.4 (Second proof). *Every prime of the form $p = 4m + 1$ is a sum of two squares, that is, it can be written as $p = x^2 + y^2$ for some natural numbers $x, y \in \mathbb{N}$.*

Proof. We study the set

$$S := \{(x, y, z) \in \mathbb{Z}^3 : 4xy + z^2 = p, \ x > 0, \ y > 0\}.$$

This set is finite. Indeed, $x \geq 1$ and $y \geq 1$ implies $y \leq \frac{p}{4}$ and $x \leq \frac{p}{4}$. So there are only finitely many possible values for x and y , and given x and y , there are at most two values for z .

1. The first linear involution is given by

$$f : S \rightarrow S, \quad (x, y, z) \mapsto (y, x, -z),$$

that is, “interchange x and y , and negate z .” This clearly maps S to itself, and it is an *involution*: Applied twice, it yields the identity. Also, f has no fixed points, since $z = 0$ would imply $p = 4xy$, which is impossible. Furthermore, f maps the solutions in

$$T := \{(x, y, z) \in S : z > 0\}$$

to the solutions in $S \setminus T$, which satisfy $z < 0$. Also, f reverses the signs of $x - y$ and of z , so it maps the solutions in

$$U := \{(x, y, z) \in S : (x - y) + z > 0\}$$

to the solutions in $S \setminus U$. For this we have to see that there is no solution with $(x - y) + z = 0$, but there is none since this would give $p = 4xy + z^2 = 4xy + (x - y)^2 = (x + y)^2$.

What do we get from the study of f ? The main observation is that since f maps the sets T and U to their complements, it also interchanges the elements in $T \setminus U$ with these in $U \setminus T$. That is, there is the same number of solutions in U that are not in T as there are solutions in T that are not in U — so T and U have the same cardinality.

2. The second involution that we study is an involution on the set U :

$$g : U \rightarrow U, \quad (x, y, z) \mapsto (x - y + z, y, 2y - z).$$

First we check that indeed this is a well-defined map: If $(x, y, z) \in U$, then $x - y + z > 0$, $y > 0$ and $4(x - y + z)y + (2y - z)^2 = 4xy + z^2$, so $g(x, y, z) \in S$. By $(x - y + z) - y + (2y - z) = x > 0$ we find that indeed $g(x, y, z) \in U$.

Also g is an involution: $g(x, y, z) = (x - y + z, y, 2y - z)$ is mapped by g to $((x - y + z) - y + (2y - z), y, 2y - (2y - z)) = (x, y, z)$.

And finally g has exactly one fixed point:

$$(x, y, z) = g(x, y, z) = (x - y + z, y, 2y - z)$$

implies that $y = z$, but then $p = 4xy + y^2 = (4x + y)y$, which holds only for $y = z = 1$ and $x = \frac{p-1}{4}$.

But if g is an involution on U that has exactly one fixed point, then *the cardinality of U is odd*.

3. The third, trivial, involution that we study is the involution on T that interchanges x and y :

$$h : T \rightarrow T, \quad (x, y, z) \mapsto (y, x, z).$$

This map is clearly well-defined, and an involution. We combine now our knowledge derived from the other two involutions: The cardinality of T is equal to the cardinality of U , which is odd. But if h is an involution on a finite set of odd cardinality, then it *has a fixed point*: There is a point $(x, y, z) \in T$ with $x = y$, that is, a solution of

$$p = 4x^2 + z^2 = (2x)^2 + z^2.$$

□

Proposition 4.5 (Third proof). *Every prime of the form $p = 4m + 1$ is a sum of two squares, that is, it can be written as $p = x^2 + y^2$ for some natural numbers $x, y \in \mathbb{N}$.*

Proof. Again we fix a prime number $p = 4n + 1$ and consider the set of solutions

$$T = \{(x, y, z) \in \mathbb{N}^3 : 4xy + z^2 = p\}.$$

Each element of this set gives rise to a *winged square*: This is the figure consisting of a square and four rectangles in the plane that you get if you start with a square of side length z and at each vertex attach a rectangle of side-lengths x and y in a rotation-symmetric way, such that the edge of length x points away from the square, while the edge of length y runs along the side of the square.

We consider two winged squares “the same” if they are congruent. One way to make this unique, such that the representation of the winged square depends only on its boundary curve, is to require that the L formed by the two edges in the upper right-hand corner is at least as high as it is wide. If this condition is not satisfied, then a mirror image (reflected, e.g., in a vertical axis), will repair this. So each solution in T corresponds to a *unique* winged square of area $4xy + z^2 = p$, and indeed this is reversible: From each winged square we can read off a solution.

Taking the union of the square and the four rectangles, we get for each winged square what we will call a *unique winged shape*: This is a polyomino of area p with four-fold rotation symmetry,

which has twelve vertices: eight convex ones with inner right angle and four non-convex ones with outer right angle. (We can't get a square shape, since p is a prime, so it can't be a square number.) Again we will consider winged shapes "the same" if they are congruent, so we might assume that the L shape in the upper right-hand corner is at least as high as it is wide.

Now we are getting very close to the punch line: For each winged shape we get *either one or two* winged squares, by simultaneously drawing, in a rotation-symmetric way, vertical and horizontal lines to the interior starting at the non-convex vertices. We get *only one* solution if the shape has the symmetry of a square, that is, if the two arms of the L shapes have the same length. This happens exactly if $y = z$, but then $p = 4xz + z^2 = (4x + z)z$; assuming that p is a prime, this implies that $z = 1$ and $x = n$. In other words: Exactly one winged shape yields a single winged square, while all other winged shapes yield two winged squares each. Consequently, *the number $|T|$ of winged squares is odd.*

However, the winged squares with non-square rectangles (with $x \neq y$) come in pairs, as we can always flip the four rectangular wings between vertical and horizontal format (that is, exchange x and y). As $|T|$ is odd, this implies that there is an odd number of winged squares whose wings are squares, that is, T contains an odd number of triples (x, y, z) with $x = y$, and hence at least one, and this yields a solution to $(2x)^2 + z^2 = p$. $\square$

Theorem 4.6. *A natural number n can be represented as a sum of two squares if and only if every prime factor of the form $p = 4m + 3$ appears with an even exponent in the prime decomposition of n .*

Proof. Call a number n *representable* if it is a sum of two squares, that is, if $n = x^2 + y^2$ for some $x, y \in \mathbb{N}_0$. The theorem is a consequence of the following five facts.

- (1) $1 = 1^2 + 0^2$ and $2 = 1^2 + 1^2$ are representable. Every prime of the form $p = 4m + 1$ is representable.
- (2) The product of any two representable numbers $n_1 = x_1^2 + y_1^2$ and $n_2 = x_2^2 + y_2^2$ is representable: $n_1 n_2 = (x_1 x_2 + y_1 y_2)^2 + (x_1 y_2 - x_2 y_1)^2$.
- (3) If n is representable, $n = x^2 + y^2$, then also $n z^2$ is representable, by $n z^2 = (x z)^2 + (y z)^2$.

Facts (1), (2) and (3) together yield the "if" part of the theorem.

- (4) If $p = 4m + 3$ is a prime that divides a representable number $n = x^2 + y^2$, then p divides both x and y , and thus p^2 divides n . In fact, if we had $x \not\equiv 0 \pmod{p}$, then we could find $\bar{x}$ such that $x\bar{x} \equiv 1 \pmod{p}$, multiply the equation $x^2 + y^2 \equiv 0$ by $\bar{x}^2$, and thus we would obtain that $1 + y^2 \bar{x}^2 = 1 + (\bar{x}y)^2 \equiv 0 \pmod{p}$, which is impossible for $p = 4m + 3$ by Lemma 1.
- (5) If n is representable, and $p = 4m + 3$ divides n , then p^2 divides n , and n/p^2 is representable. This follows from (4), and completes the proof. $\square$

Chapter 5

The law of quadratic reciprocity

Theorem 5.1 (Fermat's little theorem). *For $a \not\equiv 0 \pmod{p}$,*

$$a^{p-1} \equiv 1 \pmod{p}$$

Proof. Since $\mathbb{Z}_p^* = \mathbb{Z}_p \setminus \{0\}$ is a group with multiplication, the set $\{1a, 2a, 3a, \dots, (p-1)a\}$ runs again through all nonzero residues,

$$(1a)(2a) \dots ((p-1)a) \equiv 1 \cdot 2 \dots (p-1) \pmod{p}$$

and hence by dividing by $(p-1)!$, we get $a^{p-1} \equiv 1 \pmod{p}$. □

Theorem 5.2 (Euler's criterion). *For $a \not\equiv 0 \pmod{p}$,*

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$$

Proof. From Fermat's little theorem, the polynomial $x^{p-1} - 1 \in \mathbb{Z}_p[x]$ has as roots all nonzero residues. Next we note that

$$x^{p-1} - 1 = \left(x^{\frac{p-1}{2}} - 1\right) \left(x^{\frac{p-1}{2}} + 1\right).$$

Suppose $a \equiv b^2 \pmod{p}$ is a quadratic residue. Then by Fermat's little theorem $a^{\frac{p-1}{2}} \equiv b^{p-1} \equiv 1 \pmod{p}$. Hence the quadratic residues are precisely the roots of the first factor $x^{\frac{p-1}{2}} - 1$, and the $\frac{p-1}{2}$ nonresidues must thus be the roots of the second factor $x^{\frac{p-1}{2}} + 1$. Comparing this to the definition of the Legendre symbol, we obtain

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}. \quad \square$$

Theorem 5.3 (Product Rule).

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \cdot \left(\frac{b}{p}\right) \tag{5.1}$$

Proof. This obviously holds for the right-hand side of Euler's criterion. □

Theorem 5.4 (Lemma of Gauss). *Suppose $a \not\equiv 0 \pmod{p}$. Take the numbers $1a, 2a, \dots, \frac{p-1}{2}a$ and reduce them modulo p to the residue system smallest in absolute value, $ia \equiv r_i \pmod{p}$ with $-\frac{p-1}{2} \leq r_i \leq \frac{p-1}{2}$ for all i . Then*

$$\left(\frac{a}{p}\right) = (-1)^s, \quad \text{where } s = \#\{i : r_i < 0\}.$$

Proof. Suppose $u_1, \dots, u_s$ are the residues smaller than 0, and that $v_1, \dots, v_{\frac{p-1}{2}-s}$ are those greater than 0. If $-u_i = v_j$, then $u_i + v_j \equiv 0 \pmod{p}$. Now $u_i \equiv ka, v_j \equiv \ell a \pmod{p}$ implies $p \mid (k+\ell)a$. As p and a are relatively prime, p must divide $k+\ell$ which is impossible, since $k+\ell \leq p-1$. Thus the numbers $-u_1, \dots, -u_s$ are between 1 and $\frac{p-1}{2}$, and are all different from the v_j 's; hence $\{-u_1, \dots, -u_s, v_1, \dots, v_{\frac{p-1}{2}-s}\} = \{1, 2, \dots, \frac{p-1}{2}\}$. Therefore

$$\prod_i (-u_i) \prod_j v_j = \left(\frac{p-1}{2}\right)!,$$

which implies

$$(-1)^s \prod_i u_i \prod_j v_j \equiv \left(\frac{p-1}{2}\right)! \pmod{p}.$$

Now remember how we obtained the numbers u_i and v_j ; they are the residues of $1a, \dots, \frac{p-1}{2}a$. Hence

$$\left(\frac{p-1}{2}\right)! \equiv (-1)^s \prod_i u_i \prod_j v_j \equiv (-1)^s \left(\frac{p-1}{2}\right)! a^{\frac{p-1}{2}} \pmod{p}.$$

Cancelling $\left(\frac{p-1}{2}\right)!$ together with Euler's criterion gives

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \equiv (-1)^s \pmod{p},$$

and therefore $\left(\frac{a}{p}\right) = (-1)^s$, since p is odd. $\square$

Theorem 5.5 (Quadratic reciprocity I). *Let p and q be different odd primes. Then*

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}}.$$

Proof. The key to our first proof is a counting formula given by Lemma of Gauss. Let p and q be odd primes, and consider $\left(\frac{q}{p}\right)$. Suppose iq is a multiple of q that reduces to negative residue $r_i < 0$ in the Lemma of Gauss. This means that there is a unique integer j such that $-\frac{p}{2} < iq - jp < 0$. Note that $0 < j < \frac{q}{2}$ since $0 < i < \frac{p}{2}$. In other words, $\left(\frac{q}{p}\right) = (-1)^s$, where s is the number of lattice points (x, y) , that is, pairs of integers x, y satisfying

$$0 < py - qx < \frac{p}{2}, \quad 0 < x < \frac{p}{2}, \quad 0 < y < \frac{q}{2}. \quad (5.2)$$

Similarly, $\left(\frac{p}{q}\right) = (-1)^t$ where t is the number of lattice points (x, y) with

$$0 < qx - py < \frac{q}{2}, \quad 0 < x < \frac{p}{2}, \quad 0 < y < \frac{q}{2}. \quad (5.3)$$

Now look at the rectangle with side lengths $\frac{p}{2}, \frac{q}{2}$, and draw the two lines parallel to the diagonal $py = qx$, $y = \frac{q}{p}x + \frac{1}{2}$ or $py - qx = \frac{p}{2}$, respectively, $y = \frac{q}{p}(x - \frac{1}{2})$ or $qx - py = \frac{q}{2}$.

The proof is now quickly completed by the following three observations:

1. There are no lattice points on the diagonal and the two parallels. This is so because $py = qx$ would imply $p \mid x$, which cannot be. For the parallels observe that $py - qx$ is an integer while $\frac{p}{2}$ and $\frac{q}{2}$ are not.
2. The lattice points observing (5.2) are precisely the points in the upper strip $0 < py - qx < \frac{p}{2}$, and those of (5.3) the points in the lower strip $0 < qx - py < \frac{q}{2}$. Hence the number of lattice points in the two strips is $s + t$.

3. The outer regions $R : py - qx > \frac{p}{2}$ and $S : qx - py > \frac{q}{2}$ contain the *same* number of points. To see this consider the map $\varphi : R \rightarrow S$ which maps (x, y) to $(\frac{p+1}{2} - x, \frac{q+1}{2} - y)$ and check that φ is an involution.

Since the total number of lattice points in the rectangle is $\frac{p-1}{2} \cdot \frac{q-1}{2}$, we infer that $s + t$ and $\frac{p-1}{2} \cdot \frac{q-1}{2}$ have the same parity, and so

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{s+t} = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}. \quad \square$$

Theorem 5.6. *The multiplicative group of a finite field is cyclic.*

Proof. Let F^* be the multiplicative group of the field F , with $|F^*| = n$. Writing $\text{ord}(a)$ for the order of an element, that is, the smallest positive integer k such that $a^k = 1$, we want to find an element $a \in F^*$ with $\text{ord}(a) = n$. If $\text{ord}(b) = d$, then by Lagrange's theorem, d divides n . Classifying the elements according to their order, we have

$$n = \sum_{d|n} \psi(d), \quad \text{where } \psi(d) = \#\{b \in F^* : \text{ord}(b) = d\}. \quad (5.4)$$

If $\text{ord}(b) = d$, then every element b^i ($i = 1, \dots, d$) satisfies $(b^i)^d = 1$ and is therefore a root of the polynomial $x^d - 1$. But, since F is a field, $x^d - 1$ has at most d roots, and so the elements $b, b^2, \dots, b^d = 1$ are precisely these roots. In particular, every element of order d is of the form b^i .

On the other hand, it is easily checked that $\text{ord}(b^i) = \frac{d}{(i,d)}$, where (i,d) denotes the greatest common divisor of i and d . Hence $\text{ord}(b^i) = d$ if and only if $(i,d) = 1$, that is, if i and d are relatively prime. Denoting *Euler's function* by $\varphi(d) = \#\{i : 1 \leq i \leq d, (i,d) = 1\}$, we thus have $\psi(d) = \varphi(d)$ whenever $\psi(d) > 0$. Looking at (5.4) we find

$$n = \sum_{d|n} \psi(d) \leq \sum_{d|n} \varphi(d).$$

But as we are going to show that

$$\sum_{d|n} \varphi(d) = n, \quad (5.5)$$

we must have $\psi(d) = \varphi(d)$ for all d . In particular, $\psi(n) = \varphi(n) \geq 1$, and so there is an element of order n .

The following (folklore) proof of (5.5) belongs in the Book as well. Consider the n fractions

$$\frac{1}{n}, \frac{2}{n}, \dots, \frac{k}{n}, \dots, \frac{n}{n},$$

reduce them to the lowest term $\frac{k}{n} = \frac{i}{d}$ with $1 \leq i \leq d$, $(i,d) = 1$, $d \mid n$, and check that the denominator d appears precisely $\varphi(d)$ times. $\square$

Theorem 5.7 (A). *Let p and q be distinct odd primes, and consider the finite field F with q^{p-1} elements. Then for any $a, b \in F$, $(a+b)^q = a^q + b^q$.*

Proof. The prime field of F is $\mathbb{Z}_q$, whence $qa = 0$ for any $a \in F$. This implies that $(a+b)^q = a^q + b^q$, since any binomial coefficient $\binom{q}{i}$ is a multiple of q for $0 < i < q$, and thus 0 in F . $\square$

Theorem 5.8 (B). *For the field F defined in (A), there exists an element $\zeta \in F$ of multiplicative order p , that is, $\zeta^p = 1$. Moreover, we have a polynomial decomposition*

$$x^p - 1 = (x - \zeta)(x - \zeta^2) \cdots (x - \zeta^p).$$

Proof. The multiplicative group $F^* = F \setminus \{0\}$ is cyclic of size $q^{p-1} - 1$. Since by Fermat's little theorem p is a divisor of $q^{p-1} - 1$, there exists an element $\zeta \in F$ of order p , that is, $\zeta^p = 1$, and ζ generates the subgroup $\{\zeta, \zeta^2, \dots, \zeta^p = 1\}$ of F^* . Note that any ζ^i ($i \neq p$) is again a generator. Hence we obtain the polynomial decomposition $x^p - 1 = (x - \zeta)(x - \zeta^2) \cdots (x - \zeta^p)$. $\square$

Theorem 5.9 (Quadratic reciprocity II). *Let p and q be different odd primes. Then*

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}}.$$

Proof. The second proof does not use Gauss' lemma, instead it employs so-called “Gauss sums” in finite fields. Gauss invented them in his study of the equation $x^p - 1 = 0$ and the arithmetical properties of the field $\mathbb{Q}(\zeta)$ (called cyclotomic field), where ζ is a p -th root of unity. They have been the starting point for the search for higher reciprocity laws in general number fields.

Consider the *Gauss sum*

$$G := \sum_{i=1}^{p-1} \left(\frac{i}{p}\right) \zeta^i \in F,$$

where $\left(\frac{i}{p}\right)$ is the Legendre symbol. For the proof we derive two different expressions for G^q and then set them equal.

First expression. We have

$$G^q = \sum_{i=1}^{p-1} \left(\frac{i}{p}\right)^q \zeta^{iq} = \sum_{i=1}^{p-1} \left(\frac{i}{p}\right) \zeta^{iq} = \left(\frac{q}{p}\right) \sum_{i=1}^{p-1} \left(\frac{iq}{p}\right) \zeta^{iq} = \left(\frac{q}{p}\right) G, \quad (5.6)$$

where the first equality follows from $(a + b)^q = a^q + b^q$, the second uses that $\left(\frac{i}{p}\right)^q = \left(\frac{i}{p}\right)$ since q is odd, the third one is derived from (5.1), which yields $\left(\frac{iq}{p}\right) = \left(\frac{q}{p}\right)\left(\frac{i}{p}\right)$, and the last one holds since iq runs with i through all nonzero residues modulo p .

Second expression. Suppose we can prove

$$G^2 = (-1)^{\frac{p-1}{2}} p, \quad (5.7)$$

then we are quickly done. Indeed,

$$G^q = G(G^2)^{\frac{q-1}{2}} = G(-1)^{\frac{p-1}{2} \frac{q-1}{2}} p^{\frac{q-1}{2}} = G\left(\frac{p}{q}\right)(-1)^{\frac{p-1}{2} \frac{q-1}{2}}. \quad (5.8)$$

Equating the expressions in (5.6) and (5.8) and cancelling G , which is nonzero by (5.7), we find $\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right)(-1)^{\frac{p-1}{2} \frac{q-1}{2}}$, and thus

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}}.$$

It remains to verify (5.7), and for this we first make two simple observations:

- $\sum_{i=1}^p \zeta^i = 0$ and thus $\sum_{i=1}^{p-1} \zeta^i = -1$. Just note that $-\sum_{i=1}^p \zeta^i$ is the coefficient of x^{p-1} in $x^p - 1 = \prod_{i=1}^p (x - \zeta^i)$, and thus 0.
- $\sum_{k=1}^{p-1} \left(\frac{k}{p}\right) = 0$ and thus $\sum_{k=1}^{p-2} \left(\frac{k}{p}\right) = -\left(\frac{-1}{p}\right)$, since there are equally many quadratic residues and nonresidues.

We have

$$G^2 = \left(\sum_{i=1}^{p-1} \left(\frac{i}{p} \right) \zeta^i \right) \left(\sum_{j=1}^{p-1} \left(\frac{j}{p} \right) \zeta^j \right) = \sum_{i,j} \left(\frac{ij}{p} \right) \zeta^{i+j}.$$

Setting $j \equiv ik \pmod{p}$ we find

$$G^2 = \sum_{i,k} \left(\frac{k}{p} \right) \zeta^{i(1+k)} = \sum_{k=1}^{p-1} \left(\frac{k}{p} \right) \sum_{i=1}^{p-1} \zeta^{(1+k)i}.$$

For $k = p-1 \equiv -1 \pmod{p}$ this gives $\left(\frac{-1}{p} \right)(p-1)$, since $\zeta^{1+k} = 1$. Move $k = p-1$ in front and write

$$G^2 = \left(\frac{-1}{p} \right)(p-1) + \sum_{k=1}^{p-2} \left(\frac{k}{p} \right) \sum_{i=1}^{p-1} \zeta^{(1+k)i}.$$

Since ζ^{1+k} is a generator of the group for $k \neq p-1$, the inner sum equals $\sum_{i=1}^{p-1} \zeta^i = -1$ for all $k \neq p-1$ by our first observation. Hence the second summand is $-\sum_{k=1}^{p-2} \left(\frac{k}{p} \right) = \left(\frac{-1}{p} \right)$ by our second observation. It follows that $G^2 = \left(\frac{-1}{p} \right)p$ and thus with Euler's criterion $G^2 = (-1)^{\frac{p-1}{2}} p$, which completes the proof. $\square$

Chapter 6

Every finite division ring is a field

Theorem 6.1 (Roots of unity). *The n -th roots of unity are*

$$\lambda_k = e^{\frac{2k\pi i}{n}} = \cos(2k\pi/n) + i \sin(2k\pi/n), \quad 0 \leq k \leq n-1.$$

Proof. Any complex number $z = x + iy$ may be written in the “polar” form

$$z = re^{i\varphi} = r(\cos \varphi + i \sin \varphi),$$

where $r = |z| = \sqrt{x^2 + y^2}$ is the distance of z to the origin, and φ is the angle measured from the positive x -axis. The n -th roots of unity are therefore of the form

$$\lambda_k = e^{\frac{2k\pi i}{n}} = \cos(2k\pi/n) + i \sin(2k\pi/n), \quad 0 \leq k \leq n-1,$$

since for all k

$$\lambda_k^n = e^{2k\pi i} = \cos(2k\pi) + i \sin(2k\pi) = 1.$$

We obtain these roots geometrically by inscribing a regular n -gon into the unit circle. Note that $\lambda_k = \zeta^k$ for all k , where $\zeta = e^{\frac{2\pi i}{n}}$. Thus the n -th roots of unity form a cyclic group $\{\zeta, \zeta^2, \dots, \zeta^{n-1}, \zeta^n = 1\}$ of order n . $\square$

Theorem 6.2 (Wedderburn’s theorem). *Every finite division ring is commutative.*

Proof. Our first ingredient comes from a blend of linear algebra and basic group theory. For an arbitrary element $s \in R$, let C_s be the set $\{x \in R : xs = sx\}$ of elements which commute with s ; C_s is called the *centralizer* of s . Clearly, C_s contains 0 and 1 and is a sub-division ring of R . The *center* Z is the set of elements which commute with all elements of R , thus $Z = \bigcap_{s \in R} C_s$. In particular, all elements of Z commute, 0 and 1 are in Z , and so Z is a *finite field*. Let us set $|Z| = q$.

We can regard R and C_s as vector spaces over the field Z and deduce that $|R| = q^n$, where n is the dimension of the vector space R over Z , and similarly $|C_s| = q^{n_s}$ for suitable integers $n_s \geq 1$.

Now let us assume that R is not a field. This means that for *some* $s \in R$ the centralizer C_s is not all of R , or, what is the same, $n_s < n$.

On the set $R^* := R \setminus \{0\}$ we consider the relation

$$r' \sim r : \Longleftrightarrow r' = x^{-1}rx \quad \text{for some } x \in R^*.$$

It is easy to check that $\sim$ is an equivalence relation. Let

$$A_s := \{x^{-1}sx : x \in R^*\}$$

be the equivalence class containing s . We note that $|A_s| = 1$ precisely when s is in the center Z . So by our assumption, there are classes A_s with $|A_s| \geq 2$. Consider now for $s \in R^*$ the map $f_s : x \mapsto x^{-1}sx$ from R^* onto A_s . For $x, y \in R^*$ we find

$$x^{-1}sx = y^{-1}sy \iff (yx^{-1})s = s(yx^{-1}) \iff yx^{-1} \in C_s^* \iff y \in C_s^*x,$$

for $C_s^* := C_s \setminus \{0\}$, where $C_s^*x = \{zx : z \in C_s^*\}$ has size $|C_s^*|$. Hence any element $x^{-1}sx$ is the image of precisely $|C_s^*| = q^{n_s} - 1$ elements in R^* under the map f_s , and we deduce $|R^*| = |A_s||C_s^*|$. In particular, we note that

$$\frac{|R^*|}{|C_s^*|} = \frac{q^n - 1}{q^{n_s} - 1} = |A_s| \quad \text{is an integer for all } s.$$

We know that the equivalence classes partition R^* . We now group the central elements Z^* together and denote by $A_1, \dots, A_t$ the equivalence classes containing more than one element. By our assumption we know $t \geq 1$. Since $|R^*| = |Z^*| + \sum_{k=1}^t |A_k|$, we have proved the so-called *class formula*

$$q^n - 1 = q - 1 + \sum_{k=1}^t \frac{q^n - 1}{q^{n_k} - 1}, \quad (6.1)$$

where we have $1 < \frac{q^n - 1}{q^{n_k} - 1} \in \mathbb{N}$ for all k .

With (6.1) we have left abstract algebra and are back to the natural numbers. Next we claim that $q^{n_k} - 1 \mid q^n - 1$ implies $n_k \mid n$. Indeed, write $n = an_k + r$ with $0 \leq r < n_k$, then $q^{n_k} - 1 \mid q^{an_k+r} - 1$ implies

$$q^{n_k} - 1 \mid (q^{an_k+r} - 1) - (q^{n_k} - 1) = q^{n_k}(q^{(a-1)n_k+r} - 1),$$

and thus $q^{n_k} - 1 \mid q^{(a-1)n_k+r} - 1$, since q^{n_k} and $q^{n_k} - 1$ are relatively prime. Continuing in this way we find $q^{n_k} - 1 \mid q^r - 1$ with $0 \leq r < n_k$, which is only possible for $r = 0$, that is, $n_k \mid n$. In summary, we note

$$n_k \mid n \quad \text{for all } k. \quad (6.2)$$

Now comes the second ingredient: the complex numbers $\mathbb{C}$. Consider the polynomial $x^n - 1$. Its roots in $\mathbb{C}$ are called the n -th roots of unity. Since $\lambda^n = 1$, all these roots λ have $|\lambda| = 1$ and lie therefore on the unit circle of the complex plane. In fact, they are precisely the numbers $\lambda_k = e^{\frac{2k\pi i}{n}} = \cos(2k\pi/n) + i\sin(2k\pi/n)$, $0 \leq k \leq n-1$. Some of the roots λ satisfy $\lambda^d = 1$ for $d < n$; for example, the root $\lambda = -1$ satisfies $\lambda^2 = 1$. For a root λ , let d be the smallest positive exponent with $\lambda^d = 1$, that is, d is the order of λ in the group of the roots of unity. Then $d \mid n$, by Lagrange's theorem ("the order of every element of a group divides the order of the group"). Note that there are roots of order n , such as $\lambda_1 = e^{\frac{2\pi i}{n}}$.

Now we group all roots of order d together and set

$$\phi_d(x) := \prod_{\lambda \text{ of order } d} (x - \lambda).$$

Note that the definition of $\phi_d(x)$ is independent of n . Since every root has some order d , we conclude that

$$x^n - 1 = \prod_{d \mid n} \phi_d(x). \quad (6.3)$$

Here is the crucial observation: The coefficients of the polynomials $\phi_n(x)$ are integers (that is, $\phi_n(x) \in \mathbb{Z}[x]$ for all n), where in addition the constant coefficient is either 1 or -1 . Let us carefully verify this claim. For $n = 1$ we have 1 as the only root, and so $\phi_1(x) = x - 1$. Now we proceed by induction, where we assume $\phi_d(x) \in \mathbb{Z}[x]$ for all $d < n$, and that the constant coefficient of $\phi_d(x)$ is 1 or -1 . By (6.3),

$$x^n - 1 = p(x)\phi_n(x) \quad (6.4)$$

where $p(x) = \prod_{d|n, d < n} \phi_d(x) = \sum_{j=0}^{n-\ell} p_j x^j$, $\phi_n(x) = \sum_{k=0}^{\ell} a_k x^k$, with $p_0 = 1$ or $p_0 = -1$. Since $-1 = p_0 a_0$, we see $a_0 \in \{1, -1\}$. Suppose we already know that $a_0, a_1, \dots, a_{k-1} \in \mathbb{Z}$. Computing the coefficient of x^k on both sides of (6.4) we find

$$\sum_{j=0}^k p_j a_{k-j} = \sum_{j=1}^k p_j a_{k-j} + p_0 a_k \in \mathbb{Z}.$$

By assumption, all $a_0, \dots, a_{k-1}$ (and all p_j) are in $\mathbb{Z}$. Thus $p_0 a_k$ and hence a_k must also be integers, since p_0 is 1 or -1 .

We are ready for the *coup de grâce*. Let $n_k \mid n$ be one of the numbers appearing in (6.1). Then

$$x^n - 1 = \prod_{d|n} \phi_d(x) = (x^{n_k} - 1)\phi_n(x) \prod_{d|n, d \nmid n_k, d \neq n} \phi_d(x).$$

We conclude that in $\mathbb{Z}$ we have the divisibility relations

$$\phi_n(q) \mid q^n - 1 \quad \text{and} \quad \phi_n(q) \mid \frac{q^n - 1}{q^{n_k} - 1}. \quad (6.5)$$

Since (6.5) holds for all k , we deduce from the class formula (6.1)

$$\phi_n(q) \mid q - 1,$$

but this cannot be. Why? We know $\phi_n(x) = \prod (x - \lambda)$ where λ runs through all roots of $x^n - 1$ of order n . Let $\lambda = a + ib$ be one of those roots. By $n > 1$ (because of $R \neq \mathbb{Z}$) we have $\lambda \neq 1$, which implies that the real part a is smaller than 1. Now $|\lambda|^2 = a^2 + b^2 = 1$, and hence

$$\begin{aligned} |q - \lambda|^2 &= |q - a - ib|^2 = (q - a)^2 + b^2 \\ &= q^2 - 2aq + a^2 + b^2 = q^2 - 2aq + 1 \\ &> q^2 - 2q + 1 \quad (\text{because of } a < 1) \\ &= (q - 1)^2, \end{aligned}$$

and so $|q - \lambda| > q - 1$ holds for all roots of order n . This implies

$$|\phi_n(q)| = \prod_{\lambda} |q - \lambda| > q - 1,$$

which means that $\phi_n(q)$ cannot be a divisor of $q - 1$, contradiction and end of proof. $\square$

Chapter 7

The spectral theorem and Hadamard's determinant problem

We start with some preliminary facts. Let $O(n) \subseteq \mathbb{R}^{n \times n}$ be the set of real orthogonal matrices of order n . Since

$$(PQ)^{-1} = Q^{-1}P^{-1} = Q^T P^T = (PQ)^T$$

for $P, Q \in O(n)$, we see that the set $O(n)$ is a group. Regarding any matrix in $\mathbb{R}^{n \times n}$ as a vector in $\mathbb{R}^{n^2}$, we find that $O(n)$ is a compact set. Indeed, as the columns of an orthogonal matrix $Q = (q_{ij})$ are unit vectors, we have $|q_{ij}| \leq 1$ for all i and j , thus $O(n)$ is bounded. Furthermore, the set $O(n)$ is defined as a subset of $\mathbb{R}^{n^2}$ by the equations

$$x_{i1}x_{j1} + x_{i2}x_{j2} + \cdots + x_{in}x_{jn} = \delta_{ij} \quad \text{for } 1 \leq i, j \leq n,$$

hence it is closed, and thus compact.

For any real square matrix A let $\text{Od}(A) = \sum_{i \neq j} a_{ij}^2$ be the sum of the squares of the off-diagonal entries.

Lemma 7.1. *If A is a real symmetric $n \times n$ matrix that is not diagonal, that is, $\text{Od}(A) > 0$, then there exists $U \in O(n)$ such that $\text{Od}(U^T A U) < \text{Od}(A)$.*

Proof. We use a very clever method attributed to Carl Gustav Jacob Jacobi. Suppose that $a_{rs} \neq 0$ for some $r \neq s$. Then we claim that the matrix U that agrees with the identity matrix except that $u_{rr} = u_{ss} = \cos \vartheta$, $u_{rs} = \sin \vartheta$, $u_{sr} = -\sin \vartheta$ does the job, for some choice of the (real) angle ϑ :

$$U = \begin{pmatrix} & & r & & s & & \\ & 1 & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & \cos \vartheta & & \sin \vartheta & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \\ & & & -\sin \vartheta & & \cos \vartheta & \\ & & & & & & \ddots & \\ & & & & & & & 1 & \\ & & & & & & & & \ddots & \\ & & & & & & & & & 1 \end{pmatrix} \begin{matrix} \\ r \\ \\ s \end{matrix}$$

Clearly, U is orthogonal for any ϑ .

Now let us compute the (k, ℓ) -entry $b_{k\ell}$ of $U^T AU$. We have

$$b_{k\ell} = \sum_{i,j} u_{ik} a_{ij} u_{j\ell}. \quad (7.1)$$

For $k, \ell \notin \{r, s\}$ we get $b_{k\ell} = a_{k\ell}$. Furthermore, we have

$$\begin{aligned} b_{kr} &= \sum_{i=1}^n u_{ik} \sum_{j=1}^n a_{ij} u_{jr} \\ &= \sum_{i=1}^n u_{ik} (a_{ir} \cos \vartheta - a_{is} \sin \vartheta) \\ &= a_{kr} \cos \vartheta - a_{ks} \sin \vartheta \quad (\text{for } k \neq r, s). \end{aligned}$$

Similarly, one computes

$$b_{ks} = a_{kr} \sin \vartheta + a_{ks} \cos \vartheta \quad (\text{for } k \neq r, s).$$

It follows that

$$\begin{aligned} b_{kr}^2 + b_{ks}^2 &= a_{kr}^2 \cos^2 \vartheta - 2a_{kr}a_{ks} \cos \vartheta \sin \vartheta + a_{ks}^2 \sin^2 \vartheta \\ &\quad + a_{kr}^2 \sin^2 \vartheta + 2a_{kr}a_{ks} \sin \vartheta \cos \vartheta + a_{ks}^2 \cos^2 \vartheta \\ &= a_{kr}^2 + a_{ks}^2, \end{aligned}$$

and by symmetry

$$b_{r\ell}^2 + b_{s\ell}^2 = a_{r\ell}^2 + a_{s\ell}^2 \quad (\text{for } \ell \neq r, s).$$

We conclude that the function Od , which sums the squares of the off-diagonal values, agrees for A and $U^T AU$ except for the entries at (r, s) and (s, r) , for *any* ϑ . To conclude the proof we now show that ϑ_0 can be chosen suitably as to make $b_{rs} = 0$, which will result in

$$\text{Od}(U^T AU) = \text{Od}(A) - 2a_{rs}^2 < \text{Od}(A)$$

as required.

Using (7.1) we find

$$b_{rs} = (a_{rr} - a_{ss}) \sin \vartheta \cos \vartheta + a_{rs} (\cos^2 \vartheta - \sin^2 \vartheta).$$

For $\vartheta = 0$ this becomes a_{rs} , while for $\vartheta = \pi/2$ it is $-a_{rs}$. Hence by the intermediate value theorem there is some ϑ_0 between 0 and $\pi/2$ such that $b_{rs} = 0$, and we are through. $\square$

Theorem 7.2. *For every real symmetric matrix A there is a real orthogonal matrix Q such that $Q^T A Q$ is diagonal.*

Proof. The theorem follows in three quick steps. Let A be a real symmetric $n \times n$ matrix.

- (A) Consider the map $f_A : O(n) \rightarrow \mathbb{R}^{n \times n}$ with $f_A(P) := P^T A P$. The map f_A is continuous on the compact set $O(n)$, and so the image $f_A(O(n))$ is compact.
- (B) The function $\text{Od} : f_A(O(n)) \rightarrow \mathbb{R}$ is continuous, hence it assumes a minimum, say at $D = Q^T A Q \in f_A(O(n))$.
- (C) The value $\text{Od}(D)$ must be zero, and hence D is a diagonal matrix as required.

Indeed, if $\text{Od}(D) > 0$, then applying the Lemma we find $U \in O(n)$ with $\text{Od}(U^T D U) < \text{Od}(D)$. But

$$U^T D U = U^T Q^T A Q U = (Q U)^T A (Q U)$$

is in $f_A(O(n))$ (remember $O(n)$ is a group!) with Od -value smaller than that of D — contradiction, and end of proof. $\square$

Theorem 7.3 (Hadamard's inequality). *For any real $n \times n$ matrix $A = (a_{ij})$ with $|a_{ij}| \leq 1$,*

$$|\det A| \leq n^{n/2}.$$

Proof. The problem to find the maximum value of $\det A$ on the set of all real $n \times n$ matrices $A = (a_{ij})$ with $|a_{ij}| \leq 1$ is unsolved. Since the determinant is a continuous function in the a_{ij} (considered as variables) and the matrices form a compact set in $\mathbb{R}^{n^2}$, this maximum must exist. Furthermore, the maximum is attained for some matrix all of whose entries are $+1$ or -1 , because the function $\det A$ is linear in each single entry a_{ij} (if we keep all other entries fixed). Thus we can start with any matrix A and move one entry after the other to $+1$ or to -1 , in every single step not decreasing the determinant, until we arrive at a ± 1 -matrix. In the search for the largest determinant we may thus assume that all entries of A are ± 1 .

Here is the trick: Instead of A we consider the matrix $B = A^T A = (b_{ij})$. That is, if $c_j = (a_{1j}, a_{2j}, \dots, a_{nj})^T$ denotes the j -th column vector of A , then $b_{ij} = \langle c_i, c_j \rangle$, the inner product of c_i and c_j . In particular,

$$b_{ii} = \langle c_i, c_i \rangle = n \quad \text{for all } i,$$

and

$$\text{trace } B = \sum_{i=1}^n b_{ii} = n^2, \tag{7.2}$$

which will come in handy in a moment.

Now we can go to work. First of all, from $B = A^T A$ we get $|\det A| = \sqrt{\det B}$. Since multiplication of a column of A by -1 turns $\det A$ into $-\det A$, we see that the maximum problem for $\det A$ is the same as for $\det B$. Furthermore, we may assume that A is nonsingular, and hence that B is nonsingular as well.

Since $B = A^T A$ is a symmetric matrix the spectral theorem tells us that for some $Q \in O(n)$,

$$Q^T B Q = Q^T A^T A Q = (A Q)^T (A Q) = \begin{pmatrix} \lambda_1 & & & \\ & \ddots & & \\ & & O & \\ & & & \ddots \\ O & & & & \ddots \\ & & & & & \lambda_n \end{pmatrix}, \tag{7.3}$$

where the λ_i are the eigenvalues of B . Now, if d_j denotes the j -th column vector of $A Q$ (which is nonzero since A is nonsingular), then

$$\lambda_j = \langle d_j, d_j \rangle = \sum_{i=1}^n d_{ij}^2 > 0.$$

Thus $\lambda_1, \dots, \lambda_n$ are positive real numbers and

$$\det B = \lambda_1 \cdots \lambda_n, \quad \text{trace } B = \sum_{i=1}^n \lambda_i.$$

Whenever such a product and sum of positive numbers turn up, it is always a good idea to try the arithmetic-geometric mean inequality. In our case this gives with (7.2)

$$\det B = \lambda_1 \cdots \lambda_n \leq \left(\frac{\sum_{i=1}^n \lambda_i}{n} \right)^n = \left(\frac{\text{trace } B}{n} \right)^n = n^n, \quad (7.4)$$

and out comes Hadamard's upper bound

$$|\det A| \leq n^{n/2}. \quad (7.5)$$

When do we have equality in (7.5) or, what is the same, in (7.4)? Easy enough: if and only if the geometric mean of the λ_i 's equals the arithmetic mean, or equivalently, if and only if $\lambda_1 = \cdots = \lambda_n = \lambda$. But then $\text{trace } B = n\lambda = n^2$, and so $\lambda_1 = \cdots = \lambda_n = n$. Looking at (7.3) this means $Q^T B Q = nI_n$, where I_n is the $n \times n$ identity matrix. Now recall $Q^T = Q^{-1}$, multiply by Q on the left, by Q^{-1} on the right, to obtain

$$B = nI_n.$$

Going back to A this means that

$$|\det A| = n^{n/2} \iff \langle c_i, c_j \rangle = 0 \quad \text{for } i \neq j.$$

Matrices A with ± 1 -entries that achieve equality in (7.5) are aptly called *Hadamard matrices*. So an $n \times n$ matrix A with ± 1 -entries is a Hadamard matrix if and only if

$$A^T A = A A^T = nI_n.$$

□

Theorem 7.4. *If a Hadamard matrix of size $n \times n$ exists for $n > 2$, then n must be a multiple of 4.*

Proof. A short argument shows that if n is greater than 2, then it must be a multiple of 4. Indeed, suppose that A is an $n \times n$ Hadamard matrix, $n \geq 2$, whose rows are the vectors $r_1, \dots, r_n$. Clearly, multiplication of any row or column by -1 gives another Hadamard matrix. So we may assume that the first row consists of 1's only. Since $\langle r_1, r_i \rangle = 0$ for $i \neq 1$, every other row must contain $n/2$ 1's and $n/2$ -1 's; in particular, n must be even. Assume now that $n > 2$ and consider rows r_2 and r_3 , and denote by a, b, c, d the numbers of columns that have $\begin{pmatrix} +1 \\ +1 \end{pmatrix}$, $\begin{pmatrix} +1 \\ -1 \end{pmatrix}$, $\begin{pmatrix} -1 \\ +1 \end{pmatrix}$, and $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$ in rows 2 and 3, respectively. Then from $\langle r_1, r_2 \rangle = 0$ and $\langle r_1, r_3 \rangle = 0$ we get

$$a + b = c + d = a + c = b + d = n/2,$$

which gives $b = c, a = d$. But from $\langle r_2, r_3 \rangle = 0$ we also have $a + d = b + c$, resulting in $2a = 2b$. We conclude that $a = b = c = d = n/4$. Thus the order of the Hadamard matrix is either $n = 1$ or $n = 2$, or $n = a + b + c + d = 4a$, a multiple of 4. □

Theorem 7.5. *Hadamard matrices exist for all $n = 2^m$.*

Proof. Consider an m -set X and index the 2^m subsets $C \subseteq X$ in any way $C_1, \dots, C_{2^m}$. The matrix $A = (a_{ij})$ is defined as

$$a_{ij} = (-1)^{|C_i \cap C_j|}.$$

We want to verify $\langle r_i, r_j \rangle = 0$ for $i \neq j$. From the definition,

$$\langle r_i, r_j \rangle = \sum_k (-1)^{|C_i \cap C_k| + |C_j \cap C_k|}.$$

Now, as $C_i \neq C_j$ there exists an element $a \in X$ with $a \in C_i \setminus C_j$ or $a \in C_j \setminus C_i$; suppose $a \in C_i \setminus C_j$. Half the subsets of X contain a , and half do not. Let C run through all subsets that contain a , then the pairs $\{C, C \setminus \{a\}\}$ will comprise all subsets of X . But for each such pair $\{C, C \setminus \{a\}\}$, $|C_i \cap C| + |C_j \cap C|$ and $|C_i \cap (C \setminus \{a\})| + |C_j \cap (C \setminus \{a\})|$ have different parity, and so the corresponding terms in the sum will sum to 0. But then the whole sum is 0, as required. $\square$

Theorem 7.6. *There exists an $n \times n$ matrix with entries ± 1 whose determinant is greater than $\sqrt{n!}$.*

Proof. Let us look at all 2^{n^2} matrices with ± 1 -entries and consider some averages of the determinant. The arithmetic mean $\frac{1}{2^{n^2}} \sum_A \det A$ is 0 (clear?), so this is no big help. But if we consider the *mean square average* instead,

$$D_n := \sqrt{\frac{\sum_A (\det A)^2}{2^{n^2}}},$$

then things brighten up. Clearly,

$$\max_A \det A \geq D_n,$$

so this will give us a lower bound for the maximum.

The following stunningly simple calculation of D_n^2 probably appeared first in an article by George Szekeres and Paul Turán. We learnt it from a beautiful paper of Herb Wilf who heard it from Mark Kac. In the words of Mark Kac: “*Just write $(\det A)^2$ out twice, interchange summation, and everything simplifies.*” So we want to do just that.

From the definition of the determinant we get

$$\begin{aligned} D_n^2 &= \frac{1}{2^{n^2}} \sum_A \left(\sum_{\pi} (\text{sign } \pi) a_{1\pi(1)} a_{2\pi(2)} \cdots a_{n\pi(n)} \right)^2 \\ &= \frac{1}{2^{n^2}} \sum_A \sum_{\sigma} \sum_{\tau} (\text{sign } \sigma) (\text{sign } \tau) a_{1\sigma(1)} a_{1\tau(1)} \cdots a_{n\sigma(n)} a_{n\tau(n)}, \end{aligned}$$

where σ and τ run independently through all permutations of $\{1, \dots, n\}$. Interchange of summation yields

$$D_n^2 = \frac{1}{2^{n^2}} \sum_{\sigma, \tau} (\text{sign } \sigma) (\text{sign } \tau) \left(\sum_A a_{1\sigma(1)} a_{1\tau(1)} \cdots a_{n\sigma(n)} a_{n\tau(n)} \right).$$

This doesn't look too promising, but wait. Look at a fixed pair (σ, τ) . The inner sum $\sum_A$ is really a summation over n^2 variables, one for each a_{ij} :

$$\sum_{a_{11}=\pm 1} \sum_{a_{12}=\pm 1} \cdots \sum_{a_{nn}=\pm 1} a_{1\sigma(1)} a_{1\tau(1)} \cdots a_{n\sigma(n)} a_{n\tau(n)}. \quad (7.6)$$

Suppose $\sigma(i) = k \neq \tau(i)$. Then every summand contains a_{ik} , and therefore the whole sum has the *factor* $\sum_{a_{ik}=\pm 1} a_{ik} = 0$, and hence is 0 as well. The only way that the sum fails to be 0 is

when $\sigma = \tau$, and everything simplifies indeed: For $\sigma = \tau$, the inner product is 1 as is the term $(\text{sign } \sigma)^2$. The sum in (7.6) is therefore

$$\sum_{a_{11}=\pm 1} \cdots \sum_{a_{nn}=\pm 1} 1 = 2^{n^2},$$

and wrapping things up we obtain

$$D_n^2 = \frac{1}{2^{n^2}} \sum_{\sigma} 2^{n^2} = n!,$$

and thus the result. □

Chapter 8

Some irrational numbers

Theorem 8.1. *e is irrational.*

Proof. To start with, it is rather easy to see (as did Fourier in 1815) that $e = \sum_{k \geq 0} \frac{1}{k!}$ is irrational. Indeed, if we had $e = \frac{a}{b}$ for integers a and $b > 0$, then we would get

$$n!be = n!a$$

for *every* $n \geq 0$. But this cannot be true, because on the right-hand side we have an integer, while the left-hand side with

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} + \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} + \cdots$$

decomposes into an integral part

$$bn! \left(1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} \right)$$

and a second part

$$b \left(\frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} + \cdots \right)$$

which is *approximately* $\frac{b}{n}$, so that for large n it certainly cannot be integral: It is larger than $\frac{b}{n+1}$ and smaller than $\frac{b}{n}$, as one can see from a comparison with a geometric series:

$$\begin{aligned} \frac{1}{n+1} &< \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} + \cdots \\ &< \frac{1}{n+1} + \frac{1}{(n+1)^2} + \frac{1}{(n+1)^3} + \cdots = \frac{1}{n}. \end{aligned}$$

□

Theorem 8.2. *e^2 is irrational.*

Proof. Now one might be led to think that this simple multiply-by- $n!$ trick is not sufficient to show that e^2 is irrational. This is a stronger statement: $\sqrt{2}$ is an example of a number which is irrational, but whose square is not. From John Cosgrave we have learned that with two nice

ideas/observations (let's call them "tricks") one can get two steps further nevertheless: Each of the tricks is sufficient to show that e^2 is irrational, the combination of both of them even yields the same for e^4 . The first trick may be found in a one page paper by J. Liouville from 1840 — and the second one in a two page "addendum" which Liouville published on the next two journal pages.

Why is e^2 irrational? What can we derive from $e^2 = \frac{a}{b}$? According to Liouville we should write this as

$$be = ae^{-1},$$

substitute the series

$$e = 1 + \frac{1}{1} + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + \dots$$

and

$$e^{-1} = 1 - \frac{1}{1} + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} \pm \dots,$$

and then multiply by $n!$, for a sufficiently large even n . Then we see that $n!be$ is nearly integral:

$$n!b \left(1 + \frac{1}{1} + \frac{1}{2} + \frac{1}{6} + \dots + \frac{1}{n!} \right)$$

is an integer, and the rest

$$n!b \left(\frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \dots \right)$$

is approximately $\frac{b}{n}$: It is larger than $\frac{b}{n+1}$ but smaller than $\frac{b}{n}$, as we have seen above.

At the same time $n!ae^{-1}$ is nearly integral as well: Again we get a large integral part, and then a rest

$$(-1)^{n+1}n!a \left(\frac{1}{(n+1)!} - \frac{1}{(n+2)!} + \frac{1}{(n+3)!} \mp \dots \right),$$

and this is approximately $(-1)^{n+1}\frac{a}{n}$. More precisely: for even n the rest is larger than $-\frac{a}{n}$, but smaller than

$$-a \left(\frac{1}{n+1} - \frac{1}{(n+1)^2} - \frac{1}{(n+1)^3} - \dots \right) = -\frac{a}{n+1} \left(1 - \frac{1}{n} \right) < 0.$$

But this cannot be true, since for large even n it would imply that $n!ae^{-1}$ is just a bit smaller than an integer, while $n!be$ is a bit larger than an integer, so $n!ae^{-1} = n!be$ cannot hold. $\square$

Theorem 8.3 (Little Lemma). *For any $n \geq 1$ the integer $n!$ contains the prime factor 2 at most $n-1$ times — with equality if (and only if) n is a power of two, $n = 2^m$.*

Proof. This lemma is not hard to show: $\lfloor \frac{n}{2} \rfloor$ of the factors of $n!$ are even, $\lfloor \frac{n}{4} \rfloor$ of them are divisible by 4, and so on. So if 2^k is the largest power of two which satisfies $2^k \leq n$, then $n!$ contains the prime factor 2 exactly

$$\left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{4} \right\rfloor + \dots + \left\lfloor \frac{n}{2^k} \right\rfloor \leq \frac{n}{2} + \frac{n}{4} + \dots + \frac{n}{2^k} = n \left(1 - \frac{1}{2^k} \right) \leq n-1$$

times, with equality in both inequalities exactly if $n = 2^k$. $\square$

Theorem 8.4. e^4 is irrational.

Proof. In order to show that e^4 is irrational, we now courageously assume that $e^4 = \frac{a}{b}$ were rational, and write this as

$$be^2 = ae^{-2}.$$

We could now try to multiply this by $n!$ for some large n , and collect the non-integral summands, but this leads to nothing useful: The sum of the remaining terms on the left-hand side will be approximately $b\frac{2^{n+1}}{n}$, on the right side $(-1)^{n+1}a\frac{2^{n+1}}{n}$, and both will be very large if n gets large.

So one has to examine the situation a bit more carefully, and make two little adjustments to the strategy: First we will not take an *arbitrary* large n , but a large power of two, $n = 2^m$; and secondly we will not multiply by $n!$, but by $\frac{n!}{2^{n-1}}$. Then we need the little lemma 8.3, a special case of Legendre's theorem (see page 10).

Let's get back to $be^2 = ae^{-2}$. We are looking at

$$b\frac{n!}{2^{n-1}}e^2 = a\frac{n!}{2^{n-1}}e^{-2} \quad (8.1)$$

and substitute the series

$$e^2 = 1 + \frac{2}{1} + \frac{4}{2} + \frac{8}{6} + \dots + \frac{2^r}{r!} + \dots$$

and

$$e^{-2} = 1 - \frac{2}{1} + \frac{4}{2} - \frac{8}{6} \pm \dots + (-1)^r \frac{2^r}{r!} + \dots$$

For $r \leq n$ we get integral summands on both sides, namely

$$b\frac{n!}{2^{n-1}}\frac{2^r}{r!} \quad \text{resp.} \quad (-1)^r a\frac{n!}{2^{n-1}}\frac{2^r}{r!},$$

where for $r > 0$ the denominator $r!$ contains the prime factor 2 at most $r - 1$ times, while $n!$ contains it *exactly* $n - 1$ times. (So for $r > 0$ the summands are even.)

And since n is even (we assume that $n = 2^m$), the series that we get for $r \geq n + 1$ are

$$2b \left(\frac{2}{n+1} + \frac{4}{(n+1)(n+2)} + \frac{8}{(n+1)(n+2)(n+3)} + \dots \right)$$

resp.

$$2a \left(-\frac{2}{n+1} + \frac{4}{(n+1)(n+2)} - \frac{8}{(n+1)(n+2)(n+3)} \pm \dots \right).$$

These series will for large n be roughly $\frac{4b}{n}$ resp. $-\frac{4a}{n}$, as one sees again by comparison with geometric series. For large $n = 2^m$ this means that the left-hand side of (8.1) is a bit larger than an integer, while the right-hand side is a bit smaller — contradiction! $\square$

Lemma 8.5. For some fixed $n \geq 1$, let

$$f(x) = \frac{x^n(1-x)^n}{n!}.$$

- (i) The function $f(x)$ is a polynomial of the form $f(x) = \frac{1}{n!} \sum_{i=n}^{2n} c_i x^i$, where the coefficients c_i are integers.

Proof. Part (i) is clear. $\square$

Lemma 8.6. (ii) For $0 < x < 1$ we have $0 < f(x) < \frac{1}{n!}$.

Proof. Part (ii) is also clear. $\square$

Lemma 8.7. (iii) *The derivatives $f^{(k)}(0)$ and $f^{(k)}(1)$ are integers for all $k \geq 0$.*

Proof. For (iii) note that by (i) the k -th derivative $f^{(k)}$ vanishes at $x = 0$ unless $n \leq k \leq 2n$, and in this range $f^{(k)}(0) = \frac{k!}{n!} c_k$ is an integer. From $f(x) = f(1-x)$ we get $f'(x) = (-1)f'(1-x)$ for all x , and hence $f^{(k)}(1) = (-1)^k f^{(k)}(0)$, which is an integer. $\square$

Theorem 8.8. *e^r is irrational for every $r \in \mathbb{Q} \setminus \{0\}$.*

Proof. It suffices to show that e^s cannot be rational for a positive integer s (if $e^{\frac{s}{t}}$ were rational, then $(e^{\frac{s}{t}})^t = e^s$ would be rational, too). Assume that $e^s = \frac{a}{b}$ for integers $a, b > 0$, and let n be so large that $n! > as^{2n+1}$. Put

$$F(x) := s^{2n}f(x) - s^{2n-1}f'(x) + s^{2n-2}f''(x) \mp \dots + f^{(2n)}(x),$$

where $f(x)$ is the function of the lemma. $F(x)$ may also be written as an infinite sum

$$F(x) = s^{2n}f(x) - s^{2n-1}f'(x) + s^{2n-2}f''(x) \mp \dots,$$

since the higher derivatives $f^{(k)}(x)$, for $k > 2n$, vanish. From this we see that the polynomial $F(x)$ satisfies the identity

$$F'(x) = -sF(x) + s^{2n+1}f(x).$$

Thus differentiation yields

$$\frac{d}{dx}[e^{sx}F(x)] = se^{sx}F(x) + e^{sx}F'(x) = s^{2n+1}e^{sx}f(x)$$

and hence

$$N := b \int_0^1 s^{2n+1}e^{sx}f(x)dx = b[e^{sx}F(x)]_0^1 = aF(1) - bF(0).$$

This is an integer, since part (iii) of the lemma implies that $F(0)$ and $F(1)$ are integers. However, part (ii) of the lemma yields estimates for the size of N from below and from above,

$$0 < N = b \int_0^1 s^{2n+1}e^{sx}f(x)dx < bs^{2n+1}e^s \frac{1}{n!} = \frac{as^{2n+1}}{n!} < 1,$$

which shows that N cannot be an integer: contradiction. $\square$

Theorem 8.9. *π^2 is irrational.*

Proof. Assume that $\pi^2 = \frac{a}{b}$ for integers $a, b > 0$. We now use the polynomial

$$F(x) := b^n (\pi^{2n}f(x) - \pi^{2n-2}f^{(2)}(x) + \pi^{2n-4}f^{(4)}(x) \mp \dots),$$

which satisfies $F''(x) = -\pi^2 F(x) + b^n \pi^{2n+2}f(x)$.

From part (iii) of the lemma we get that $F(0)$ and $F(1)$ are integers. Elementary differentiation rules yield

$$\begin{aligned} \frac{d}{dx}[F'(x) \sin \pi x - \pi F(x) \cos \pi x] &= (F''(x) + \pi^2 F(x)) \sin \pi x \\ &= b^n \pi^{2n+2}f(x) \sin \pi x \\ &= \pi^2 a^n f(x) \sin \pi x, \end{aligned}$$

and thus we obtain

$$\begin{aligned} N &:= \pi \int_0^1 a^n f(x) \sin \pi x \, dx = \left[\frac{1}{\pi} F'(x) \sin \pi x - F(x) \cos \pi x \right]_0^1 \\ &= F(0) + F(1), \end{aligned}$$

which is an integer. Furthermore N is positive since it is defined as the integral of a function that is positive (except on the boundary). However, if we choose n so large that $\frac{\pi a^n}{n!} < 1$, then from part (ii) of the lemma we obtain

$$0 < N = \pi \int_0^1 a^n f(x) \sin \pi x \, dx < \frac{\pi a^n}{n!} < 1,$$

a contradiction. □

Theorem 8.10. *For every odd integer $n \geq 3$, the number*

$$A(n) := \frac{1}{\pi} \arccos \left(\frac{1}{\sqrt{n}} \right)$$

is irrational.

Proof. We use the addition theorem

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

from elementary trigonometry, which for $\alpha = (k+1)\varphi$ and $\beta = (k-1)\varphi$ yields

$$\cos(k+1)\varphi = 2 \cos \varphi \cos k\varphi - \cos(k-1)\varphi. \quad (8.2)$$

For the angle $\varphi_n = \arccos \left(\frac{1}{\sqrt{n}} \right)$, which is defined by $\cos \varphi_n = \frac{1}{\sqrt{n}}$ and $0 \leq \varphi_n \leq \pi$, this yields representations of the form

$$\cos k\varphi_n = \frac{A_k}{\sqrt{n}^k},$$

where A_k is an integer that is not divisible by n , for all $k \geq 0$. In fact, we have such a representation for $k = 0, 1$ with $A_0 = A_1 = 1$, and by induction on k using (8.2) we get for $k \geq 1$

$$\cos(k+1)\varphi_n = 2 \frac{1}{\sqrt{n}} \frac{A_k}{\sqrt{n}^k} - \frac{A_{k-1}}{\sqrt{n}^{k-1}} = \frac{2A_k - nA_{k-1}}{\sqrt{n}^{k+1}}.$$

Thus we obtain $A_{k+1} = 2A_k - nA_{k-1}$. If $n \geq 3$ is odd, and A_k is not divisible by n , then we find that A_{k+1} cannot be divisible by n , either.

Now assume that

$$A(n) = \frac{1}{\pi} \varphi_n = \frac{k}{\ell}$$

is rational (with integers $k, \ell > 0$). Then $\ell\varphi_n = k\pi$ yields

$$\pm 1 = \cos k\pi = \frac{A_\ell}{\sqrt{n}^\ell}.$$

Thus $\sqrt{n}^\ell = \pm A_\ell$ is an integer, with $\ell \geq 2$, and hence $n | \sqrt{n}^\ell$. With $\sqrt{n}^\ell | A_\ell$ we find that n divides A_ℓ , a contradiction. □

Chapter 9

Four times $\pi^2/6$

Theorem 9.1 (Euler's series: Proof 1).

$$\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Proof. The proof consists in two different evaluations of the double integral

$$I := \int_0^1 \int_0^1 \frac{1}{1-xy} dx dy.$$

For the first one, we expand $\frac{1}{1-xy}$ as a geometric series, decompose the summands as products, and integrate effortlessly:

$$\begin{aligned} I &= \int_0^1 \int_0^1 \sum_{n \geq 0} (xy)^n dx dy = \sum_{n \geq 0} \int_0^1 \int_0^1 x^n y^n dx dy \\ &= \sum_{n \geq 0} \left(\int_0^1 x^n dx \right) \left(\int_0^1 y^n dy \right) = \sum_{n \geq 0} \frac{1}{n+1} \frac{1}{n+1} \\ &= \sum_{n \geq 0} \frac{1}{(n+1)^2} = \sum_{n \geq 1} \frac{1}{n^2} = \zeta(2). \end{aligned}$$

This evaluation also shows that the double integral (over a positive function with a pole at $x = y = 1$) is finite. Note that the computation is also easy and straightforward if we read it backwards — thus the evaluation of $\zeta(2)$ leads one to the double integral I .

The second way to evaluate I comes from a change of coordinates: in the new coordinates given by $u := \frac{y+x}{2}$ and $v := \frac{y-x}{2}$ the domain of integration is a square of side length $\frac{1}{2}\sqrt{2}$, which we get from the old domain by first rotating it by 45° and then shrinking it by a factor of $\sqrt{2}$. Substitution of $x = u - v$ and $y = u + v$ yields

$$\frac{1}{1-xy} = \frac{1}{1-u^2+v^2}.$$

To transform the integral, we have to replace $dx dy$ by $2 du dv$, to compensate for the fact that our coordinate transformation reduces areas by a constant factor of 2 (which is the Jacobi determinant of the transformation). The new domain of integration, and the function to be integrated, are

symmetric with respect to the u -axis, so we just need to compute two times (another factor of 2 arises here!) the integral over the upper half domain, which we split into two parts in the most natural way:

$$I = 4 \int_0^{1/2} \left(\int_0^u \frac{dv}{1-u^2+v^2} \right) du + 4 \int_{1/2}^1 \left(\int_0^{1-u} \frac{dv}{1-u^2+v^2} \right) du.$$

Using $\int \frac{dx}{a^2+x^2} = \frac{1}{a} \arctan \frac{x}{a} + C$, this becomes

$$\begin{aligned} I &= 4 \int_0^{1/2} \frac{1}{\sqrt{1-u^2}} \arctan \left(\frac{u}{\sqrt{1-u^2}} \right) du \\ &\quad + 4 \int_{1/2}^1 \frac{1}{\sqrt{1-u^2}} \arctan \left(\frac{1-u}{\sqrt{1-u^2}} \right) du. \end{aligned}$$

These integrals can be simplified and finally evaluated by substituting $u = \sin \theta$ resp. $u = \cos \theta$. But we proceed more directly, by computing that the derivative of $g(u) := \arctan \left(\frac{u}{\sqrt{1-u^2}} \right)$ is $g'(u) = \frac{1}{\sqrt{1-u^2}}$, while the derivative of $h(u) := \arctan \left(\frac{1-u}{\sqrt{1-u^2}} \right) = \arctan \left(\sqrt{\frac{1-u}{1+u}} \right)$ is $h'(u) = -\frac{1}{2} \frac{1}{\sqrt{1-u^2}}$. So we may use $\int_a^b f'(x)f(x)dx = [\frac{1}{2}f(x)^2]_a^b = \frac{1}{2}f(b)^2 - \frac{1}{2}f(a)^2$ and get

$$\begin{aligned} I &= 4 \int_0^{1/2} g'(u)g(u)du + 4 \int_{1/2}^1 -2h'(u)h(u)du \\ &= 2 [g(u)^2]_0^{1/2} - 4 [h(u)^2]_{1/2}^1 \\ &= 2g\left(\frac{1}{2}\right)^2 - 2g(0)^2 - 4h(1)^2 + 4h\left(\frac{1}{2}\right)^2 \\ &= 2\left(\frac{\pi}{6}\right)^2 - 0 - 0 + 4\left(\frac{\pi}{6}\right)^2 = \frac{\pi^2}{6}. \square \end{aligned}$$

Theorem 9.2 (Euler's series: Proof 2).

$$\sum_{k \geq 0} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$$

Proof. As above, we may express this as a double integral, namely

$$J = \int_0^1 \int_0^1 \frac{1}{1-x^2y^2} dx dy = \sum_{k \geq 0} \frac{1}{(2k+1)^2}.$$

So we have to compute this integral J . And for this Beukers, Calabi and Kolk proposed the new coordinates

$$u := \arccos \sqrt{\frac{1-x^2}{1-x^2y^2}} \quad v := \arccos \sqrt{\frac{1-y^2}{1-x^2y^2}}.$$

To compute the double integral, we may ignore the boundary of the domain, and consider x, y in the range $0 < x < 1$ and $0 < y < 1$. Then u, v will lie in the triangle $u > 0, v > 0, u+v < \pi/2$. The coordinate transformation can be inverted explicitly, which leads one to the substitution

$$x = \frac{\sin u}{\cos v} \quad \text{and} \quad y = \frac{\sin v}{\cos u}.$$

It is easy to check that these formulas define a bijective coordinate transformation between the interior of the unit square $S = \{(x, y) : 0 \leq x, y \leq 1\}$ and the interior of the triangle $T = \{(u, v) : u, v \geq 0, u + v \leq \pi/2\}$. Now we have to compute the Jacobi determinant of the coordinate transformation, and magically it turns out to be

$$\det \begin{pmatrix} \frac{\cos u}{\sin u \sin v} & \frac{\sin u \sin v}{\cos^2 v} \\ \frac{\cos v}{\sin u \sin v} & \frac{\sin u \sin v}{\cos^2 u} \end{pmatrix} = 1 - \frac{\sin^2 u \sin^2 v}{\cos^2 u \cos^2 v} = 1 - x^2 y^2.$$

But this means that the integral that we want to compute is transformed into

$$J = \int_0^{\pi/2} \int_0^{\pi/2-u} 1 dv du,$$

which is just the area $\frac{1}{2}(\frac{\pi}{2})^2 = \frac{\pi^2}{8}$ of the triangle T . □

Theorem 9.3 (Euler's series: Proof 3).

$$\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Proof. The first step is to establish a remarkable relation between values of the (squared) cotangent function. Namely, for all $m \geq 1$ one has

$$\cot^2 \left(\frac{\pi}{2m+1} \right) + \cot^2 \left(\frac{2\pi}{2m+1} \right) + \dots + \cot^2 \left(\frac{m\pi}{2m+1} \right) = \frac{2m(2m-1)}{6}. \quad (9.1)$$

To establish this, we start with the relation $e^{ix} = \cos x + i \sin x$. Taking the n -th power $e^{inx} = (e^{ix})^n$, we get

$$\cos nx + i \sin nx = (\cos x + i \sin x)^n.$$

The imaginary part of this is

$$\sin nx = \binom{n}{1} \sin x \cos^{n-1} x - \binom{n}{3} \sin^3 x \cos^{n-3} x \pm \dots \quad (9.2)$$

Now we let $n = 2m + 1$, while for x we will consider the m different values $x = \frac{r\pi}{2m+1}$, for $r = 1, 2, \dots, m$. For each of these values we have $nx = r\pi$, and thus $\sin nx = 0$, while $0 < x < \frac{\pi}{2}$ implies that for $\sin x$ we get m distinct positive values.

In particular, we can divide (9.2) by $\sin^n x$, which yields

$$0 = \binom{n}{1} \cot^{n-1} x - \binom{n}{3} \cot^{n-3} x \pm \dots,$$

that is,

$$0 = \binom{2m+1}{1} \cot^{2m} x - \binom{2m+1}{3} \cot^{2m-2} x \pm \dots$$

for each of the m distinct values of x . Thus for the polynomial of degree m

$$p(t) := \binom{2m+1}{1} t^m - \binom{2m+1}{3} t^{m-1} \pm \dots + (-1)^m \binom{2m+1}{2m+1}$$

we know m distinct roots

$$a_r = \cot^2 \left(\frac{r\pi}{2m+1} \right) \quad \text{for } r = 1, 2, \dots, m.$$

The roots are distinct because $\cot^2 x = \cot^2 y$ implies $\sin^2 x = \sin^2 y$ and thus $x = y$ for $x, y \in \{\frac{r\pi}{2m+1} : 1 \leq r \leq m\}$.

Hence the polynomial coincides with

$$p(t) = \binom{2m+1}{1} \left(t - \cot^2 \left(\frac{\pi}{2m+1}\right)\right) \dots \left(t - \cot^2 \left(\frac{m\pi}{2m+1}\right)\right).$$

Comparison of the coefficients of t^{m-1} in $p(t)$ now yields that the sum of the roots is

$$a_1 + \dots + a_m = \frac{\binom{2m+1}{3}}{\binom{2m+1}{1}} = \frac{2m(2m-1)}{6},$$

which proves (9.1).

We also need a second identity, of the same type,

$$\csc^2 \left(\frac{\pi}{2m+1}\right) + \csc^2 \left(\frac{2\pi}{2m+1}\right) + \dots + \csc^2 \left(\frac{m\pi}{2m+1}\right) = \frac{2m(2m+2)}{6}, \quad (9.3)$$

for the cosecant function $\csc x = \frac{1}{\sin x}$. But

$$\csc^2 x = \frac{1}{\sin^2 x} = \frac{\cos^2 x + \sin^2 x}{\sin^2 x} = \cot^2 x + 1,$$

so we can derive (9.3) from (9.1) by adding m to both sides of the equation.

Now the stage is set, and everything falls into place. We use that in the range $0 < y < \frac{\pi}{2}$ we have

$$0 < \sin y < y < \tan y,$$

and thus

$$0 < \cot y < \frac{1}{y} < \csc y,$$

which implies

$$\cot^2 y < \frac{1}{y^2} < \csc^2 y.$$

Now we take this double inequality, apply it to each of the m distinct values of x , and add the results. Using (9.1) for the left-hand side, and (9.3) for the right-hand side, we obtain

$$\frac{2m(2m-1)}{6} < \left(\frac{2m+1}{\pi}\right)^2 + \left(\frac{2m+1}{2\pi}\right)^2 + \dots + \left(\frac{2m+1}{m\pi}\right)^2 < \frac{2m(2m+2)}{6},$$

that is,

$$\frac{\pi^2}{6} \frac{2m}{2m+1} \frac{2m-1}{2m+1} < \frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{m^2} < \frac{\pi^2}{6} \frac{2m}{2m+1} \frac{2m+2}{2m+1}.$$

Both the left-hand and the right-hand side converge to $\frac{\pi^2}{6}$ for $m \rightarrow \infty$: end of proof. $\square$

Theorem 9.4 (Euler's series: Proof 4).

$$\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Proof. The first trick in this proof is to consider the Gregory–Leibniz series in doubly-infinite form $\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{2n+1}$. As for negative $n = -k < 0$ we get the same terms as for $n = k - 1 \geq 0$, since $\frac{(-1)^{-k}}{2(-k)+1} = \frac{(-1)^k}{-(2k-1)} = \frac{(-1)^{k-1}}{2(k-1)+1}$, we infer that $\sum_{n=-N}^N \frac{(-1)^n}{2n+1}$ converges to $\pi/2$ with $N \rightarrow \infty$, and thus the square of this sum converges to $\pi^2/4$. You may write this as

$$\lim_{N \rightarrow \infty} \sum_{m,n=-N}^N \frac{(-1)^m}{2m+1} \frac{(-1)^n}{2n+1} = \frac{\pi^2}{4}.$$

The double sum may be interpreted as the sum of all entries of a square matrix of size $(2N+1) \times (2N+1)$, and we know that for $N \rightarrow \infty$ this sum of all entries tends to $\pi^2/4$. We want to know, however, that the sum of only the *diagonal* entries, for $m = n$, also tends to $\pi^2/4$,

$$\lim_{N \rightarrow \infty} \sum_{n=-N}^N \frac{1}{(2n+1)^2} = \frac{\pi^2}{4},$$

because then $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \pi^2/8$ will follow, and this, as we know, is equivalent to Euler's theorem. So let's show that the sum of all off-diagonal terms tends to 0! We write δ_N for this sum, and use a prime to denote that the diagonal terms with $m = n$ are deleted, so

$$\begin{aligned} \delta_N &= \sum'_{m,n=-N}^N \frac{(-1)^{m+n}}{(2m+1)(2n+1)} \\ &= \sum'_{m,n=-N}^N (-1)^{m+n} \left(\frac{1}{2m-2n} \frac{1}{2m+1} - \frac{1}{2m-2n} \frac{1}{2n+1} \right) \\ &= \sum'_{m,n=-N}^N (-1)^{m+n} \left(\frac{1}{2m-2n} \frac{1}{2m+1} - \frac{1}{2n-2m} \frac{1}{2m+1} \right) \\ &= \sum'_{m,n=-N}^N (-1)^{m+n} \frac{1}{m-n} \frac{1}{2m+1} \\ &= \sum_{m=-N}^N \frac{1}{2m+1} \left(\sum'_{n=-N}^N \frac{(-1)^{m-n}}{m-n} \right). \end{aligned}$$

We only need to show that the terms

$$c_{m,N} := \sum'_{n=-N}^N \frac{(-1)^{m-n}}{m-n}$$

are small enough in absolute value. What do we know about them? It is easy to see that $c_{-m,N} = -c_{m,N}$, so in particular $c_{0,N} = 0$. Thus we may assume that $m > 0$, and note that the summands for $n = m+k$ and $n = m-k$ cancel as long as they are in the range between $-N$ and N , that is, for $1 \leq k \leq N-m$. Thus $c_{m,N}$ equals the alternating sum of fractions of decreasing size given by the remaining terms, where the largest one occurs for $n = m - (N-m) - 1 = 2m - N - 1$, that is $m - n = N - m + 1$. Hence

$$c_{m,N} = (-1)^{N-m+1} \left(\frac{1}{N-m+1} - \frac{1}{N-m+2} \pm \cdots \pm \frac{1}{m+N} \right),$$

which implies that

$$|c_{m,N}| \leq \frac{1}{N-m+1}.$$

This finally yields

$$\begin{aligned} |\delta_N| &\leq \sum_{m=-N}^N \left| \frac{1}{2m+1} \right| |c_{m,N}| \leq \sum_{m=-N}^N \frac{1}{2|m|-1} |c_{m,N}| \\ &\leq 2 \sum_{m=1}^N \frac{1}{m} |c_{m,N}| \leq 2 \sum_{m=1}^N \frac{1}{m} \frac{1}{N-m+1} \\ &= 2 \sum_{m=1}^N \frac{1}{N+1} \left(\frac{1}{m} + \frac{1}{N-m+1} \right) \\ &= 2 \frac{1}{N+1} (H_N + H_N) < 4 \frac{\log N + 1}{N+1}, \end{aligned}$$

and this goes to 0 as N goes to infinity. □

Theorem 9.5 (Four proofs of Euler's series). *Collecting the proofs from the chapter.*

Proof. See theorems in this chapter. □

Chapter 10

Hilbert's third problem: decomposing polyhedra

Lemma 10.1 (Pearl Lemma). *If P and Q are equidecomposable, then one can place a positive number of pearls (that is, assign positive integers) to all the segments of the decompositions $P = P_1 \cup \dots \cup P_n$ and $Q = Q_1 \cup \dots \cup Q_n$ in such a way that each edge of a piece P_k receives the same number of pearls as the corresponding edge of Q_k .*

Proof. Assign a variable x_i to each segment in the decomposition of P and a variable y_j to each segment in the decomposition of Q . Now we have to find positive *integer* values for the variables x_i and y_j in such a way that the x_i -variables corresponding to the segments of any edge of some P_k yield the same sum as the y_j -variables assigned to the segments of the corresponding edge of Q_k . This yields conditions that require that “some x_i -variables have the same sum as some y_j -values”, namely

$$\sum_{i:s_i \subseteq e} x_i - \sum_{j:s'_j \subseteq e'} y_j = 0$$

where the edge $e \subseteq P_k$ decomposes into the segments s_i , while the corresponding edge $e' \subseteq Q_k$ decomposes into the segments s'_j . This is a linear equation with integer coefficients.

We note, however, that positive *real* values satisfying all these requirements exist, namely the (real) lengths of the segments! Thus we are done, in view of the following lemma. $\square$

Lemma 10.2 (Cone Lemma). *If a system of homogeneous linear equations with integer coefficients has a positive real solution, then it also has a positive integer solution.*

Proof. The name of this lemma stems from the interpretation that the set

$$C = \{\mathbf{x} \in \mathbb{R}^N : A\mathbf{x} = \mathbf{0}, \mathbf{x} > \mathbf{0}\}$$

given by an integer matrix $A \in \mathbb{Z}^{M \times N}$ describes a (relatively open) rational cone. We have to show that if this is nonempty, then it also contains integer points: $C \cap \mathbb{N}^N \neq \emptyset$.

If C is nonempty, then so is $\overline{C} := \{\mathbf{x} \in \mathbb{R}^N : A\mathbf{x} = \mathbf{0}, \mathbf{x} \geq \mathbf{1}\}$, since for any positive vector a suitable multiple will have all coordinates equal to or larger than 1. (Here $\mathbf{1}$ denotes the vector with all coordinates equal to 1.) It suffices to verify that $\overline{C} \subseteq C$ contains a point with *rational* coordinates, since then multiplication with a common denominator for all coordinates will yield an integer point in $\overline{C} \subseteq C$.

There are many ways to prove this. We follow a well-trodden path that was first explored by Fourier and Motzkin [8, Lecture 1]: By “Fourier-Motzkin elimination” we show that the lexicographically smallest solution to the system

$$A\mathbf{x} = \mathbf{0}, \mathbf{x} \geq \mathbf{1}$$

exists, and that it is rational if the matrix A is integral.

Indeed, any linear equation $\mathbf{a}^T \mathbf{x} = 0$ can be equivalently enforced by two inequalities $\mathbf{a}^T \mathbf{x} \geq 0, -\mathbf{a}^T \mathbf{x} \geq 0$. (Here $\mathbf{a}$ denotes a column vector and $\mathbf{a}^T$ its transpose.) Thus it suffices to prove that any system of the type

$$A\mathbf{x} \geq \mathbf{b}, \mathbf{x} \geq \mathbf{1}$$

with integral A and $\mathbf{b}$ has a lexicographically smallest solution, which is rational, provided that the system has any real solution at all.

For this we argue with induction on N . The case $N = 1$ is clear. For $N > 1$ look at all the inequalities that involve x_N . If $\mathbf{x}' = (x_1, \dots, x_{N-1})$ is fixed, these inequalities give lower bounds on x_N (among them $x_N \geq 1$) and possibly also upper bounds. So we form a new system $A'\mathbf{x}' \geq \mathbf{b}, \mathbf{x}' \geq \mathbf{1}$ in $N - 1$ variables, which contains all the inequalities from the system $A\mathbf{x} \geq \mathbf{b}$ that do not involve x_N , as well as all the inequalities obtained by requiring that all upper bounds on x_N (if there are any) are larger or equal to all the lower bounds on x_N (which include $x_N \geq 1$). This system in $N - 1$ variables has a solution, and thus by induction it has a lexicographically minimal solution $\mathbf{x}'_*$, which is rational. And then the smallest x_N compatible with this solution $\mathbf{x}'_*$ is easily found, it is determined by a linear equation or inequality with integer coefficients, and thus it is rational as well. $\square$

Theorem 10.3 (Bricard’s condition). *TODO*

Proof. *TODO* $\square$

Theorem 10.4 (Example 1). *TODO*

Proof. *TODO* $\square$

Theorem 10.5 (Example 2). *TODO*

Proof. *TODO* $\square$

Theorem 10.6 (Example 3). *TODO*

Proof. *TODO* $\square$

Theorem 10.7 (Hilbert’s third problem). *TODO*

Proof. $\square$

Chapter 11

Lines in the plane and decompositions of graphs

Theorem 11.1. *In any configuration of n points in the plane, not all on a line, there is a line which contains exactly two of the points.*

Proof. TODO

□

Theorem 11.2. *Let P be a set of $n \geq 3$ points in the plane, not all on a line. Then the set $\mathcal{L}$ of lines passing through at least two points contains at least n lines.*

Proof. TODO

□

Theorem 11.3. *Let X be a set of $n \geq 3$ elements, and let $A_1, \dots, A_m$ be proper subsets of X , such that every pair of elements of X is contained in precisely one set A_i . Then $m \geq n$ holds.*

Proof. TODO

□

Theorem 11.4. *If K_n is decomposed into complete bipartite subgraphs $H_1, \dots, H_m$, then $m \geq n - 1$.*

Proof. TODO

□

Chapter 12

The slope problem

Theorem 12.1. *If $n \geq 3$ points in the plane do not lie on one single line, then they determine at least $n - 1$ different slopes, where equality is possible only if n is odd and $n \geq 5$.*

Proof. 1. TODO

2. TODO

3. TODO

4. TODO

5. TODO

6. TODO

□

Chapter 13

Three applications of Euler's formula

Theorem 13.1 (Euler's formula). *If G is a connected plane graph with n vertices, e edges and f faces, then*

$$n - e + f = 2.$$

Proof. TODO □

Proposition 13.2. *Let G be any simple plane graph with $n > 2$ vertices. Then G has at most $3 * n - 6$ edges.*

Proof. TODO □

Proposition 13.3. *Let G be any simple plane graph with $n > 2$ vertices. Then G has a vertex of degree at most 5.*

Proof. TODO □

Proposition 13.4. *Let G be any simple plane graph with $n > 2$ vertices. If the edges of G are two-colored, then there is a vertex of G with at most two color-changes in the cyclic order of the edges around the vertex.*

Proof. TODO □

Theorem 13.5 (Sylvester-Gallai). *Given any set of $n \geq 3$ points in the plane, not all on one line, there is always a line that contains exactly two of the points.*

Proof. TODO □

Theorem 13.6 (Monochromatic lines). *Given any finite configuration of “black” and “white” points in the plane, not all on one line, there is always a “monochromatic” line: a line that contains at least two points of one color and none of the other.*

Proof. TODO □

Lemma 13.7. *Every elementary triangle $\Delta = \text{conv}\{p_0, p_1, p_2\} \subset \mathbb{R}^2$ has area $A(\Delta) = 12$*

Proof. TODO □

Theorem 13.8 (Pick's theorem). *The area of any (not necessarily convex) polygon $Q \subset \mathbb{R}^2$ with integral vertices is given by*

$$A(Q) = n_{int} + \frac{1}{2}n_{bd} - 1$$

where n_{int} and n_{bd} are the numbers of integral points in the interior respectively on the boundary of Q .

Proof. TODO

□

Chapter 14

Cauchy's rigidity theorem

Lemma 14.1 (Cauchy's arm lemma). *TODO*

Proof. TODO

□

Theorem 14.2 (Cauchy's rigidity). *If two 3-dimensional convex polyhedra P and P' are combinatorially equivalent with corresponding pairs of adjacent congruent, then also the angles between corresponding pairs of adjacent facets are equal (and thus P is congruent to P').*

Proof. TODO

□

Chapter 15

The Borromean rings don't exist

Theorem 15.1. *If a link consists of disjoint perfect circles that are pairwise not linked, then the link is trivial*

Proof. TODO □

Theorem 15.2. *The Borromean rings are nontrivial, and they are also not equivalent to Tait's link No. 18*

Proof. TODO □

Theorem 15.3. *The Borromean rings cannot be build from three perfect circles*

Proof. TODO □

Chapter 16

Touching simplices

Theorem 16.1. *For every $d \geq 2$, there is a family of 2^d pairwise touching d -simplices in $\mathbb{R}^d$ together with a transversal line that hits the interior of every single one of them.*

Proof. TODO

□

Theorem 16.2. *For all $d \geq 1$, we have $f(d) < 2^{d+1}$.*

Proof. TODO

□

Chapter 17

Every large point set has an obtuse angle

Theorem 17.1. *For every d , one has the following chain of inequalities:*

$$2^d \leq_{(1)} \max \left\{ \#S \mid S \subseteq \mathbb{R}^d, \angle(s_i, s_j, s_k) \leq \frac{\pi}{2} \text{ for every } \{s_i, s_j, s_k\} \subseteq S \right\} \quad (17.1)$$

$$\leq_{(2)} \max \left\{ \#S \mid S \subseteq \mathbb{R}^d \text{ such that for any two points } \{s_i, s_j\} \subseteq S, \right.$$

$$\left. \text{there is a strip } S(i, j) \text{ that contains } S, \text{ with } s_i \text{ and } s_j \text{ lying in the parallel boundary hyperplanes of } S(i, j) \right\} \quad (17.2)$$

$$=_{(3)} \max \left\{ \#S \mid S \subseteq \mathbb{R}^d \text{ such that the translates } P - s_i, s_i \in S, \text{ of the convex hull } P := \text{conv}(S) \right.$$

$$\left. \text{intersect in a common point, but they only touch} \right\} \quad (17.3)$$

$$\leq_{(4)} \max \left\{ \#S \mid S \subseteq \mathbb{R}^d \text{ such that the translates } Q + s_i \text{ of some } d\text{-dimensional convex polytope } Q \subseteq \mathbb{R}^d \text{ touch pairwise} \right\} \quad (17.4)$$

$$=_{(5)} \max \left\{ \#S \mid S \subseteq \mathbb{R}^d \text{ such that the translates } Q^* + s_i \text{ of some } d\text{-dimensional centrally symmetric convex polytope } Q^* \subseteq \mathbb{R}^d \text{ touch pairwise} \right\} \quad (17.5)$$

$$\leq_{(6)} 2^d. \quad (17.6)$$

Proof. TODO □

Theorem 17.2. *For every $d \geq 2$, there is a set $S \subset \{0, 1\}^d$ of $2 \lfloor \frac{\sqrt{6}}{9} (\frac{2}{\sqrt{3}})^d \rfloor$ points in $\mathbb{R}^n$ (vertices of the unit d -cube) that determine only acute angles. In particular, in dimension $d = 34$ there is a set of $72 > 2 * 34 - 1$ points with only acute angles.*

Proof. TODO □

Chapter 18

Borsuk's conjecture

Theorem 18.1 (Borsuk's conjecture). *Let $q = p^m$ be a prime power, $n := 4q - 2$, and $d := \binom{n}{2} = (2q - 1)(4q - 3)$. Then there is a set $S \subseteq \{+1, -1\}^d$ of 2^{n-2} points in $\mathbb{R}^d$ such that every partition of S , whose parts have smaller diameter than S , has at least*

$$\frac{2^{n-2}}{\sum_{i=0}^{q-2} \binom{n-1}{i}}$$

parts. For $q = 9$ this implies that the Borsuk conjecture is false in dimension $d = 561$. Furthermore, $f(d) > (1.2)\sqrt{d}$ holds for all large enough d .

Proof. TODO

□

Chapter 19

Sets, functions, and the continuum hypothesis

Theorem 19.1. *The set of $\mathbb{Q}$ of rational numbers is countable.*

Proof. TODO □

Theorem 19.2. *The set $\mathbb{R}$ of real numbers is not countable*

Proof. TODO □

Theorem 19.3. *The set $\mathbb{R}^2$ of all ordered pairs of real numbers (that is, the real plane) has the same size as $\mathbb{R}$.*

Proof. TODO □

Theorem 19.4. *If each of two sets M and N can be mapped injectively into the other, then there is a bijection from M to N , that is $|M| = |N|$.*

Proof. TODO □

Theorem 19.5. *If $c > \aleph_1$, then every family $\{f_\alpha\}$ satisfying (P_0) is countable. If, on the other hand, $c = \aleph_1$, then there exists some family $\{f_\alpha\}$ with property P_0 which has size c .*

Proof. TODO □

Appendix: On cardinal and ordinal numbers

Proposition 19.6. *Let μ be an ordinal number and denote by W_μ the set of ordinal numbers smaller than μ . Then the following holds:*

1. *The elements of W_μ are pairwise comparable.*
2. *If we order W_μ according to their magnitude, then W_μ is well-ordered and has ordinal number μ .*

Proof. TODO □

Proposition 19.7. *Any two ordinal numbers μ and ν satisfy precisely one of the relations $\mu < \nu$, $\mu = \nu$, or $\mu > \nu$.*

Proof. TODO □

Proposition 19.8. *Every set of ordinal numbers (ordered according to magnitude) is well-ordered.*

Proof. TODO □

Proposition 19.9. *For every cardinal number $\mathfrak{m}$, there is a definite next larger cardinal number.*

Proof. TODO □

Proposition 19.10. *Let the infinite set M have cardinality $\mathfrak{m}$, and let M be well ordered according to the initial ordinal number $\omega_{\mathfrak{m}}$. Then M has no last element.*

Proof. Indeed, if M had a last element m , then the segment M_m would have an ordinal number $\mu < \omega_{\mathfrak{m}}$ with $|\mu| = \mathfrak{m}$, contradicting the definition of $\omega_{\mathfrak{m}}$. □

Proposition 19.11. *Suppose $\{A_{\alpha}\}$ is a family of size $\mathfrak{m}$ of countable sets A_{α} , where $\mathfrak{m}$ is an infinite cardinal. Then the union $\bigcup_{\alpha} A_{\alpha}$ has size at most $\mathfrak{m}$.*

Proof. TODO □

Chapter 20

In praise of inequalities

Theorem 20.1. Let $\langle a, b \rangle$ be an inner product on a real vector space V (with the norm $|a|^2 := \langle a, a \rangle$). Then

$$\langle a, b \rangle^2 \leq |a|^2 |b|^2$$

holds for all vectors $a, b \in V$, with equality if and only if a and b are linearly dependent.

Proof. The following (folklore) proof is probably the shortest. Consider the quadratic function

$$|xa + b|^2 = x^2 |a|^2 + 2x \langle a, b \rangle + |b|^2$$

in the variable x . We may assume $a \neq 0$. If $b = \lambda a$, then clearly

$$\langle a, b \rangle^2 = |a|^2 |b|^2.$$

If, on the other hand, a and b are linearly independent, then $|xa + b|^2 > 0$ for all x , and thus the discriminant $\langle a, b \rangle^2 - |a|^2 |b|^2$ is less than 0. $\square$

Theorem 20.2 (First proof). Let $a_1, \dots, a_n$ be positive real numbers, then

$$\frac{n}{\frac{1}{a_1} + \dots + \frac{1}{a_n}} \leq \sqrt[n]{a_1 a_2 \dots a_n} \leq \frac{a_1 + \dots + a_n}{n}$$

with equality in both cases if and only if all a_i 's are equal.

Proof. The following beautiful nonstandard induction proof is attributed to Cauchy. Let $P(n)$ be the statement of the second inequality, written in the form

$$a_1 a_2 \dots a_n \leq \left(\frac{a_1 + \dots + a_n}{n} \right)^n.$$

For $n = 2$, we have $a_1 a_2 \leq \left(\frac{a_1 + a_2}{2} \right)^2 \iff (a_1 - a_2)^2 \geq 0$, which is true. Now we proceed in the following two steps:

$$(A) \quad P(n) \implies P(n-1)$$

$$(B) \quad P(n) \text{ and } P(2) \implies P(2n)$$

which will clearly imply the full result.

To prove (A), set $A := \frac{\sum_{k=1}^{n-1} a_k}{n-1}$, then

$$\left(\prod_{k=1}^{n-1} a_k \right) A \stackrel{P(n)}{\leq} \left(\frac{\sum_{k=1}^{n-1} a_k + A}{n} \right)^n = \left(\frac{(n-1)A + A}{n} \right)^n = A^n$$

and hence $\prod_{k=1}^{n-1} a_k \leq A^{n-1}$.

For (B), we see

$$\prod_{k=1}^{2n} a_k = \left(\prod_{k=1}^n a_k \right) \left(\prod_{k=n+1}^{2n} a_k \right) \stackrel{P(n)}{\leq} \left(\frac{\sum_{k=1}^n a_k}{n} \right)^n \left(\frac{\sum_{k=n+1}^{2n} a_k}{n} \right)^n \stackrel{P(2)}{\leq} \left(\frac{\sum_{k=1}^{2n} a_k}{2n} \right)^{2n}.$$

The condition for equality is derived just as easily.

The left-hand inequality, between the harmonic and the geometric mean, follows now by considering $\frac{1}{a_1}, \dots, \frac{1}{a_n}$. $\square$

Theorem 20.3 (Another Proof). *Let $a_1, \dots, a_n$ be positive real numbers, then*

$$\frac{n}{\frac{1}{a_1} + \dots + \frac{1}{a_n}} \leq \sqrt[n]{a_1 a_2 \dots a_n} \leq \frac{a_1 + \dots + a_n}{n}$$

with equality in both cases if and only if all a_i 's are equal.

Proof. Of the many other proofs of the arithmetic-geometric mean inequality, let us single out a particularly striking one by Horst Alzer, with some shortenings due to France Dacar. As a matter of fact, this proof yields the stronger inequality

$$a_1^{p_1} a_2^{p_2} \dots a_n^{p_n} \leq p_1 a_1 + p_2 a_2 + \dots + p_n a_n$$

for any positive numbers $a_1, \dots, a_n, p_1, \dots, p_n$ with $\sum_{i=1}^n p_i = 1$. Let us denote the expression on the left side by G , and on the right side by A . Fix $c > 0$ and define the function $f(t) := \frac{1}{c} - \frac{1}{t}$ on $\mathbb{R}_{>0}$. Since $f(t) < 0$ for $t < c$ and $f(t) > 0$ for $t > c$, we get the inequality

$$\int_c^x f(t) dt \geq 0$$

for every $x > 0$, with equality if and only if $x = c$. Now

$$0 \leq \int_c^x f(t) dt = \left[\frac{t}{c} - \log t \right]_c^x = \frac{x}{c} - 1 - \log \frac{x}{c},$$

and setting $c = G$ and $x = a_i$ we conclude that

$$\frac{a_i}{G} - 1 \geq \log a_i - \log G \quad \text{for } i = 1, 2, \dots, n.$$

Multiplying this inequality by p_i and summing over all i gives

$$\sum_{i=1}^n p_i \frac{a_i}{G} - \sum_{i=1}^n p_i \geq \sum_{i=1}^n p_i \log a_i - \sum_{i=1}^n p_i \log G.$$

With $\sum p_i = 1$, the left side equals $\frac{A}{G} - 1$, while the right side is $\log(\prod a_i^{p_i}) - \log G = \log G - \log G = 0$. We conclude $\frac{A}{G} - 1 \geq 0$, which is $A \geq G$. In the case of equality, all inequalities must be equalities, which implies $a_1 = \dots = a_n = G$. $\square$

Theorem 20.4 (Still another Proof). *Let $a_1, \dots, a_n$ be positive real numbers, then*

$$\frac{n}{\frac{1}{a_1} + \dots + \frac{1}{a_n}} \leq \sqrt[n]{a_1 a_2 \dots a_n} \leq \frac{a_1 + \dots + a_n}{n}$$

with equality in both cases if and only if all a_i 's are equal.

Proof. There is another nice proof, due to Michael D. Hirschhorn. It uses Bernoulli's inequality, which says

$$(1+t)^{n+1} \geq 1 + (n+1)t \quad \text{for real } t \geq -1.$$

Suppose $a_1, a_2, \dots, a_{n+1} > 0$ and set

$$t = \frac{\frac{a_1 + \dots + a_{n+1}}{n+1}}{\frac{a_1 + \dots + a_n}{n}} - 1.$$

By Bernoulli,

$$\left(\frac{\frac{a_1 + \dots + a_{n+1}}{n+1}}{\frac{a_1 + \dots + a_n}{n}} \right)^{n+1} \geq 1 + (n+1) \left(\frac{\frac{a_1 + \dots + a_{n+1}}{n+1}}{\frac{a_1 + \dots + a_n}{n}} - 1 \right) = \frac{n a_{n+1}}{a_1 + \dots + a_n},$$

which translates into

$$\left(\frac{a_1 + \dots + a_{n+1}}{n+1} \right)^{n+1} \geq a_{n+1} \left(\frac{a_1 + \dots + a_n}{n} \right)^n,$$

and the arithmetic-geometric mean inequality follows by induction. $\square$

Theorem 20.5. *Suppose all roots of the polynomial $x^n + a_{n-1}x^{n-1} + \dots + a_0$ are real. Then the roots are contained in the interval with the endpoints*

$$-\frac{a_{n-1}}{n} \pm \frac{n-1}{n} \sqrt{a_{n-1}^2 - \frac{2n}{n-1} a_{n-2}}.$$

Proof. Let y be one of the roots and $y_1, \dots, y_{n-1}$ the others. Then the polynomial is $(x-y)(x-y_1) \dots (x-y_{n-1})$. Thus by comparing coefficients

$$\begin{aligned} -a_{n-1} &= y + y_1 + \dots + y_{n-1}, \\ a_{n-2} &= y(y_1 + \dots + y_{n-1}) + \sum_{i < j} y_i y_j, \end{aligned}$$

and so $a_{n-1}^2 - 2a_{n-2} - y^2 = \sum_{i=1}^{n-1} y_i^2$.

By Cauchy's inequality applied to $(y_1, \dots, y_{n-1})$ and $(1, \dots, 1)$,

$$(a_{n-1} + y)^2 = (y_1 + \dots + y_{n-1})^2 \leq (n-1) \sum_{i=1}^{n-1} y_i^2 = (n-1)(a_{n-1}^2 - 2a_{n-2} - y^2),$$

or

$$y^2 + \frac{2a_{n-1}}{n}y + \frac{2(n-1)}{n}a_{n-2} - \frac{n-2}{n}a_{n-1}^2 \leq 0.$$

Thus y (and hence all y_i) lie between the two roots of the quadratic function, and these roots are our bounds. $\square$

Theorem 20.6. Let $f(x)$ be a real polynomial of degree $n \geq 2$ with only real roots, such that $f(x) > 0$ for $-1 < x < 1$ and $f(-1) = f(1) = 0$. Then

$$\frac{2}{3}T \leq A \leq \frac{2}{3}R,$$

and equality holds in both cases only for $n = 2$.

Proof. **Proof of $\frac{2}{3}T \leq A$.** Since $f(x)$ has only real roots, and none of them in the open interval $(-1, 1)$, it can be written (apart from a constant positive factor which cancels out) in the form

$$f(x) = (1 - x^2) \prod_i (\alpha_i - x) \prod_j (\beta_j + x)$$

with $\alpha_i \geq 1, \beta_j \geq 1$. Hence $A = \int_{-1}^1 f(x) dx$.

By making the substitution $x \rightarrow -x$, we find that also $A = \int_{-1}^1 (1 - x^2) \prod_i (\alpha_i + x) \prod_j (\beta_j - x) dx$, and hence by the inequality of the arithmetic and the geometric mean (note that all factors are ≥ 0):

$$\begin{aligned} A &= \int_{-1}^1 \frac{1}{2} \left[(1 - x^2) \prod_i (\alpha_i - x) \prod_j (\beta_j + x) + (1 - x^2) \prod_i (\alpha_i + x) \prod_j (\beta_j - x) \right] dx \\ &\geq \int_{-1}^1 (1 - x^2) \left[\prod_i (\alpha_i^2 - x^2) \prod_j (\beta_j^2 - x^2) \right]^{1/2} dx \\ &\geq \int_{-1}^1 (1 - x^2) \left[\prod_i (\alpha_i^2 - 1) \prod_j (\beta_j^2 - 1) \right]^{1/2} dx \\ &= \frac{4}{3} \left[\prod_i (\alpha_i^2 - 1) \prod_j (\beta_j^2 - 1) \right]^{1/2}. \end{aligned}$$

Computing $f'(1)$ and $f'(-1)$ from the product form, we find $f'(1) = -2 \prod_i (\alpha_i - 1) \prod_j (\beta_j + 1)$ and $f'(-1) = 2 \prod_i (\alpha_i + 1) \prod_j (\beta_j - 1)$, hence $-f'(1)f'(-1) = 4 \prod_i (\alpha_i^2 - 1) \prod_j (\beta_j^2 - 1)$, so

$$A \geq \frac{2}{3} (-f'(1)f'(-1))^{1/2}.$$

Applying the inequality of the harmonic and geometric mean to $-f'(1)$ and $f'(-1)$, we arrive by formula (2) for T at

$$A \geq \frac{2}{3} \cdot \frac{2}{\frac{1}{-f'(1)} + \frac{1}{f'(-1)}} = \frac{4}{3} \cdot \frac{f'(1)f'(-1)}{f'(1) - f'(-1)} = \frac{2}{3}T.$$

The inequality $A \leq \frac{2}{3}R$: The book notes “The reader is invited to search for an equally inspired proof of the second inequality in Theorem 2.” This half is not formalized in Lean. $\square$

Theorem 20.7. Suppose G is a graph on n vertices without triangles. Then G has at most $\frac{n^2}{4}$ edges, and equality holds only when n is even and G is the complete bipartite graph $K_{n/2, n/2}$.

Proof. This proof, using Cauchy’s inequality, is due to Mantel. Let $V = \{1, \dots, n\}$ be the vertex set and E the edge set of G . By d_i we denote the degree of i , hence $\sum_{i \in V} d_i = 2|E|$ (see chapter 28). Suppose ij is an edge. Since G has no triangles, we find $d_i + d_j \leq n$ since no vertex is a neighbor of both i and j .

It follows that

$$\sum_{ij \in E} (d_i + d_j) \leq n|E|.$$

Note that d_i appears exactly d_i times in the sum, so we get

$$n|E| \geq \sum_{ij \in E} (d_i + d_j) = \sum_{i \in V} d_i^2,$$

and hence with Cauchy's inequality applied to the vectors $(d_1, \dots, d_n)$ and $(1, \dots, 1)$,

$$n|E| \geq \sum_{i \in V} d_i^2 \geq \frac{(\sum d_i)^2}{n} = \frac{4|E|^2}{n},$$

and the result follows. In the case of equality we find $d_i = d_j$ for all i, j , and further $d_i = \frac{n}{2}$ (since $d_i + d_j = n$). Since G is triangle-free, $G = K_{n/2, n/2}$ is immediately seen from this. $\square$

Theorem 20.8. *Suppose G is a graph on n vertices without triangles. Then G has at most $\frac{n^2}{4}$ edges, and equality holds only when n is even and G is the complete bipartite graph $K_{n/2, n/2}$.*

Proof. The following proof of Theorem 3, using the inequality of the arithmetic and the geometric mean, is a folklore Book Proof. Let α be the size of a largest independent set A , and set $\beta = n - \alpha$. Since G is triangle-free, the neighbors of a vertex i form an independent set, and we infer $d_i \leq \alpha$ for all i .

The set $B = V \setminus A$ of size β meets every edge of G . Counting the edges of G according to their endvertices in B , we obtain $|E| \leq \sum_{i \in B} d_i$. The inequality of the arithmetic and geometric mean now yields

$$|E| \leq \sum_{i \in B} d_i \leq \alpha\beta \leq \left(\frac{\alpha + \beta}{2}\right)^2 = \frac{n^2}{4},$$

and again the case of equality is easily dealt with. $\square$

Chapter 21

The fundamental theorem of algebra

Lemma 21.1. *Let $p(z) = \sum_{k=0}^n c_k z^k$ be a complex polynomial of degree $n \geq 1$. If $p(a) \neq 0$, then every disk D around a contains an interior point b with $|p(b)| < |p(a)|$*

Proof. TODO □

Theorem 21.2. *Every nonconstant polynomial with complex coefficients has at least one root in the field of complex numbers.*

Proof. The rest is easy. Clearly, $p(z)z^{-n}$ approaches the leading coefficient c_n of $p(z)$ as $|z|$ goes to infinity. Hence $|p(z)|$ goes to infinity as well with $|z| \rightarrow \infty$. Consequently, there exists $R_1 > 0$ such that $|p(z)| > |p(0)|$ for all points z on the circle $\{z : |z| = R_1\}$. Furthermore, our third fact (C) tells us that in the compact set $D_1 = \{z : |z| \leq R_1\}$ the continuous real-valued function $|p(z)|$ attains the minimum value at some point z_0 . Because of $|p(z)| > |p(0)|$ for z on the boundary of D_1 , z_0 must lie in the interior. But by d'Alembert's lemma [21.1](#) this minimum value $|p(z_0)|$ must be 0 — and this is the whole proof. □

Chapter 22

One square and an odd number of triangles

Definition 22.1 (valuation on $\mathbb{R}$).

Definition 22.2 (Three-coloring of plane). TODO

Definition 22.3 (Rainbow triangle). TODO

Lemma 22.4. *For any blue point $p_0 = (x_b, y_b)$, green point (x_g, y_g) , and red point (x_r, y_r) , the v -value of the determinant*

$$\det \begin{bmatrix} x_b & y_b & 1 \\ x_g & y_g & 1 \\ x_r & y_r & 1 \end{bmatrix}$$

is at least 1.

Proof. TODO □

Corollary 22.5. *Any line of the plane receives at most two different colors. The area of a rainbow triangle cannot be 0, and it cannot be $\frac{1}{n}$ for odd n .*

Proof. Follow from [22.4](#) □

Lemma 22.6. *Every dissection of the unit square $S = [0, 1]^2$ into finitely many triangles contains an odd number of rainbow triangles, and thus at least one.*

Proof. TODO □

Theorem 22.7 (Monsky's theorem). *It is not possible to dissect a square into an odd number of triangles of equal algebra area.*

Proof. TODO □

Appendix: Extending valuations

Lemma 22.8. *A proper subring $R \subset K$ is a valuation ring with respect to some valuation v into some ordered group G if and only if $K = R \cup R^{-1}$.*

Proof. TODO

□

Theorem 22.9. *The field of real numbers $\mathbb{R}$ has a non-Archimedean valuation to an ordered abelian group*

$$v : \mathbb{R} \rightarrow \{0\} \cup G$$

such that $v(\frac{1}{2}) > 1$.

Proof. TODO

□

Chapter 23

A theorem of Pólya on polynomials

Theorem 23.1. Let $f(z)$ be a complex polynomial of degree at least 1 and leading coefficient 1. Set $C = \{z \in \mathbb{C} : |f(z)| \leq 2\}$ and let $\mathcal{R}$ be the orthogonal projection of C onto the real axis. Then there are intervals $I_1, \dots, I_t$ on the real line which together cover $\mathcal{R}$ and satisfy

$$\ell(I_1) + \dots + \ell(I_t) \leq 4.$$

Proof. □

Theorem 23.2. Let $p(x)$ be a real polynomial of degree $n \geq 1$ with leading coefficient 1, and all roots real. Then the set $\mathcal{P} = \{x \in \mathbb{R} : |p(x)| \leq 2\}$ can be covered by intervals of total length at most 4.

Proof. □

Corollary 23.3. Let $p(x)$ be a real polynomial of degree $n \geq 1$ with leading coefficient 1, and suppose that $|p(x)| \leq 2$ for all x in the interval $[a, b]$. Then $b - a \leq 4$.

Proof. TODO □

23.1 Appendix: Chebyshev's theorem

Theorem 23.4 (Chebyshev's theorem). Let $p(x)$ be a real polynomial of degree $n \geq 1$ with leading coefficient 1. Then

$$\max_{-1 \leq x \leq 1} |p(x)| \geq \frac{1}{2^{n-1}}.$$

Proof. TODO □

Theorem 23.5 (Fact 1). If b is a multiple root of $p'(x)$, then b is also a root of $p(x)$.

Proof. Let $b_1 < \dots < b_r$ be the roots of $p(x)$ with multiplicities $s_1, \dots, s_r$, $\sum_{j=1}^r s_j = n$. From $p(x) = (x - b_j)^{s_j} h(x)$ we infer that b_j is a root of $p'(x)$ if $s_j \geq 2$, and the multiplicity of b_j in $p'(x)$ is $s_j - 1$. Furthermore, there is a root of $p'(x)$ between b_1 and b_2 , another root between b_2 and $b_3, \dots$, and one between b_{r-1} and b_r , and all these roots must be single roots, since $\sum_{j=1}^r (s_j - 1) + (r - 1)$ counts already up to the degree $n - 1$ of $p'(x)$. Consequently, the multiple roots of $p'(x)$ can only occur among the roots of $p(x)$. □

Theorem 23.6 (Fact 2). *We have $p'(x)^2 \geq p(x)p''(x)$ for all $x \in \mathbb{R}$.*

Proof. If $x = a_i$ is a root of $p(x)$, then there is nothing to show. Assume then x is not a root. The product rule of differentiation yields

$$p'(x) = \sum_{k=1}^n \frac{p(x)}{x - a_k}, \quad \text{that is,} \quad \frac{p'(x)}{p(x)} = \sum_{k=1}^n \frac{1}{x - a_k}.$$

Differentiating this again we have

$$\frac{p''(x)p(x) - p'(x)^2}{p(x)^2} = - \sum_{k=1}^n \frac{1}{(x - a_k)^2} < 0.$$

□

Chapter 24

Van der Waerden's permanent conjecture

Theorem 24.1. *Let $M = (m_{ij})$ be a doubly stochastic $n \times n$ matrix. Then*

$$\text{per } M \geq \frac{n!}{n^n}$$

and equality holds if and only if $m_{ij} = \frac{1}{n}$

Proof. TODO

□

Proposition 24.2 (Gurvit's proposition). *If $p(x) \in \mathbb{R}_+[x_1, \dots, x_n]$ is a H -stable and homogeneous of degree n , then either $p' \cong 0$, or p' is H -stable and homogeneous of degree $n - 1$. In either case*

$$\text{cap}(p') \geq \text{cap} \cdot g(\deg_n p).$$

Proof. TODO

□

Chapter 25

On a lemma of Littlewood and Offord

Theorem 25.1. *Let $a_1, \dots, a_n$ be vectors in $\mathbb{R}^d$, each of length at least 1, and let $R_1, \dots, R_k$ be k open regions of $\mathbb{R}^d$, where $|x - y| < 2$ for any x, y that lie in the same region R_i . Then the number of linear combinations $\sum_{i=1}^n \epsilon_i a_i$, $\epsilon_i \in \{1, -1\}$, that can lie in the union $\bigcup_i R_i$ of the regions is at most the sum of the k largest binomial coefficients $\binom{n}{j}$.*

In particular, we get the bound $\binom{\lfloor n/2 \rfloor}{n}$ for $k = 1$.

Proof. TODO

□

Chapter 26

Cotangent and the Herglotz trick

Lemma 26.1 (A). *The functions f and g are defined for all non-integral values and are continuous there.*

Proof. TODO □

Lemma 26.2 (B). *Both f and g are periodic of period 1, that is $f(x+1) = f(x)$ and $g(x+1) = g(x)$ hold for all $x \in \mathbb{R} \setminus \mathbb{Z}$.*

Proof. TODO □

Lemma 26.3 (C). *Both f and g are odd functions, that is we have $f(-x) = -f(x)$ and $g(-x) = -g(x)$ for all $x \in \mathbb{R} \setminus \mathbb{Z}$.*

Proof. TODO □

Lemma 26.4 (D). *The two functions f and g satisfy the same functional equation: $f(\frac{x}{2}) + f(\frac{x+1}{2}) = 2f(x)$ and $g(\frac{x}{2}) + g(\frac{x+1}{2}) = gf(x)$.*

Proof. TODO □

Lemma 26.5 (E). *By setting $h(x) := 0$ for $x \in \mathbb{Z}$, h becomes a continuous function on all of $\mathbb{R}$ that shares the properties given in [26.2](#), [26.3](#), [26.4](#).*

Proof. TODO □

Theorem 26.6.

$$\pi \cot \pi x = \frac{1}{x} + \sum_{n=1}^{\infty} \left(\frac{1}{x+n} + \frac{1}{x-n} \right)$$

for $x \in \mathbb{R} \setminus \mathbb{Z}$.

Proof. □

Chapter 27

Buffon's needle problem

Theorem 27.1 (Buffon's needle problem). *If a short needle, of length ℓ , is dropped on paper that is ruled with equally spaced lines of distance $d \geq \ell$, then the probability that the needle comes to lie in a position where it crosses one of the lines is exactly*

$$p = \frac{2\ell}{\pi d}.$$

Proof. TODO

□

Chapter 28

Pigeon-hole and double counting

Some mathematical principles, such as the two in the title of this chapter, are so obvious that you might think they would only produce equally obvious results. To convince you that “It ain’t necessarily so” we illustrate them with examples that were suggested by Paul Erdős to be included in The Book. We will encounter instances of them also in later chapters.

Theorem 28.1 (Pigeon-hole principle). *If n objects are placed in r boxes, where $r < n$, then at least one of the boxes contains more than one object.*

Well, this is indeed obvious, there is nothing to prove. In the language of mappings our principle reads as follows: Let N and R be two finite sets with $|N| = n > r = |R|$, and let $f : N \rightarrow R$ be a mapping. Then there exists some $a \in R$ with $|f^{-1}(a)| \geq 2$. We may even state a stronger inequality: There exists some $a \in R$ with

$$|f^{-1}(a)| \geq \left\lceil \frac{n}{r} \right\rceil. \quad (1)$$

In fact, otherwise we would have $|f^{-1}(a)| < \frac{n}{r}$ for all a , and hence $n = \sum_{a \in R} |f^{-1}(a)| < r \cdot \frac{n}{r} = n$, which cannot be.

Proof. Obvious. □

28.1 Numbers

Theorem 28.2 (Claim). *Consider the numbers $1, 2, 3, \dots, 2n$, and take any $n + 1$ of them. Then there are two among these $n + 1$ numbers which are relatively prime.*

Proof. This is again obvious. There must be two numbers which are only 1 apart, and hence relatively prime. □

But let us now turn the condition around.

Theorem 28.3 (Claim). *Suppose again $A \subset \{1, 2, \dots, 2n\}$ with $|A| = n + 1$. Then there are always two numbers in A such that one divides the other.*

Proof. Write every number $a \in A$ in the form $a = 2^k m$, where m is an odd number between 1 and $2n - 1$. Since there are $n + 1$ numbers in A , but only n different odd parts, there must be two numbers in A with the same odd part. Hence one is a multiple of the other. □

28.2 Sequences

Theorem 28.4 (Claim (Erdős–Szekeres)). *In any sequence $a_1, a_2, \dots, a_{mn+1}$ of $mn + 1$ distinct real numbers, there exists an increasing subsequence*

$$a_{i_1} < a_{i_2} < \dots < a_{i_{m+1}} \quad (i_1 < i_2 < \dots < i_{m+1})$$

of length $m + 1$, or a decreasing subsequence

$$a_{j_1} > a_{j_2} > \dots > a_{j_{n+1}} \quad (j_1 < j_2 < \dots < j_{n+1})$$

of length $n + 1$, or both.

Proof. This time the application of the pigeon-hole principle is not immediate. Associate to each a_i the number t_i , which is the length of a longest increasing subsequence starting at a_i . If $t_i \geq m + 1$ for some i , then we have an increasing subsequence of length $m + 1$. Suppose then that $t_i \leq m$ for all i . The function $f : a_i \mapsto t_i$ mapping $\{a_1, \dots, a_{mn+1}\}$ to $\{1, \dots, m\}$ tells us by (1) that there is some $s \in \{1, \dots, m\}$ such that $f(a_i) = s$ for $\frac{mn}{m} + 1 = n + 1$ numbers a_i . Let $a_{j_1}, a_{j_2}, \dots, a_{j_{n+1}}$ ($j_1 < \dots < j_{n+1}$) be these numbers. Now look at two consecutive numbers $a_{j_i}, a_{j_{i+1}}$. If $a_{j_i} < a_{j_{i+1}}$, then we would obtain an increasing subsequence of length s starting at a_{j_i} , and consequently an increasing subsequence of length $s + 1$ starting at a_{j_i} , which cannot be since $f(a_{j_i}) = s$. We thus obtain a decreasing subsequence $a_{j_1} > a_{j_2} > \dots > a_{j_{n+1}}$ of length $n + 1$. $\square$

28.3 Sums

Theorem 28.5 (Claim). *Suppose we are given n integers $a_1, \dots, a_n$, which need not be distinct. Then there is always a set of consecutive numbers $a_{k+1}, a_{k+2}, \dots, a_\ell$ whose sum $\sum_{i=k+1}^\ell a_i$ is a multiple of n .*

Proof. For the proof we set $N = \{0, 1, \dots, n\}$ and $R = \{0, 1, \dots, n-1\}$. Consider the map $f : N \rightarrow R$, where $f(m)$ is the remainder of $a_1 + \dots + a_m$ upon division by n . Since $|N| = n + 1 > n = |R|$, it follows that there are two sums $a_1 + \dots + a_k, a_1 + \dots + a_\ell$ ($k < \ell$) with the same remainder, where the first sum may be the empty sum denoted by 0. It follows that

$$\sum_{i=k+1}^\ell a_i = \sum_{i=1}^\ell a_i - \sum_{i=1}^k a_i$$

has remainder 0 — end of proof. $\square$

Let us turn to the second principle: counting in two ways. By this we mean the following.

Theorem 28.6 (Double counting). *Suppose that we are given two finite sets R and C and a subset $S \subseteq R \times C$. Whenever $(p, q) \in S$, then we say p and q are incident. If r_p denotes the number of elements that are incident to $p \in R$, and c_q denotes the number of elements that are incident to $q \in C$, then*

$$\sum_{p \in R} r_p = |S| = \sum_{q \in C} c_q. \quad (3)$$

Proof. Again, there is nothing to prove. The first sum classifies the pairs in S according to the first entry, while the second sum classifies the same pairs according to the second entry. $\square$

There is a useful way to picture the set S . Consider the matrix $A = (a_{pq})$, the incidence matrix of S , where the rows and columns of A are indexed by the elements of R and C , respectively, with

$$a_{pq} = \begin{cases} 1 & \text{if } (p, q) \in S, \\ 0 & \text{if } (p, q) \notin S. \end{cases}$$

With this set-up, r_p is the sum of the p -th row of A and c_q is the sum of the q -th column. Hence the first sum in (3) adds the entries of A (that is, counts the elements in S) by rows, and the second sum by columns.

28.4 Numbers again

Look at the incidence matrix for $R = C = \{1, 2, \dots, n\}$ and $S = \{(i, j) : i \mid j\}$. The number of 1's in column j is precisely the number of divisors of j ; let us denote this number by $t(j)$. Let us ask how large this number $t(j)$ is on the average when j ranges from 1 to n . Thus, we ask for the quantity

$$\bar{t}(n) = \frac{1}{n} \sum_{j=1}^n t(j).$$

How large is $\bar{t}(n)$ for arbitrary n ? At first glance, this seems hopeless. For prime numbers p we have $t(p) = 2$, while for 2^k we obtain a large number $t(2^k) = k + 1$. So, $t(n)$ is a wildly jumping function, and we surmise that the same is true for $\bar{t}(n)$. Wrong guess, the opposite is true! Counting in two ways provides an unexpected and simple answer.

Theorem 28.7 (Average number of divisors). *For any positive integer n ,*

$$\sum_{j=1}^n t(j) = \sum_{i=1}^n \left\lfloor \frac{n}{i} \right\rfloor.$$

Consequently,

$$\log n - 1 < \bar{t}(n) \leq H_n < \log n + 1,$$

where $H_n = \sum_{i=1}^n \frac{1}{i}$ is the n -th harmonic number.

Proof. Consider the matrix A (as above) for the integers 1 up to n . Counting by columns we get $\sum_{j=1}^n t(j)$. How many 1's are in row i ? Easy enough, the 1's correspond to the multiples of i : $1 \cdot i, 2 \cdot i, \dots$, and the last multiple not exceeding n is $\lfloor n/i \rfloor \cdot i$. Hence we obtain

$$\bar{t}(n) = \frac{1}{n} \sum_{j=1}^n t(j) = \frac{1}{n} \sum_{i=1}^n \left\lfloor \frac{n}{i} \right\rfloor \leq \frac{1}{n} \sum_{i=1}^n \frac{n}{i} = \sum_{i=1}^n \frac{1}{i} = H_n,$$

where the error in each summand, when passing from $\lfloor n/i \rfloor$ to n/i , is less than 1. Together with the standard estimates $\log n < H_n < \log n + 1$ this gives $\log n - 1 < \bar{t}(n) \leq H_n < \log n + 1$.

Thus we have proved the remarkable result that, while $t(n)$ is totally erratic, the average $\bar{t}(n)$ behaves beautifully: It differs from $\log n$ by less than 1. $\square$

28.5 Graphs

Let G be a finite simple graph with vertex set V and edge set E . The degree $d(v)$ of a vertex v is the number of edges which have v as an end-vertex.

Almost every book in graph theory starts with the following result:

Lemma 28.8 (Handshaking). *Let G be a finite simple graph with vertex set V and edge set E and let $d(v)$ denote the degree of a vertex v . Then*

$$\sum_{v \in V} d(v) = 2|E|. \quad (4)$$

Proof. For the proof consider $S \subseteq V \times E$, where S is the set of pairs (v, e) such that $v \in V$ is an end-vertex of $e \in E$. Counting S in two ways gives on the one hand $\sum_{v \in V} d(v)$, since every vertex contributes $d(v)$ to the count, and on the other hand $2|E|$, since every edge has two ends. $\square$

We want to single out the following beautiful application to an extremal problem on graphs. Here is the problem:

Suppose $G = (V, E)$ has n vertices and contains no cycle of length 4 (denoted by C_4). How many edges can G have at most?

Let us tackle the general problem. Let G be a graph on n vertices without a 4-cycle. We count the following set S in two ways: S is the set of pairs $(u, \{v, w\})$ where u is adjacent to v and to w , with $v \neq w$. In other words, we count all occurrences of a “cherry” (path of length 2 centered at u).

Summing over u , we find $|S| = \sum_{u \in V} \binom{d(u)}{2}$. On the other hand, every pair $\{v, w\}$ has at most one common neighbor (by the C_4 -condition). Hence $|S| \leq \binom{n}{2}$, and we conclude

$$\sum_{u \in V} \binom{d(u)}{2} \leq \binom{n}{2}$$

or

$$\sum_{u \in V} d(u)^2 \leq n(n-1) + \sum_{u \in V} d(u). \quad (5)$$

Next we apply the Cauchy–Schwarz inequality to the vectors $(d(u_1), \dots, d(u_n))$ and $(1, 1, \dots, 1)$, obtaining

$$\left(\sum_{u \in V} d(u) \right)^2 \leq n \sum_{u \in V} d(u)^2,$$

and hence by (5)

$$\left(\sum_{u \in V} d(u) \right)^2 \leq n^2(n-1) + n \sum_{u \in V} d(u).$$

Invoking (4) we find

$$4|E|^2 \leq n^2(n-1) + 2n|E|$$

or

$$|E|^2 - \frac{n}{2}|E| - \frac{n^2(n-1)}{4} \leq 0.$$

Solving the corresponding quadratic equation we thus obtain the following result of István Reiman.

Theorem 28.9 (Reiman). *If the graph G on n vertices contains no 4-cycles, then*

$$|E| \leq \left\lfloor \frac{n}{4} (1 + \sqrt{4n-3}) \right\rfloor. \quad (6)$$

Proof. The proof follows from the chain of inequalities above: the key combinatorial step $\sum \binom{d(u)}{2} \leq \binom{n}{2}$ is established by double counting (Theorem 28.10), and the final bound follows by Cauchy–Schwarz and the quadratic formula. $\square$

Theorem 28.10 (Cherry counting). *If the graph G on n vertices contains no 4-cycles, then*

$$\sum_{v \in V} \binom{d(v)}{2} \leq \binom{n}{2}.$$

Proof. Each pair $(v, \{u, w\})$ with $u, w \in N(v)$ maps to $\{u, w\}$. The C_4 -free condition ensures this map is injective on the second component: if both v_1 and v_2 are common neighbors of $\{u, w\}$ with $v_1 \neq v_2$, then v_1 – u – v_2 – w forms a 4-cycle, a contradiction. $\square$

28.6 Sperner’s Lemma

In 1912, Luitzen Brouwer published his famous fixed point theorem:

Every continuous function $f : B^n \rightarrow B^n$ of an n -dimensional ball to itself has a fixed point (a point $x \in B^n$ with $f(x) = x$).

For dimension 1, that is for an interval, this follows easily from the intermediate value theorem, but for higher dimensions Brouwer’s proof needed some sophisticated machinery. It was therefore quite a surprise when in 1928 young Emanuel Sperner produced a simple combinatorial result from which both Brouwer’s fixed point theorem and the invariance of the dimension under continuous bijective maps could be deduced. And what’s more, Sperner’s ingenious lemma is matched by an equally beautiful proof — it is just double counting.

We discuss Sperner’s lemma, and Brouwer’s theorem as a consequence, for the first interesting case, that of dimension $n = 2$.

Lemma 28.11 (Sperner’s Lemma). *Suppose that some “big” triangle with vertices V_1, V_2, V_3 is triangulated (that is, decomposed into a finite number of “small” triangles that fit together edge-by-edge). Assume that the vertices in the triangulation get “colors” from the set $\{1, 2, 3\}$ such that V_i receives the color i (for each i), and only the colors i and j are used for vertices along the edge from V_i to V_j (for $i \neq j$), while the interior vertices are colored arbitrarily with 1, 2, or 3. Then in the triangulation there must be a small “tricolored” triangle, which has all three different vertex colors.*

Proof. We will prove a stronger statement: The number of tricolored triangles is not only nonzero, it is always *odd*.

Consider the dual graph to the triangulation, but don’t take all its edges — only those which cross an edge that has endvertices with the (different) colors 1 and 2. Thus we get a “partial dual graph” which has degree 1 at all vertices that correspond to tricolored triangles, degree 2 for all triangles in which the two colors 1 and 2 appear, and degree 0 for triangles that do not have both colors 1 and 2. Thus only the tricolored triangles correspond to vertices of odd degree (of degree 1).

However, the vertex of the dual graph which corresponds to the outside of the triangulation has odd degree: in fact, along the big edge from V_1 to V_2 , there is an odd number of changes between 1 and 2. Thus an odd number of edges of the partial dual graph crosses this big edge, while the other big edges cannot have both 1 and 2 occurring as colors.

Now since the number of odd-degree vertices in any finite graph is even (by equation (4)), we find that the number of small triangles with three different colors (corresponding to odd inside vertices of our dual graph) is odd. $\square$

With this lemma, it is easy to derive Brouwer's theorem.

Theorem 28.12 (Brouwer's Fixed Point Theorem (for $n = 2$)). *Every continuous function $f : B^2 \rightarrow B^2$ of a 2-dimensional ball to itself has a fixed point (a point $x \in B^2$ with $f(x) = x$).*

Proof. Let Δ be the triangle in $\mathbb{R}^3$ with vertices $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, and $e_3 = (0, 0, 1)$. It suffices to prove that every continuous map $f : \Delta \rightarrow \Delta$ has a fixed point, since Δ is homeomorphic to B^2 .

We use $\delta(T)$ to denote the maximal length of an edge in a triangulation T . One can easily construct an infinite sequence of triangulations $T_1, T_2, \dots$ of Δ such that the sequence of maximal diameters $\delta(T_k)$ converges to 0 (for example by iterated barycentric subdivision).

For each of these triangulations, we define a 3-coloring of their vertices v by setting $\lambda(v) := \min\{i : f(v)_i < v_i\}$, that is, $\lambda(v)$ is the smallest index i such that the i -th coordinate of $f(v) - v$ is negative. If this smallest index i does not exist, then we have found a fixed point and are done: To see this, note that every $v \in \Delta$ lies in the plane $x_1 + x_2 + x_3 = 1$, hence $\sum_i v_i = 1$. So if $f(v) \neq v$, then at least one of the coordinates of $f(v) - v$ must be negative (and at least one must be positive).

Let us check that this coloring satisfies the assumptions of Sperner's lemma. First, the vertex e_i must receive color i , since the only possible negative component of $f(e_i) - e_i$ is the i -th component. Moreover, if v lies on the edge opposite to e_i , then $v_i = 0$, so the i -th component of $f(v) - v$ cannot be negative, and hence v does not get the color i .

Sperner's lemma now tells us that in each triangulation T_k there is a tricolored triangle $\{v^{k:1}, v^{k:2}, v^{k:3}\}$ with $\lambda(v^{k:i}) = i$. Since the simplex Δ is compact, some subsequence of $(v^{k:1})_{k \geq 1}$ has a limit point v . The distances of $v^{k:2}$ and $v^{k:3}$ from $v^{k:1}$ are at most the mesh length $\delta(T_k) \rightarrow 0$, so all three sequences converge to the same point v .

But where is $f(v)$? We know that the first coordinate of $f(v^{k:1})$ is smaller than that of $v^{k:1}$ for all k . Now since f is continuous, we derive that the first coordinate of $f(v)$ is smaller or equal to that of v . The same reasoning works for the second and third coordinates. Thus none of the coordinates of $f(v) - v$ is positive — and we have already seen that this contradicts the assumption $f(v) \neq v$. $\square$

Chapter 29

Tiling rectangles

Theorem 29.1 (First proof). *Whenever a rectangle is tiled by rectangles all of which have at least one side of integer length, then the tiled rectangle has at least one side of integer length.*

Proof. TODO

□

Theorem 29.2 (Second proof). *Whenever a rectangle is tiled by rectangles all of which have at least one side of integer length, then the tiled rectangle has at least one side of integer length.*

Proof. TODO

□

Theorem 29.3 (Third proof). *Whenever a rectangle is tiled by rectangles all of which have at least one side of integer length, then the tiled rectangle has at least one side of integer length.*

Proof. TODO

□

Chapter 30

Three famous theorems on finite sets

Theorem 30.1 (Sperner’s theorem). *The size of a largest antichain of an n -set is $\binom{n}{\lfloor n/2 \rfloor}$.*

Proof. We follow Lubell’s elegant proof via the LYM inequality. Consider all $n!$ permutations σ of $\{1, \dots, n\}$, each generating a maximal chain $\{\sigma(1)\} \subset \{\sigma(1), \sigma(2)\} \subset \dots \subset \{1, \dots, n\}$. A k -element set A appears in exactly $k!(n-k)!$ of these chains. Since an antichain $\mathcal{F}$ meets each chain in at most one set, summing over $\mathcal{F}$ gives

$$\sum_{A \in \mathcal{F}} k_A!(n-k_A)! \leq n!,$$

i.e. $\sum_{A \in \mathcal{F}} \frac{1}{\binom{n}{|A|}} \leq 1$ (the *LYM inequality*). Since each summand is at least $1/\binom{n}{\lfloor n/2 \rfloor}$, we conclude $|\mathcal{F}| \leq \binom{n}{\lfloor n/2 \rfloor}$. $\square$

Lemma 30.2 (Katona’s arc lemma). *Let $n \geq 2k$, and suppose we are given t distinct arcs $A_1, \dots, A_t$ of length k on a circle of n points, such that any two arcs share at least one point. Then $t \leq k$.*

Proof. Consider the n points arranged on a circle. Each arc of length k consists of k consecutive points. Two arcs of length k on a circle of $n \geq 2k$ points are disjoint if and only if their starting points are at distance $\geq k$. Since there are n possible starting points and each arc “blocks out” $2k-1$ starting points for intersecting arcs, the maximum number of pairwise intersecting arcs is at most k . (This lemma is subsumed by the Kruskal–Katona machinery in Mathlib’s proof of EKR.) $\square$

Theorem 30.3 (Erdős–Ko–Rado). *Let $n \geq 2k$. The largest size of an intersecting k -uniform family in an n -set is $\binom{n-1}{k-1}$.*

Proof. We follow Katona’s cyclic permutation argument. Arrange $\{1, \dots, n\}$ on a circle. Among the n arcs of k consecutive elements, at most k can be pairwise intersecting (by the arc lemma). For each of the $(n-1)!$ circular permutations, an intersecting family $\mathcal{F}$ contributes at most k arcs. Each k -set appears as an arc in exactly $k!(n-k)!$ circular permutations. Double counting gives

$$|\mathcal{F}| \cdot k!(n-k)! \leq k \cdot (n-1)!,$$

so $|\mathcal{F}| \leq \binom{n-1}{k-1}$. $\square$

Theorem 30.4 (Hall's marriage theorem). *Let $A_1, \dots, A_n$ be a collection of subsets of a finite set X . Then there exists a system of distinct representatives if and only if the union of any m sets A_i contains at least m elements, for $1 \leq m \leq n$.*

Proof. Necessity is clear: the m distinct representatives of any m sets must lie in their union.

For sufficiency, we use induction on n . **Case 1:** If Hall's condition holds strictly ($|\bigcup_{i \in S} A_i| \geq |S| + 1$ for every proper nonempty $S \subsetneq \{1, \dots, n\}$), pick any $a_1 \in A_1$ as representative, remove a_1 from all sets, and verify that Hall's condition still holds for $A_2 \setminus \{a_1\}, \dots, A_n \setminus \{a_1\}$.

Case 2: If some proper nonempty subset S is "tight" ($|\bigcup_{i \in S} A_i| = |S|$), first find an SDR for the subfamily $(A_i)_{i \in S}$ by induction. Then show that the remaining sets $(A_j)_{j \notin S}$, with the chosen representatives removed, still satisfy Hall's condition (using tightness of S), and apply induction again. $\square$

Corollary 30.5 (k systems of distinct representatives). *Suppose the sets $A_1, \dots, A_n$ all have size $k \geq 1$ and suppose further that no element is contained in more than k sets. Then there exist k SDR's such that for any i the k representatives of A_i are distinct and thus together form the set A_i .*

Chapter 31

Shuffling cards

Lemma 31.1. *Let $\mathbb{Q} : \mathfrak{S}_n \rightarrow \mathbb{R}$ be any probability distribution that defines a shuffling process $\mathbb{Q}^*k$ with a strong uniform stopping rule whose stopping time is T . Then for all $k \geq 0$,*

$$\|\mathbb{Q}^*k - \mathbb{U}\| \leq \text{Prob}[T > k].$$

Proof. If X is a random variable with values in $\mathfrak{S}_n$, with probability distribution $\mathbb{Q}$, then we write $\mathbb{Q}(S)$ for the probability that X takes a value in $S \subseteq \mathfrak{S}_n$. Thus $\mathbb{Q}(S) = \text{Prob}[X \in S]$, and in the case of the uniform distribution $\mathbb{Q} = \mathbb{U}$ we get

$$\mathbb{U}(S) = \text{Prob}[X \in S] = \frac{|S|}{n!}.$$

For every subset $S \subseteq \mathfrak{S}_n$, we get the probability that after k steps our deck is ordered according to a permutation in S as

$$\begin{aligned} \mathbb{Q}^*k(S) &= \text{Prob}[X_k \in S] = \sum_{j \leq k} \text{Prob}[X_k \in S \wedge T = j] + \text{Prob}[X_k \in S \wedge T > k] \\ &= \sum_{j \leq k} \mathbb{U}(S) \cdot \text{Prob}[T = j] + \text{Prob}[X_k \in S | T > k] \cdot \text{Prob}[T > k] \\ &= \mathbb{U}(S)(1 - \text{Prob}[T > k]) + \text{Prob}[X_k \in S | T > k] \cdot \text{Prob}[T > k] \\ &= \mathbb{U}(S) + (\text{Prob}[X_k \in S | T > k] - \mathbb{U}(S)) \cdot \text{Prob}[T > k]. \end{aligned}$$

This yields

$$|\mathbb{Q}^*k(S) - \mathbb{U}(S)| \leq \text{Prob}[T > k]$$

since

$$\text{Prob}[X_k \in S | T > k] - \mathbb{U}(S)$$

is a difference of two probabilities, so it has absolute value at most 1. □

Theorem 31.2. *Let $c \geq 0$ and $k := \lceil n \log n + cn \rceil$. Then after performing k top-in-at-random shuffles on a deck of n cards, the variation distance from the uniform distribution satisfies*

$$d(k) := \|\text{Top}^*k - \mathbb{U}\| \leq e^{-c}.$$

Proof. TODO □

Theorem 31.3. *After performing k riffle shuffles on a deck of n cards, the variation distance from a uniform distribution satisfies*

$$||Rif^k - \mathbb{U}|| \leq 1 - \prod_{i=1}^{n-1} \left(1 - \frac{i}{2^k}\right).$$

Proof. TODO

□

Chapter 32

Lattice paths and determinants

Lemma 32.1. *Let $G = (V, E)$ be a finite weighted acyclic directed graph, $\mathcal{A} = \{A_1, \dots, A_n\}$ and $\mathcal{B} = \{B_1, \dots, B_n\}$ two n -sets of vertices, and M the path matrix from $\mathcal{A}$ to $\mathcal{B}$. Then*

$$\det M = \sum_{\mathcal{P} \text{ vertex-disjoint path system}} \text{sign}(\mathcal{P}) w(\mathcal{P}). \quad (3)$$

Proof. TODO

□

Theorem 32.2. *If P is an $r \times s$ matrix and Q an $s \times r$ matrix, $r \leq s$, then*

$$\det(PQ) = \sum_{\mathcal{Z}} (\det P_{\mathcal{Z}})(\det Q_{\mathcal{Z}}),$$

where $P_{\mathcal{Z}}$ is the $r \times r$ submatrix of P with column-set $\mathcal{Z}$, and $Q_{\mathcal{Z}}$ the $r \times r$ submatrix of Q with corresponding rows $\mathcal{Z}$.

Proof. TODO

□

Chapter 33

Cayley's formula for the number of trees

Theorem 33.1 (First proof (bijection)). *There are n^{n-2} different labeled trees on n nodes.*

Proof. TODO □

Theorem 33.2 (Second proof (Linear Algebra)). *There are n^{n-2} different labeled trees on n nodes.*

Proof. TODO □

Theorem 33.3 (Second proof (Recursion)). *There are n^{n-2} different labeled trees on n nodes.*

Proof. TODO □

Theorem 33.4 (Second proof (Double Counting)). *There are n^{n-2} different labeled trees on n nodes.*

Proof. TODO □

Chapter 34

Identities versus bijections

Theorem 34.1.

$$\prod_{k \geq 1} (1 - x^k) = 1 + \sum_{j \geq 1} (-1)^j (x^{\frac{3j^2-j}{2}} + x^{3j^2+j2}).$$

Proof. TODO

□

Chapter 35

The finite Kakeya problem

Let F be a finite field.

Lemma 35.1. *Every nonzero polynomial $p(x) \in F[x_1, \dots, x_n]$ of degree d has at most dq^{n-1} roots in F^n .*

Proof. We use induction on n , with fact (1) above as the starting case $n = 1$. Let us split $p(x)$ into summands according to the powers of x_n ,

$$p(x) = g_0 + g_1x_n + g_2x_n^2 + \dots + g_\ell x_n^\ell,$$

where $g_i \in F[x_1, \dots, x_{n-1}]$ for $0 \leq i \leq \ell \leq d$, and g_ℓ is nonzero. We write every $v \in F^n$ in the form $v = (a, b)$ with $a \in F^{n-1}$, $b \in F$, and estimate the number of roots $p(a, b) = 0$.

Case 1. Roots (a, b) with $g_\ell(a) = 0$. Since $g_\ell \neq 0$ and $\deg g_\ell \leq d - \ell$, by induction the polynomial g_ℓ has at most $(d - \ell)q^{n-2}$ roots in F^{n-1} , and for each a there are at most q different choices for b , which gives at most $(d - \ell)q^{n-1}$ such roots for $p(x)$ in F^n .

Case 2. Roots (a, b) with $g_\ell(a) \neq 0$. Here $p(a, x_n) \in F[x_n]$ is not the zero polynomial in the single variable x_n , it has degree ℓ , and hence for each a by (1) there are at most ℓ elements b with $p(a, b) = 0$. Since the number of a 's is at most q^{n-1} we get at most ℓq^{n-1} roots for $p(x)$ in this way.

Summing the two cases gives at most

$$(d - \ell)q^{n-1} + \ell q^{n-1} = dq^{n-1}$$

roots for $p(x)$, as asserted. □

Lemma 35.2. *For every set $E \subseteq F^n$ of size $|E| < \binom{n+d}{d}$ there is a nonzero polynomial $p(x) \in F[x_1, \dots, x_n]$ of degree at most d that vanishes on E .*

Proof. Consider the vector space V_d of all polynomials in $F[x_1, \dots, x_n]$ of degree at most d . A basis for V_d is provided by the monomials $x_1^{s_1} \dots x_n^{s_n}$ with $\sum s_i \leq d$:

$$1, x_1, \dots, x_n, x_1^2, x_1x_2, \dots, x_1^3, \dots, x_n^d.$$

The following pleasing argument shows that the number of monomials $x_1^{s_1} \dots x_n^{s_n}$ of degree at most d equals the binomial coefficient $\binom{n+d}{d}$. What we want to count is the number of n -tuples $(s_1, \dots, s_n)$ of nonnegative integers with $s_1 + \dots + s_n \leq d$. To do this, we map every n -tuple $(s_1, \dots, s_n)$ to the increasing sequence

$$s_1 + 1 < s_1 + s_2 + 2 < \dots < s_1 + \dots + s_n + n,$$

which determines an n -subset of $\{1, 2, \dots, d + n\}$. The map is bijective, so the number of monomials is $\binom{n+d}{d}$.

Next look at the vector space F^E of all functions $f : E \rightarrow F$; it has dimension $|E|$, which by assumption is less than $\binom{n+d}{d} = \dim V_d$. The evaluation map $p(x) \mapsto (p(a))_{a \in E}$ from V_d to F^E is a linear map of vector spaces. We conclude that it has a nonzero kernel, containing as desired a nonzero polynomial that vanishes on E . $\square$

Theorem 35.3 (finite Kakeya problem). *Let $K \subseteq F^n$ be a Kakeya set. Then*

$$|K| \geq \binom{|F| + n - 1}{n} \geq \frac{|F|^n}{n!}.$$

Proof. The second inequality is clear from the definition of binomial coefficients. For the first, set again $q = |F|$ and suppose for a contradiction that

$$|K| < \binom{q + n - 1}{n} = \binom{n + q - 1}{q - 1}.$$

By Lemma 35.2 there exists a nonzero polynomial $p(x) \in F[x_1, \dots, x_n]$ of degree $d \leq q - 1$ that vanishes on K . Let us write

$$p(x) = p_0(x) + p_1(x) + \dots + p_d(x), \tag{1}$$

where $p_i(x)$ is the sum of the monomials of degree i ; in particular, $p_d(x)$ is nonzero. Since $p(x)$ vanishes on the nonempty set K , we have $d > 0$. Take any $v \in F^n \setminus \{0\}$. By the Kakeya property for this v there exists a $w \in F^n$ such that

$$p(w + tv) = 0 \quad \text{for all } t \in F.$$

Here comes the trick: Consider $p(w + tv)$ as a polynomial in the single variable t . It has degree at most $d \leq q - 1$ but vanishes on all q points of F , whence $p(w + tv)$ is the zero polynomial in t . Looking at (1) above we see that the coefficient of t^d in $p(w + tv)$ is precisely $p_d(v)$, which must therefore be 0. But $v \in F^n \setminus \{0\}$ was arbitrary and $p_d(0) = 0$ since $d > 0$, and we conclude that $p_d(x)$ vanishes on all of F^n . Since

$$dq^{n-1} \leq (q - 1)q^{n-1} < q^n,$$

Lemma 35.1, however, tells us that $p_d(x)$ must then be the zero polynomial — contradiction and end of the proof. $\square$

Chapter 36

Completing Latin squares

Lemma 36.1. *Any $(r \times n)$ -Latin rectangle, $r < n$, can be extended to an $((r + 1) \times n)$ -Latin rectangle and hence can be completed to a Latin square.*

Proof. We apply Hall's theorem 30.4 (see Chapter 30). Let A_j be the set of numbers that do not appear in column j . An admissible $(r + 1)$ -st row corresponds then precisely to a system of distinct representatives for the collection $A_1, \dots, A_n$. To prove the lemma we therefore have to verify Hall's condition (H). Every set A_j has size $n - r$, and every element is in precisely $n - r$ sets A_j (since it appears r times in the rectangle). Any m of the sets A_j contain together $m(n - r)$ elements and therefore at least m different ones, which is just condition (H). $\square$

Lemma 36.2. *Let P be a partial Latin square of order n with at most $n - 1$ cells filled and at most $\frac{n}{2}$ distinct elements, then P can be completed to a Latin square of order n .*

Proof. TODO $\square$

Theorem 36.3 (Smetaniuk's theorem). *Any partial Latin square of order n with at most $n - 1$ filled cells can be completed to a Latin square of the same order.*

Proof. $\square$

Chapter 37

Permanents and the power of entropy

Theorem 37.1. Let $M = (m_{ij})$ be an $n \times n$ matrix with entries in $\{0, 1\}$, and let $d_1, \dots, d_n$ be the row sums of M , that is, $d_i = \sum_{j=1}^n m_{ij}$. Then

$$\text{per } M \leq \prod_{i=1}^n (d_i!)^{1/d_i}.$$

Proof. TODO

□

Theorem 37.2. The number $L(n)$ of Latin squares of order n is bounded by

$$\frac{n!^{2n}}{n^{n^2}} \leq L(n) \leq \prod_{k=1}^n k!^{n/k}$$

Proof. TODO

□

37.1 Appendix: More about entropy

Theorem 37.3 (Fact A).

$$H(X) \leq \log_2(|\text{supp } X|).$$

Proof. TODO

□

Theorem 37.4 (Fact B).

$$H(X, Y) = H(X) + H(Y|X).$$

Proof. TODO

□

Theorem 37.5 (Fact B).

$$H(Y|X) \leq \sum_{j=1}^d \text{Prop}(X \in E_j) \log_2 j.$$

Proof. TODO

□

Chapter 38

The Dinitz problem

Definition 38.1. Let $\vec{G} = (V, E)$ be a directed graph. A *kernel* $K \subseteq V$ is a subset of the vertices such that

- (i) K is independent in G , and
- (ii) for every $u \notin K$ there exists a vertex $v \in K$ with an edge $u \rightarrow v$.

Lemma 38.2. Let $\vec{G} = (V, E)$ be a directed graph, and suppose that for each vertex $v \in V$ we have a color set $C(v)$ that is larger than the outdegree, $|C(v)| \geq d^+(v) + 1$. If every induced subgraph of $\vec{G}$ possesses a kernel, then there exists a list coloring of G with a color from $C(v)$ for each v .

Proof. We proceed by induction on $|V|$. For $|V| = 1$ there is nothing to prove. Choose a color $c \in \mathcal{C} = \bigcup_{v \in V} C(v)$ and set

$$A(c) := \{v \in V : c \in C(v)\}.$$

By hypothesis, the induced subgraph $G_{A(c)}$ possesses a kernel $K(c)$. Now we color all $v \in K(c)$ with the color c (this is possible since $K(c)$ is independent), and delete $K(c)$ from G and c from $\mathcal{C}$. Let G' be the induced subgraph of G on $V \setminus K(c)$ with $C'(v) = C(v) \setminus \{c\}$ as the new list of color sets. Notice that for each $v \in A(c) \setminus K(c)$, the outdegree $d^+(v)$ is decreased by at least 1 (due to condition (ii) of a kernel). So $d^+(v) + 1 \leq |C'(v)|$ still holds in $\vec{G}'$. The same condition also holds for the vertices outside $A(c)$, since in this case the color sets $C(v)$ remain unchanged. The new graph G' contains fewer vertices than G , and we are done by induction. $\square$

Definition 38.3. A matching M of $G = (X \cup Y, E)$ is called *stable* if the following condition holds: Whenever $uv \in E \setminus M$, $u \in X$, $v \in Y$, then either $uy \in M$ with $y > v$ in $N(u)$ or $xv \in M$ with $x > u$ in $N(v)$, or both.

Lemma 38.4. A stable matching always exists.

Proof. Consider the following algorithm. In the first stage all men $u \in X$ propose to their top choice. If a girl receives more than one proposal she picks the one she likes best and keeps him on a string, and if she receives just one proposal she keeps that one on a string. The remaining men are rejected and form the reservoir R . In the second stage all men in R propose to their next choice. The women compare the proposals (together with the one on the string, if there is one), pick their favorite and put him on the string. The rest is rejected and forms the new set R . Now the men in R propose to their next choice, and so on. A man who has proposed to his last

choice and is again rejected drops out from further consideration (as well as from the reservoir). Clearly, after some time the reservoir R is empty, and at this point the algorithm stops.

Claim. When the algorithm stops, then the men on the strings together with the corresponding girls form a stable matching.

Notice first that the men on the string of a particular girl move there in increasing preference (of the girl) since at each stage the girl compares the new proposals with the present mate and then picks the new favorite. Hence if $uv \in E$ but $uv \notin M$, then either u never proposed to v in which case he found a better mate before he even got around to v , implying $uy \in M$ with $y > v$ in $N(u)$, or u proposed to v but was rejected, implying $xv \in M$ with $x > u$ in $N(v)$. But this is exactly the condition of a stable matching. $\square$

Theorem 38.5. *We have $\chi_\ell(S_n) = n$ for all n .*

Proof. As before we denote the vertices of S_n by (i, j) , $1 \leq i, j \leq n$. Thus (i, j) and (r, s) are adjacent if and only if $i = r$ or $j = s$. Take any Latin square L with letters from $\{1, 2, \dots, n\}$ and denote by $L(i, j)$ the entry in cell (i, j) . Next make S_n into a directed graph $\vec{S}_n$ by orienting the horizontal edges $(i, j) \rightarrow (i, j')$ if $L(i, j) < L(i, j')$ and the vertical edges $(i, j) \rightarrow (i', j)$ if $L(i, j) > L(i', j)$. Thus, horizontally we orient from the smaller to the larger element, and vertically the other way around. (In the margin we have an example for $n = 3$.)

Notice that we obtain $d^+(i, j) = n - 1$ for all (i, j) . In fact, if $L(i, j) = k$, then $n - k$ cells in row i contain an entry larger than k , and $k - 1$ cells in column j have an entry smaller than k .

By Lemma 38.2 it remains to show that every induced subgraph of $\vec{S}_n$ possesses a kernel. Consider a subset $A \subseteq V$, and let X be the set of rows of L , and Y the set of its columns. Associate to A the bipartite graph $G = (X \cup Y, A)$, where every $(i, j) \in A$ is represented by the edge ij with $i \in X, j \in Y$. In the example in the margin the cells of A are shaded.

The orientation on $\vec{S}_n$ naturally induces a ranking on the neighborhoods in $G = (X \cup Y, A)$ by setting $j' > j$ in $N(i)$ if $(i, j) \rightarrow (i, j')$ in $\vec{S}_n$ respectively $i' > i$ in $N(j)$ if $(i, j) \rightarrow (i', j)$. By Lemma 38.4, $G = (X \cup Y, A)$ possesses a stable matching M . This M , viewed as a subset of A , is our desired kernel! To see why, note first that M is independent in A since for edges in $G = (X \cup Y, A)$ they do not share an endvertex i or j . Secondly, if $(i, j) \in A \setminus M$, then by the definition of a stable matching there either exists $(i, j') \in M$ with $j' > j$ or $(i', j) \in M$ with $i' > i$, which for $\vec{S}_n$ means $(i, j) \rightarrow (i, j') \in M$ or $(i, j) \rightarrow (i', j) \in M$, and the proof is complete. $\square$

Chapter 39

Five-coloring plane graphs

Theorem 39.1. *All planar graphs G can be 5-colored:*

$$\chi_\ell(G) \leq 5.$$

Proof. TODO

□

Chapter 40

How to guard a museum

Theorem 40.1. *For any museum with n walls, $\lfloor \frac{n}{3} \rfloor$ guards suffice.*

Proof. First of all, let us draw $n - 3$ noncrossing diagonals between corners of the walls until the interior is triangulated. For example, we can draw 9 diagonals in the museum depicted in the margin to produce a triangulation. It does not matter which triangulation we choose, any one will do. Now think of the new figure as a plane graph with the corners as vertices and the walls and diagonals as edges.

Claim. *This graph is 3-colorable.*

For $n = 3$ there is nothing to prove. Now for $n > 3$ pick any two vertices u and v which are connected by a diagonal. This diagonal will split the graph into two smaller triangulated graphs both containing the edge uv . By induction we may color each part with 3 colors where we may choose color 1 for u and color 2 for v in each coloring. Pasting the colorings together yields a 3-coloring of the whole graph.

The rest is easy. Since there are n vertices, at least one of the color classes, say the vertices colored 1, contains at most $\lfloor \frac{n}{3} \rfloor$ vertices, and this is where we place the guards. Since every triangle contains a vertex of color 1 we infer that every triangle is guarded, and hence so is the whole museum.

□

Chapter 41

Turán's graph theorem

Theorem 41.1 (First Proof). *If a graph $G = (V, E)$ on n vertices has no p -clique, $p \geq 2$, then*

$$|E| \leq \left(1 - \frac{1}{p-1}\right) \frac{n^2}{2}. \quad (1)$$

Proof. We use induction on n . One easily computes that (1) is true for $n < p$. Let G be a graph on $V = \{v_1, \dots, v_n\}$ without p -cliques with a maximal number of edges, where $n \geq p$. G certainly contains $(p-1)$ -cliques, since otherwise we could add edges. Let A be a $(p-1)$ -clique, and set $B := V \setminus A$.

A contains $\binom{p-1}{2}$ edges, and we now estimate the edge-number e_B in B and the edge-number $e_{A,B}$ between A and B . By induction, we have $e_B \leq \frac{1}{2} \left(1 - \frac{1}{p-1}\right) (n-p+1)^2$. Since G has no p -clique, every $v_j \in B$ is adjacent to at most $p-2$ vertices in A , and we obtain $e_{A,B} \leq (p-2)(n-p+1)$. Altogether, this yields

$$|E| \leq \binom{p-1}{2} + \frac{1}{2} \left(1 - \frac{1}{p-1}\right) (n-p+1)^2 + (p-2)(n-p+1),$$

which is precisely $\left(1 - \frac{1}{p-1}\right) \frac{n^2}{2}$. □

Theorem 41.2 (Second Proof). *If a graph $G = (V, E)$ on n vertices has no p -clique, $p \geq 2$, then*

$$|E| \leq \left(1 - \frac{1}{p-1}\right) \frac{n^2}{2}. \quad (1)$$

Proof. This proof makes use of the structure of the Turán graphs. Let $v_m \in V$ be a vertex of maximal degree $d_m = \max_{1 \leq j \leq n} d_j$. Denote by S the set of neighbors of v_m , $|S| = d_m$, and set $T := V \setminus S$. As G contains no p -clique, and v_m is adjacent to all vertices of S , we note that S contains no $(p-1)$ -clique.

We now construct the following graph H on V (see the figure). H corresponds to G on S and contains all edges between S and T , but no edges within T . In other words, T is an independent set in H , and we conclude that H has again no p -cliques. Let d'_j be the degree of v_j in H . If $v_j \in S$, then we certainly have $d'_j \geq d_j$ by the construction of H , and for $v_j \in T$, we see $d'_j = |S| = d_m \geq d_j$ by the choice of v_m . We infer $|E(H)| \geq |E|$, and find that among all graphs with a maximal number of edges, there must be one of the form of H . By induction, the graph induced by S has at most as many edges as a suitable graph $K_{n_1, \dots, n_{p-2}}$ on S . So $|E| \leq |E(H)| \leq E(K_{n_1, \dots, n_{p-1}})$ with $n_{p-1} = |T|$, which implies (1). □

Theorem 41.3 (Third Proof). *If a graph $G = (V, E)$ on n vertices has no p -clique, $p \geq 2$, then*

$$|E| \leq \left(1 - \frac{1}{p-1}\right) \frac{n^2}{2}. \quad (1)$$

Proof. Consider a *probability distribution* $\mathbf{w} = (w_1, \dots, w_n)$ on the vertices, that is, an assignment of values $w_i \geq 0$ to the vertices with $\sum_{i=1}^n w_i = 1$. Our goal is to maximize the function

$$f(\mathbf{w}) = \sum_{v_i v_j \in E} w_i w_j.$$

Suppose $\mathbf{w}$ is any distribution, and let v_i and v_j be a pair of nonadjacent vertices with positive weights w_i, w_j . Let s_i be the sum of the weights of all vertices adjacent to v_i , and define s_j similarly for v_j , where we may assume that $s_i \geq s_j$. Now we move the weight from v_j to v_i , that is, the new weight of v_i is $w_i + w_j$, while the weight of v_j drops to 0. For the new distribution $\mathbf{w}'$ we find

$$f(\mathbf{w}') = f(\mathbf{w}) + w_j s_i - w_j s_j \geq f(\mathbf{w}).$$

We repeat this (reducing the number of vertices with a positive weight by one in each step) until there are no nonadjacent vertices of positive weight anymore. Thus we conclude that there is an optimal distribution whose nonzero weights are concentrated on a clique, say on a k -clique. Now if, say, $w_1 \geq w_2 > 0$, then choose $w'_1 = w_1 - \varepsilon w_1 - w_2$ and change w_1 to $w_1 - \varepsilon$ and w_2 to $w_2 + \varepsilon$. The new distribution $\mathbf{w}'$ satisfies $f(\mathbf{w}') = f(\mathbf{w}) + \varepsilon(w_2 s_1 - w_1 s_2) \geq f(\mathbf{w})$, and we infer that the maximal value of $f(\mathbf{w})$ is attained for $w_i = 1/k$ on a k -clique and $w_i = 0$ otherwise. Since a k -clique contains $\binom{k}{2}$ edges, we obtain

$$f(\mathbf{w}) = \binom{k}{2} \frac{1}{k^2} = \frac{1}{2} \left(1 - \frac{1}{k}\right).$$

Since this expression is increasing in k , the best we can do is to set $k = p - 1$ (since G has no p -cliques). So we conclude

$$f(\mathbf{w}) \leq \frac{1}{2} \left(1 - \frac{1}{p-1}\right)$$

for any distribution $\mathbf{w}$. In particular, this inequality holds for the *uniform* distribution given by $w_i = \frac{1}{n}$ for all i . Thus we find

$$\frac{|E|}{n^2} = f\left(\mathbf{w} = \frac{1}{n}\right) \leq \frac{1}{2} \left(1 - \frac{1}{p-1}\right),$$

which is precisely (1). □

Theorem 41.4 (Fourth Proof). *If a graph $G = (V, E)$ on n vertices has no p -clique, $p \geq 2$, then*

$$|E| \leq \left(1 - \frac{1}{p-1}\right) \frac{n^2}{2}. \quad (1)$$

Proof. This time we use some concepts from probability theory. Let G be an arbitrary graph on the vertex set $V = \{v_1, \dots, v_n\}$. Denote the degree of v_i by d_i , and write $\omega(G)$ for the number of vertices in a largest clique, called the clique number of G .

Claim. We have $\omega(G) \geq \sum_{i=1}^n \frac{1}{n-d_i}$.

We choose a random permutation $\pi = v_1 v_2 \dots v_n$ of the vertex set V , where each permutation is supposed to appear with the same probability $\frac{1}{n!}$, and then consider the following set C_π . We

put v_i into C_π if and only if v_i is adjacent to all v_j ($j < i$) preceding v_i . By definition, C_π is a clique in G . Let $X = |C_\pi|$ be the corresponding random variable. We have $X = \sum_{i=1}^n X_i$, where X_i is the indicator random variable of the vertex v_i , that is, $X_i = 1$ or $X_i = 0$ depending on whether $v_i \in C_\pi$ or $v_i \notin C_\pi$. Note that v_i belongs to C_π with respect to the permutation $v_1 v_2 \dots v_n$ if and only if v_i appears before all $n - 1 - d_i$ vertices which are not adjacent to v_i , or in other words, if v_i is the first among v_i and its $n - 1 - d_i$ non-neighbors. The probability that this happens is $\frac{1}{n - d_i}$, hence $EX_i = \frac{1}{n - d_i}$.

Thus by linearity of expectation (see ?) we obtain

$$E(|C_\pi|) = EX = \sum_{i=1}^n EX_i = \sum_{i=1}^n \frac{1}{n - d_i}.$$

Consequently, there must be a clique of at least that size, and this was our claim. To deduce Turán's theorem from the claim we use the Cauchy-Schwarz inequality from Chapter 20,

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

Set $a_i = \sqrt{n - d_i}$, $b_i = \frac{1}{\sqrt{n - d_i}}$, then $a_i b_i = 1$, and so

$$n^2 \leq \left(\sum_{i=1}^n (n - d_i) \right) \left(\sum_{i=1}^n \frac{1}{n - d_i} \right) \leq \omega(G) \sum_{i=1}^n (n - d_i). \quad (2)$$

At this point we apply the hypothesis $\omega(G) \leq p - 1$ of Turán's theorem. Using also $\sum_{i=1}^n d_i = 2|E|$ from the chapter on double counting, inequality (2) leads to

$$n^2 \leq (p - 1)(n^2 - 2|E|),$$

and this is equivalent to Turán's inequality. □

Theorem 41.5 (Fifth Proof). *If a graph $G = (V, E)$ on n vertices has no p -clique, $p \geq 2$, then*

$$|E| \leq \left(1 - \frac{1}{p - 1} \right) \frac{n^2}{2}. \quad (1)$$

Proof. Let G be a graph on n vertices without a p -clique and with a maximal number of edges.

Claim. G does not contain three vertices u, v, w such that $vw \in E$, but $uv \notin E$, $uw \notin E$.

Suppose otherwise, and consider the following cases.

Case 1: $d(u) < d(v)$ or $d(u) < d(w)$. We may suppose that $d(u) < d(v)$. Then we duplicate v , that is, we create a new vertex v' which has exactly the same neighbors as v (but v' is not an edge), delete u , and keep the rest unchanged. The new graph G' has again no p -clique, and for the number of edges we find

$$|E(G')| = |E(G)| + d(v) - d(u) > |E(G)|,$$

a contradiction.

Case 2: $d(u) \geq d(v)$ and $d(u) \geq d(w)$. Duplicate u twice and delete v and w (as illustrated in the margin). Again, the new graph G' has no p -clique, and we compute (the -1 results from the edge vw):

$$|E(G')| = |E(G)| + 2d(u) - (d(v) + d(w) - 1) > |E(G)|.$$

So we have a contradiction once more. A moment's thought shows that the claim we have proved is equivalent to the statement that

$$u \sim v : \Longleftrightarrow uv \notin E(G)$$

defines an equivalence relation. Thus G is a complete multipartite graph, $G = K_{n_1, \dots, n_{p-1}}$, and we are finished. $\square$

Theorem 41.6 (Five proofs of Turán's graph theorem). *Collecting the proofs from the chapter...*

Proof. $\square$

Chapter 42

Communicating without errors

Theorem 42.1. *Whenever $T = \{v^{(1)}, \dots, v^{(m)}\}$ is an orthonormal representation of G with constant σ_T , then*

$$\Theta(G) \leq \frac{1}{\sigma_T}.$$

Proof. TODO

□

Chapter 43

The chromatic number of Kneser graphs

Theorem 43.1 (Lyusternik–Shnirel’man). *If the d -sphere S^d is covered by $d + 1$ sets,*

$$S^d = U_1 \cup \dots \cup U_d \cup U_{d+1},$$

such that each of the first d sets $U_1, \dots, U_d$ is either open or closed, then one of the $d + 1$ sets contains a pair of antipodal points $x^, -x^*$.*

Proof. Let a covering $S^d = U_1 \cup \dots \cup U_d \cup U_{d+1}$ be given as specified, and assume that there are no antipodal points in any of the sets U_i . We define a map $f : S^d \rightarrow \mathbb{R}^d$ by

$$f(x) := (\delta(x, U_1), \delta(x, U_2), \dots, \delta(x, U_d)).$$

Here $\delta(x, U_i)$ denotes the distance of x from U_i . Since this is a continuous function in x , the map f is continuous. Thus the Borsuk–Ulam theorem tells us that there are antipodal points $x^*, -x^*$ with $f(x^*) = f(-x^*)$. Since U_{d+1} does not contain antipodes, we get that at least one of x^* and $-x^*$ must be contained in one of the sets U_i , say in U_k ($k \leq d$). After exchanging x^* with $-x^*$ if necessary, we may assume that $x^* \in U_k$. In particular this yields $\delta(x^*, U_k) = 0$, and from $f(x^*) = f(-x^*)$ we get that $\delta(-x^*, U_k) = 0$ as well.

If U_k is closed, then $\delta(-x^*, U_k) = 0$ implies that $-x^* \in U_k$, and we arrive at the contradiction that U_k contains a pair of antipodal points.

If U_k is open, then $\delta(-x^*, U_k) = 0$ implies that $-x^*$ lies in $\overline{U_k}$, the closure of U_k . The set U_k , in turn, is contained in $S^d \setminus \overline{U_k}$, since this is a closed subset of S^d that contains U_k . But this means that $-x^*$ lies in $S^d \setminus \overline{U_k}$, so it cannot lie in $\overline{U_k}$, and x^* cannot lie in U_k , a contradiction. $\square$

Theorem 43.2 (Gale’s theorem). *There is an arrangement of $2k + d$ points on S^d such that every open hemisphere contains at least k of these points.*

Proof. $\square$

Theorem 43.3 (Kneser’s conjecture). *We have*

$$\chi(K(2k + d, k)) = d + 2.$$

Proof. For our ground set let us take $2k+d$ points in general position on the sphere S^{d+1} . Suppose the set $V(n, k)$ of all k -subsets of this set is partitioned into $d+1$ classes, $V(n, k) = V_1 \dot{\cup} \dots \dot{\cup} V_{d+1}$. We have to find a pair of disjoint k -sets A and B that belong to the same class V_i .

For $i = 1, \dots, d+1$ we set

$$O_i = \{x \in S^{d+1} : \text{the open hemisphere } H_x \text{ with pole } x \text{ contains a } k\text{-set from } V_i\}.$$

Clearly, each O_i is an open set. Together, the open sets O_i and the closed set $C = S^{d+1} \setminus (O_1 \cup \dots \cup O_{d+1})$ cover S^{d+1} . Invoking Lyusternik–Shnirel’man (43.1) we know that one of these sets contains antipodal points x^* and $-x^*$. This set cannot be C ! Indeed, if $x^*, -x^* \in C$, then by the definition of the O_i ’s, the hemispheres H_{x^*} and H_{-x^*} would contain fewer than k points. This means that at least $d+2$ points would be on the equator $H_{x^*} \cap H_{-x^*}$ with respect to the north pole x^* , that is, on a hyperplane through the origin. But this cannot be since the points are in general position. Hence some O_i contains a pair $x^*, -x^*$, so there exist k -sets A and B both in class V_i , with $A \subset H_{x^*}$ and $B \subset H_{-x^*}$.

But since we are talking about open hemispheres, H_{x^*} and H_{-x^*} are disjoint, hence A and B are disjoint, and this is the whole proof. $\square$

43.1 Appendix: A proof sketch for the Borsuk–Ulam theorem

Theorem 43.4. *For every continuous map $f : S^d \rightarrow \mathbb{R}^d$ from d -sphere to d -space, there are antipodal points $x^*, -x^*$ that are mapped to the same point $f(x^*) = f(-x^*)$.*

Proof. TODO $\square$

Chapter 44

Of friends and politicians

Theorem 44.1. *Suppose that G is a finite graph in which any two vertices have precisely one common neighbor. Then there is a vertex which is adjacent to all other vertices.*

Proof. Suppose the assertion is false, and G is a counterexample, that is, no vertex of G is adjacent to all other vertices. To derive a contradiction, we proceed in two steps. The first part is combinatorics, and the second part is linear algebra.

(1) We claim that G is a regular graph, that is, $d(u) = d(v)$ for any $u, v \in V$.

Note first that the condition of the theorem implies that there are no cycles of length 4 in G . Let us call this the C_4 -condition.

We first prove that any two *nonadjacent* vertices u and v have equal degree $d(u) = d(v)$. Suppose $d(u) = k$, where $w_1, \dots, w_k$ are the neighbors of u . Exactly one of the w_i , say w_2 , is adjacent to v , and w_2 is adjacent to exactly one of the other w_i 's, say w_1 , so that we have the situation of the figure to the left. The vertex v has with w_1 the common neighbor w_2 , and with w_i ($i \geq 2$) a common neighbor z_i ($i \geq 2$). By the C_4 -condition, all these z_i must be distinct. We conclude $d(v) \geq k = d(u)$, and thus $d(u) = d(v) = k$ by symmetry.

To finish the proof of (1), observe that any vertex different from w_2 is not adjacent to either u or v , and hence has degree k , by what we already proved. But since w_2 also has a non-neighbor, it has degree k as well, and thus G is k -regular.

Summing over the degrees of the k neighbors of u we get k^2 . Since every vertex (except u) has exactly one common neighbor with u , we have counted every vertex once, except for u , which was counted k times. So the total number of vertices of G is

$$n = k^2 - k + 1.$$

(2) The rest of the proof is a beautiful application of some standard results of linear algebra. Note first that k must be greater than 2, since for $k \leq 2$ only $G = K_1$ and $G = K_3$ are possible by (1), both of which are trivial windmill graphs. Consider the adjacency matrix $A = (a_{ij})$, as defined on page 282. By part (1), any row has exactly k 1's, and by the condition of the theorem, for any two rows there is exactly one column where they both have a 1. Note further that the main diagonal consists of 0's. Hence we have

$$A^2 = \begin{pmatrix} k & 1 & \dots & 1 \\ 1 & k & & \\ \vdots & & \ddots & \\ 1 & \dots & & k \end{pmatrix} = (k-1)I + J,$$

where I is the identity matrix, and J the matrix of all 1's. It is immediately checked that J has the eigenvalues n (of multiplicity 1) and 0 (of multiplicity $n-1$). It follows that A^2 has the eigenvalues $k-1+n=k^2$ (of multiplicity 1) and $k-1$ (of multiplicity $n-1$).

Since A is symmetric and hence diagonalizable, we conclude that A has the eigenvalues k (of multiplicity 1) and $\pm\sqrt{k-1}$. Suppose r of the eigenvalues are equal to $\sqrt{k-1}$ and s of them are equal to $-\sqrt{k-1}$, with $r+s=n-1$. Now we are almost home. Since the sum of the eigenvalues of A equals the trace (which is 0), we find

$$k + r\sqrt{k-1} - s\sqrt{k-1} = 0,$$

and, in particular, $r \neq s$, and

$$\sqrt{k-1} = \frac{k}{s-r}.$$

Now if the square root $\sqrt{m}$ of a natural number m is rational, then it is an integer! An elegant proof for this was presented by Dedekind in 1858: Let n_0 be the smallest natural number with $n_0\sqrt{m} \in \mathbb{N}$. If $\sqrt{m} \notin \mathbb{N}$, then there exists $\ell \in \mathbb{N}$ with $0 < \sqrt{m} - \ell < 1$. Setting $n_1 := n_0(\sqrt{m} - \ell)$, we find $n_1 \in \mathbb{N}$ and $n_1\sqrt{m} = n_0(\sqrt{m} - \ell)\sqrt{m} = n_0m - \ell(n_0\sqrt{m}) \in \mathbb{N}$. With $n_1 < n_0$ this yields a contradiction to the choice of n_0 .

Returning to our equation, let us set $h = \sqrt{k-1} \in \mathbb{N}$, then

$$h(s-r) = k = h^2 + 1.$$

Since h divides $h^2 + 1$ and h^2 , we find that h must be equal to 1, and thus $k = 2$, which we have already excluded. So we have arrived at a contradiction, and the proof is complete. $\square$

Chapter 45

Probability makes counting (sometimes) easy

Theorem 45.1. *Every family of at most 2^{d-1} d -sets is 2-colorable, that is, $m(d) > 2^{d-1}$.*

Proof. TODO

□

Theorem 45.2. *Every family of at most 2^{d-1} d -sets is 2-colorable, that is, $m(d) > 2^{d-1}$.*

Proof. TODO

□

Theorem 45.3. *For every $k \geq 2$, there exists a graph G with chromatic number $\chi(G) > k$ and girth $\gamma(G) > k$.*

Proof. TODO

□

Theorem 45.4. *Let G be a simple graph with n vertices and m edges, where $m \geq 4n$. Then*

$$\text{cr}(G) \geq \frac{1}{64} \frac{m^3}{n^2}.$$

Proof. TODO

□